

Settlement and Reporting Manual

BASE24-pos[®]



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Contents

Preface	ix
Conventions Used in This Manual	xiii
1: Introduction	1-1
The BASE24-pos Product Role in Settlement	1-2
BASE24-pos Settlement	1-3
Defining the Network Participants	1-4
Cardholders	1-4
Terminals	1-5
Retailers	1-5
Retailer Sponsors	1-6
Card Issuers	1-7
Interchanges	1-7
BASE24-pos Business Day	1-9
Terminal Cutover	1-10
Retailer Cutover	1-13
Institution Cutover	1-13
Logical Network Cutover	1-14
Interchange Cutover	1-14
BASE24-pos Cutover and Reporting Database	1-16
POS Transaction Log File (PTLF)	1-16
POS Institution Reporting File (PIRF)	1-17
POS Retailer Reporting File (PRRF)	1-18
Interchange Log File (ILF)	1-18
Other Database Files	1-18
BASE24-pos Settlement Processes	1-20
Main Settlement Process	1-20
Secondary Settlement Process	1-21
Timers	1-22

Introduction *continued*

BASE24-pos Report Overview	1-24
Daily Reports	1-24
Periodic Reports	1-26
Interchange Reports	1-26
ACI Commerce Gateway Support	1-27
Broadcast Notification	1-28

2: Retailer Terminal Support. 2-1

Terminal Totals	2-2
Service Totals	2-2
Terminal Totals	2-3
Clerk Totals	2-4
Updating Service/Terminal/Clerk Totals	2-5
Terminal Balancing	2-7
Balancing Transactions	2-7
Balancing Scenarios	2-9
Terminal Cutover	2-11
Retailer Cutover Window	2-12
Terminal Cutover Options	2-13
Clearing BASE24-pos Terminal Data Files Totals at Cutover	2-16
Terminal Reconciliation	2-21
The B24POS35 Report	2-22
Incomplete and Out-of-Balance Batches	2-24
BASE24-pos Draft Capture	2-28
Authorization Only	2-28
Authorization and Draft Capture	2-29
Authorization with Electronic Follow-up	2-29
Terminal-Defined Draft Capture	2-30
Transaction Identification	2-30

3: Settlement Support. 3-1

Retailer Settlement	3-2
Crediting Retailers	3-2
Card Issuer Settlement	3-6

Card Issuer Settlement <i>continued</i>	
Posting Cardholder Accounts	3-6
Settling with Acquirers	3-6
Logical Network Settlement	3-8
Interchange Settlement	3-10
4: BASE24-pos Reporting	4-1
Daily Network Settlement Report (B24POS01)	4-3
Daily Cardholder Activity Report (B24POS02)	4-7
Daily Cardholder Exception Report (B24POS02-EXPT)	4-18
Daily Clerk Totals Report (B24POS04)	4-20
Daily Disputed Transactions Report (B24POS05)	4-24
Daily Retailer Summary Report (B24POS11)	4-29
Daily Retailer Activity Report (B24POS12)	4-34
Daily Retailer Exception Detail Report (B24POS12-EXPT)	4-45
Network Referral Statistical Report (B24POS30)	4-47
Retailer Reconciliation Report (B24POS35)	4-54
Periodic Institution Summary Report (B24POS21)	4-63
Periodic Retailer Summary Report (B24POS22)	4-67
Clearings Statement Report (SETL01)	4-71
Transaction Detail Report (STAT09)	4-76
5: Configuring Cutover and Reporting	5-1
Configuration Options	5-2
LCONF Assigns and Params	5-4
Configuring the Settlement Processes	5-6
Main Settlement Process	5-6
Secondary Settlement Processes	5-7
Senior Secondary Settlement Process	5-8
Configuring Cutover	5-9
Terminal Cutover	5-9
Retailer Cutover	5-11

Configuring Cutover <i>continued</i>	
Institution Cutover	5-12
Logical Network Cutover	5-13
Interchange Cutover	5-14
Configuring Reports	5-16
PTLF Alternate Keys	5-16
Logical Network Configuration File	5-17
POS Retailer Definition File	5-21
Institution Definition File	5-21
Interchange Configuration Files	5-22
COBNAMES File	5-22
6: Cutover and Reporting Operations	6-1
Producing BASE24-pos Daily Reports	6-2
Creating the BASE24-pos Daily Report Obey File	6-2
Example Report Obey File	6-3
Manually Starting BASE24-pos Daily Reports	6-4
Suppressing BASE24-pos Daily Reports	6-4
Producing BASE24-pos Periodic Reports	6-5
Deleting PIRF and PRRF Records	6-5
POS Report Generator (PRG) Screen	6-5
Requesting a Weekly Periodic Report	6-9
Requesting a Biweekly Periodic Report	6-9
Requesting a Monthly Periodic Report	6-10
Requesting a Quarterly Periodic Report	6-10
Producing BASE24 Interchange Reports	6-11
Manually Starting BASE24 Interchange Reports	6-11
Suppressing BASE24 Interchange Reports	6-12
A: Processing Information	A-1
Process Configuration	A-2
Multiple Copies	A-2
Startup Logic	A-2
Initialization	A-4
Warmboot Processing	A-9
Warmbooting the Main Settlement Process	A-9

Warmboot Processing <i>continued</i>	
Warmbooting One Secondary Settlement Process	A-10
Warmbooting All Secondary Settlement Processes	A-10
Warmbooting the Main and Secondary Settlement Processes	A-10
Timer Processing.	A-12
Retailer Cutover Timer	A-12
Logical Network Cutover Timer.	A-18
Start Reports Timer.	A-23
Text Commands	A-26
B: Recovery Procedures	B-1
Downed Secondary Settlement Process	B-2
Scenario 1	B-2
Scenario 2	B-2
Scenario 3	B-3
Scenario 4	B-3
Scenario 5	B-4
Scenario 6	B-4
Downed Main Settlement Process	B-5
Scenario 1	B-5
Scenario 2	B-5
Scenario 3	B-6
Scenario 4	B-6
C: BASE24-pos Terminal Data Files Totals	C-1
Terminal Cutover Options.	C-3
Example 1 - Network Force Cutover	C-5
Example 2 - Device Handler Force Cutover	C-7
Example 3 - Close-Batch Cutover with Close-Batch Balancing	C-10
Example 4 - Close-Batch Cutover with Batch, Shift, and Day Balancing	C-14
Example 5 - Close-Day Cutover with Batch, Shift, and Day Balancing	C-19
Index	Index-1

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Preface

The BASE24-pos product provides settlement and reporting tools that enable institutions to settle more efficiently and accurately with each other. The settlement and reporting tools include settlement reports, statistical reports, and database reports. The BASE24-pos product also ensures that settlement entities are cut over to new business days at appropriately configured times.

This manual describes the settlement and reporting tools available with the BASE24-pos product. It also describes the functionality within the BASE24-pos product that guides and controls cutover and reporting actions.

Audience

This manual is intended for the BASE24 operational staff who monitor and control BASE24-pos cutover, settlement, and reporting. It also provides settlement initialization and cutover procedures for technical support personnel.

Additional Documentation

The BASE24 documentation set is arranged so that each BASE24 manual presents a topic or group of related topics in detail. When one BASE24 manual presents a topic that has already been covered in detail in another BASE24 manual, the topic is summarized and the reader is directed to the other manual for additional information. Information has been arranged in this manner to be more efficient for readers that do not need the additional detail and at the same time provide the source for readers that require the additional information. This manual contains references to the following BASE24 publications:

- The ***BASE24 Event Message Reference Manual*** provides detailed information about event messages.
- The ***BASE24 Logical Network Configuration File Manual*** provides information about how to access the Logical Network Configuration File and describes the assigns and params that are configured in the Logical Network Configuration File.

- The ***BASE24 Text Command Reference Manual*** provides detailed descriptions and the command syntax for warmboot commands.
- The ***BASE24-pos Address Verification Manual*** provides a listing of interchange-specific address verification processing values.
- The ***BASE24-pos Files Maintenance Manual*** provides data entry screens and field descriptions for POS terminal files.
- The ***BASE24-pos Multiple Currency Support Manual*** provides detailed information on the BASE24-pos Multiple Currency add-on product.
- The ***BASE24-pos Transaction Processing Manual*** provides descriptions about how POS transactions are processed and describes the BASE24-pos internal message (PSTM).

Software

This manual documents standard processing as of its publication date. Software that is not current and custom software modifications (CSMs) may result in processing that differs from the material presented in this manual. The customer is responsible for identifying and noting these changes.

Manual Summary

The ***BASE24-pos Settlement and Reporting Manual*** is divided into the following sections:

Section 1, “Introduction,” provides an overview of BASE24-pos settlement and reporting, the Main and Secondary Settlement processes, the network participants, and how transaction activity is reported.

Section 2, “Retailer Terminal Support,” describes PTD service, terminal, and clerk totals; terminal balancing, cutover, and reconciliation; and draft capture processing.

Section 3, “Settlement Support,” describes the support the BASE24-pos product provides for retailer, institution, logical network, and interchange settlement.

Section 4, “BASE24-pos Reporting,” describes the daily, periodic, and interchange reports produced by the BASE24-pos product.

Section 5, “Configuring Cutover and Reporting,” describes the configuration tasks that a user must perform to establish BASE24-pos cutover and reporting.

Section 6, “Cutover and Reporting Operations,” describes the operations that a user must perform when managing BASE24-pos settlement and reporting on a daily basis.

Appendix A, “Processing Information,” contains supplementary information for process configuration, a description of process initialization and warmboot, descriptions of timer processing, and a list of network control text commands that the BASE24-pos settlement process can receive.

Appendix B, “Recovery Procedures,” contains procedures to follow when the Main Settlement process or Secondary Settlement processes go down.

Appendix C, “PTD Totals,” contains examples of how PTD totals are impacted in different transaction processing situations.

Publication Identification

Three entries appearing at the bottom of each page uniquely identify this BASE24 publication. The publication number (for example, PS-AE000-06 for the ***BASE24-pos Settlement and Reporting Manual***) appears on every page to assist readers in identifying the manual from which a page of information was printed. The publication date (for example, Nov-2007 for November, 2007) indicates the issue of the manual. The software release information (for example, R6.0v8 for release 6.0, version 8) specifies the software that the manual describes. This information matches the document information on the copyright page of the manual.

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Conventions Used in This Manual

This section explains how field descriptions, blank characters, and unlabeled fields on screens are documented in this manual.

Field Descriptions

Each field appearing on an operations screen is listed by name and then described. Field descriptions briefly summarize the contents, purpose, and permissible values, as shown in the following example taken from the POS Report Generator (PRG) screen.

Example

REPORT NUMBER — The number identifying the report to be generated.
Valid values are:

- 21 = Periodic Institution Summary Report (B24POS21)
- 22 = Periodic Retailer Summary Report (B24POS22)

Field Length: 2 numeric characters
Required Field: Yes
Default Value: 21
Data Name: Not applicable

Explanation

Each field description is completed by one or more of the following items of information:

Item	Description
Example	Illustrates a possible entry for the field to further clarify the value or values that can be entered.

Item	Description
Field Length	Specifies the size of the field and the type of characters that can be entered. This length refers to the field and valid values for the file maintenance screen, not the field in the DDLs. Possible values are alphabetic, alphanumeric, hexadecimal, and numeric. The term <i>alphanumeric</i> includes all alphabetic, numeric, and special characters that can be entered from a keyboard without using a control sequence. The term <i>hexadecimal</i> includes all numbers and the letters A through F. When a field value cannot be modified by the operator, the field length is System-protected.
Occurs	Indicates the number of times the field can be displayed on the screen. This information is provided only when the field can be displayed multiple times.
Required Field	Specifies whether a value has to be entered in the field. Possible values are Yes and No. Some fields are required only under certain conditions. In this case, the entry is Yes, followed by the conditions that determine when the field is required.
Default Value	Specifies the value that is automatically placed in the field when the screen is first displayed or when the F8 key is pressed to clear the screen.
Data Name	Provides the DDL name associated with the field appearing on the screen. Data names are included in the documentation to assist in communicating screen and field issues to your technical staff. Note that screen data is not always stored in the BASE24 database as it appears on the screen or as it is described in the field description. If you need information on how screen data is actually stored, consult the DDLs.

Blank Characters

Field descriptions for internal and external messages, tokens, files maintenance screens, and reports often include lists of valid values. When the value contains a blank character, this manual uses the symbol *b* to indicate the blank character.

Unlabeled Fields on Screens

Angle brackets (< >) indicate an unlabeled field on a screen. An unlabeled field is a field that is present on a screen but is not preceded by an identifying literal label. For the purposes of documenting the field, a label has been assigned and appears inside the angle brackets. A multiple line unlabeled field is displayed as a shaded area and also has a label in angle brackets.

In the field descriptions for the screen, the unlabeled field appears according to its place on the screen and is identified by the same label.

Unlabeled fields are not included in the index by field name; they appear by subject in the main index.

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1: Introduction

This section introduces the reader to BASE24-pos cutover, settlement, and reporting procedures. The topics discussed in this section include the following:

- The BASE24-pos product role in cutover, settlement, and reporting
- The financial entities (or network participants) involved in settlement
- The BASE24-pos business day
- How transaction activity is reported
- Overviews of the BASE24-pos Main Settlement process and Secondary Settlement processes
- Descriptions of the files used in processing settlement information
- ACI Commerce Gateway support

The BASE24-pos Product Role in Settlement

The BASE24-pos product role in settlement is one of transaction management and reporting. The BASE24-pos product defines the reporting day, logs and keeps track of each day's transaction activity, provides a daily file of transactions for back-end processing by a host, and reports the funds due from and owed to the various logical network participants.

Although BASE24-pos processing and reporting is designed specifically to enhance network participants' internal settlement and accounting, participating BASE24-pos institutions are themselves responsible for the tangible aspects of settlement. They must debit and credit their cardholder and retailer accounts, settle amongst themselves, and settle with interchanges associated with the network. The BASE24-pos product facilitates this settlement effort by providing a file of transactions it has processed during the business day, the POS Transaction Log File (PTLF), along with a set of settlement and statistical reports.

BASE24-pos Settlement

The BASE24-pos network is *cut over* and *settled* each day. Settlement, in this context, is actually a reconciliation effort whereby the current day's transaction activity is closed out and reported and the system is shown to be in balance.

BASE24-pos settlement differs from the settlement carried out by its various network participants in that the BASE24-pos product does not move funds to *settle* accounts. It does, however, provide the support which allows network participants to initiate this movement of funds.

Defining the Network Participants

As an online point-of-sale (POS) system, the BASE24-pos product interacts with a number of financial entities. Each of these financial entities, or network participants, plays a unique role in the initiation and processing of POS transactions and each has its own unique settlement requirements. The network participants that the BASE24-pos product supports are listed and described on the following pages.

Cardholders

Cardholders are individuals possessing debit or credit cards that can be used to initiate purchases and payments of services at card-accepting retail outlets.

The BASE24-pos product does not consider individual cardholders to be settlement entities. This means that the BASE24-pos product does not compute a daily settlement position for each cardholder as it does for the other network participants. The BASE24-pos product does, however, log all cardholder transactions in a daily transaction file, the POS Transaction Log File (PTLF), which allows institutions to post these transactions to card-issuer accounts.

Besides the purchase and payment of services, cardholders also affect settlement in that they can dispute transactions posted to their accounts. If a cardholder thinks that a transaction has been erroneously posted from a retailer, he or she can initiate a *chargeback*. In this case, a card-issuing institution returns the disputed transaction to the retailer. In effect, the transaction is charged back to the retailer, making the retailer responsible for the payment.

If the retailer has researched a disputed transaction that has been charged back and found it to be accurately posted, the retailer can then initiate a *representation*. A representation reposts the transaction to the cardholder account. In this way, the retailer indicates that the cardholder is still responsible for payment of the transaction or service, and that the retailer's records show the transaction to be posted correctly as it stands.

Terminals

Point-of-sale (POS) terminals, or devices, are machines where retailers initiate financial transactions on behalf of cardholders. Retailers can also initiate administrative and balancing transactions at POS terminals. The POS terminals at a particular retailer “belong” to a retailer sponsor financial institution and are defined within the BASE24-pos system by a POS terminal identification number.

Retailers

Retailers are card-accepting entities in the BASE24-pos network. When a customer makes a purchase or pays for a service using a debit or credit card, the retailer initiates a transaction to authorize, log, and electronically capture (optional) the transaction for settlement.

The BASE24-pos product allows for processing retailer transactions as draft capture or non-draft capture. Draft capture transactions are electronically-captured transactions that are sufficient in themselves to enact a funds transfer for the transaction amount.

Non-draft capture transactions are transactions for which the paper ticket represents the actual settlement instrument. These paper tickets must be physically presented for deposit before the transaction can be settled. While the BASE24-pos system may have authorized the transaction, the BASE24-pos record cannot be used to effect settlement of the item.

The BASE24-pos product identifies draft capture transactions on its transaction files and includes draft capture totals on its retailer reports so that the retailer sponsor can settle with the retailer.

The BASE24 system also allows retailers the option of tracking individual operator activity occurring at a particular terminal during the business day. At the end of the business day, a report can be generated that contains individual clerk’s transaction totals within each retailer.

The BASE24-pos product also reports disputed transactions for each retailer’s customers using a disputed transactions report, which the retailer can use to monitor chargebacks and representations.

The BASE24-pos product provides the following settlement and statistical reports for retailers. These reports are documented in section 4.

- Daily Clerk Totals Report (B24POS04)
- Daily Disputed Transactions Report (B24POS05)
- Daily Retailer Activity Report (B24POS12)
- Daily Retailer Exception Detail Report (B24POS12-EXPT)
- Periodic Retailer Summary Report (B24POS22)
- Retailer Reconciliation Report (B24POS35)

Retailer Sponsors

A retailer sponsor is the institution that credits the retailer for its debit and credit card activity. A retailer sponsor is responsible for crediting retailer accounts and then clearing the retailer items through the appropriate settlement channels. Typically, retailer sponsors are banks. However, they can also be large retail organizations, with individual outlets being defined as retailers.

The BASE24-pos product creates a daily file of transaction activity, the POS Transaction Log File (PTLF), along with a set of reports that retailer sponsors can use when settling with their retailers for the day's draft capture transactions and also with the other financial entities in the BASE24-pos network. The retailer sponsor can also extract and process the PTLF to create transaction strip files for settlement with national bank card settlement networks.

The BASE24-pos product also enables retailer sponsors to keep track of disputed transactions affecting their retailers. Retailer sponsors track disputed transactions by monitoring a disputed transactions report that lists all the chargebacks and representations the BASE24-pos product has processed during the business day.

The BASE24-pos product provides the following settlement and statistical reports for retailer sponsors. These reports are documented in section 4.

- Daily Network Settlement Report (B24POS01)
- Daily Disputed Transactions Report (B24POS05)
- Daily Retailer Summary Report (B24POS11)
- Periodic Retailer Summary Report (B24POS22)
- Retailer Reconciliation Report (B24POS35)

Card Issuers

Card issuers are financial institutions that issue debit or credit cards for use by their customers.

The BASE24-pos product provides the means for posting cardholder transactions to card issuer accounts and also, using the BASE24-pos reports, for settling with the retailer organizations accepting the cards.

The BASE24-pos product logs all transaction activity to a daily file, the POS Transaction Log File (PTLF), which can be extracted and used for posting cardholder transactions. The BASE24-pos product also computes a daily settlement position for each institution in the network. For a card issuer, this figure includes the net amount the card issuer owes the network for its cardholders' transactions.

In addition, the BASE24-pos product provides an audit report containing cardholder activity for each card issuer. This report contains a daily detail listing of cardholder transactions in cardholder number order and enables card issuers to research the transactions in cases of customer service problems or cardholder disputes (disputed transactions include chargebacks and representations).

The BASE24-pos product provides the following settlement and statistical reports for card issuers. These reports are documented in section 4.

- Daily Network Settlement Report (B24POS01)
- Daily Cardholder Activity Report (B24POS02)
- Daily Cardholder Exception Report (B24POS02-EXPT)
- Periodic Institution Summary Report (B24POS21)
- Network Referral Statistical Report (B24POS30)

Interchanges

Interchanges are national or regional networks that, for a fee, enable transactions to be authorized and settled between otherwise independent retailer and card-issuing organizations.

Interchanges differ from other BASE24-pos network entities in their settlement requirements. BASE24-pos reporting takes these differences into consideration.

Some interchanges (typically debit card interchanges) accept the electronic transactions as actual settlement vehicles. These interchanges assume that an authorizing entity both authorizes and posts transactions to its cardholder accounts when the transaction is authorized. The BASE24-pos product recognizes and reports a daily net settlement position for these interchanges.

Other interchanges separate the authorization and settlement of transactions. Typically, credit card interchanges (e.g., Visa, MasterCard, and American Express) require that transactions be settled through separate bank card settlement networks. The BASE24-pos product does not recognize or report settlement positions for these interchanges. The BASE24-pos product does, however, log all transaction activity in a daily file, the POS Transaction Log File (PTLF), which allows for creating transaction strip files for electronic settlement between BASE24-pos retailer sponsors and these interchanges.

The BASE24-pos product provides the following settlement and statistical reports for interchanges. These reports are produced using the Interchange Log File (ILF), which maintains transactions according to the interchange settlement day. These reports are documented in section 4.

- Interchange Clearings Statement Report (SETL01)
- Interchange Transaction Detail Report (STAT09)

BASE24-pos Business Day

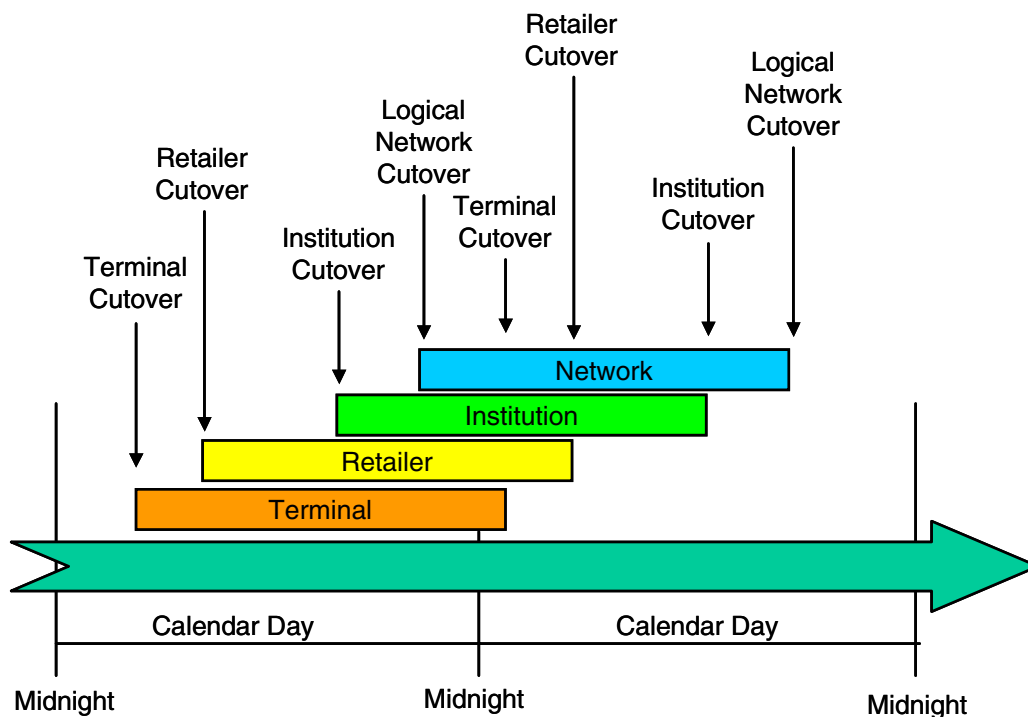
The BASE24-pos system manages transaction activity according to its business day. Transactions the BASE24-pos system processes during its defined business day are included in that day's POS Transaction Log File (PTLF) and, consequently, on that day's reports.

The BASE24-pos business day is defined by a series of daily *cutovers*. Cutover refers to the closing of the current posting day and opening of the next. It is referred to in connection with terminals, retailers, institutions, interchanges, and the BASE24-pos logical network as a whole. In all cases, cutover is the point in time at which the BASE24-pos system stops logging transactions to the current day's business and starts logging transactions to the next day's business.

Four daily cutovers exist in the BASE24-pos system that affect the business day: terminal cutover, retailer cutover, institution cutover, and logical network cutover. These cutovers enforce an orderly movement from one business day to the next. A separate cutover exists for interchanges. The BASE24 settlement processes are responsible for all four daily cutovers, whereas interchange cutover can occur at any time based on the requirements of the interchange. The BASE24 system's respective Interchange Interface processes are responsible for interchange cutovers.

The diagram on the following page illustrates how the four daily cutovers are related: a retailer's terminals must all cut over prior to the retailer cutting over, all of an institution's retailers must cut over prior to the institution cutting over, and all institutions in the logical network must cut over prior to the logical network cutting over. Interchange cutover (not pictured) is governed by each interchange.

The BASE24-pos Reporting Day



Terminal Cutover

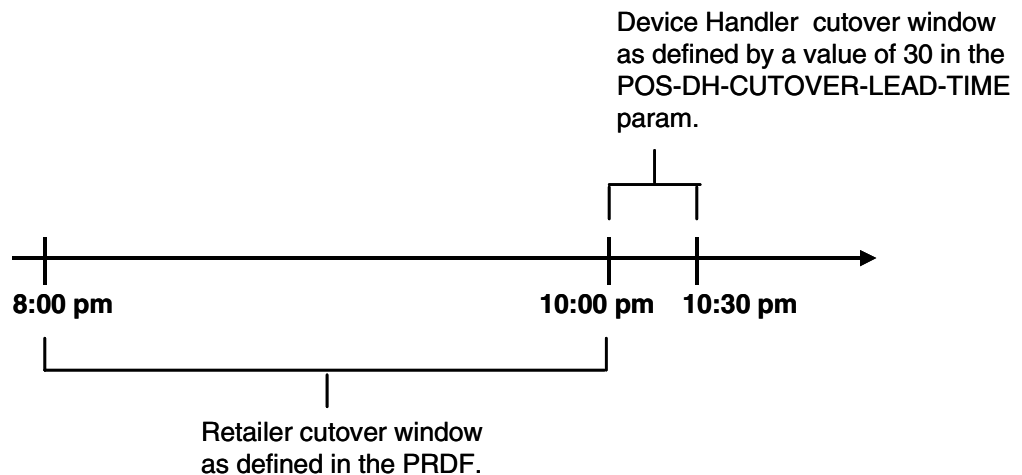
From the retailer's point of view, terminal cutover is the time at which the BASE24-pos business day begins. When a terminal is cut over, the BASE24-pos product updates the posting date in the appropriate BASE24-pos Terminal Data file record and logs all subsequent business from that terminal to the next BASE24-pos business day. Thus, as a retailer's terminals are cut over, the BASE24-pos product closes the retailer's current business day and begins the next business day.

The BASE24-pos product also provides terminal balancing support. Terminal balancing is different than terminal cutover. Terminal balancing, if supported by the terminal, can occur any number of times during a day and does not affect the terminal's business date. Terminal cutover, on the other hand, occurs only once per day and changes the terminal's business date. For more information about terminal balancing, see section 2.

Retailer terminals must be cut over each day during either the retailer cutover window or the Device Handler cutover window. Each retailer defines its own cutover window by specifying the window's start time and end time in the POS Retailer Definition File (PRDF).

The Device Handler cutover window is a time period that falls within the retailer cutover window. It is defined by the retailer cutover window's end time and the POS-DH-CUTOVER-LEAD-TIME param in the Logical Network Configuration File (LCONF). The POS-DH-CUTOVER-LEAD-TIME param defines the number of minutes prior to the end of the retailer cutover window that the Device Handler cutover window begins.

For example, a retailer could define its retailer cutover window as the time period between 8:00 pm and 10:30 pm. If the POS-DH-CUTOVER-LEAD-TIME param in the LCONF is set to a value of 30, the Device Handler cutover window would start at 10:00 pm and end at 10:30 pm. The relationship between the retailer cutover window and the Device Handler cutover window in this example is illustrated below.



The cutover window used depends on the method by which the terminal is cut over. The method by which a retailer's terminals must be cut over is defined in the BASE24-pos Terminal Data files (PTD). Each of these methods is described below.

Operator-Initiated Cutover

If a terminal is configured to cut over with a balancing transaction and the BASE24-pos system receives the specified balancing transaction during the retailer's cutover window, the Device Handler module cuts the terminal over. Once a terminal is cut over, it cannot be cut over again until the next business day.

If the BASE24-pos system does not receive the specified balancing transaction during the retailer's cutover window, the Secondary Settlement process forces the terminal to cut over at the end of the retailer's cutover window. In this case, the terminal is cut over in the same manner as described for those terminals set up to be forced to cutover automatically each day by the Secondary Settlement process.

Only terminals capable of supporting balancing transactions can be cut over by a balancing transaction. Terminals that do not support balancing transactions cannot be cut over by a balancing transaction.

Device Handler Force Cutover

If a terminal is configured to be forced to cut over by the Device Handler module, the Device Handler module cuts the terminal over when the first transaction from the terminal is received within the Device Handler cutover window. Any transaction, as long as it is the first transaction initiated within the Device Handler cutover window, causes the Device Handler module to force the terminal to cut over.

If the Device Handler module does not receive a transaction from the terminal within the Device Handler cutover window, the Secondary Settlement process forces the terminal to cut over at the end of the retailer's cutover window. In this case, the terminal is cut over in the same manner as described for those terminals set up to be forced to cut over automatically each day by the Secondary Settlement process.

Settlement Force Cutover

If a terminal is set up to be forced to cut over by the Secondary Settlement process, the Secondary Settlement process automatically cuts over the terminal at the end of the retailer's cutover window.

The Secondary Settlement process must always force cutover those terminals that do not support balancing transactions. Terminals that support balancing transactions can be forced to cut over by the Secondary Settlement process, can be cut over with a balancing transaction initiated by an operator, or can be cut over by the Device Handler module.

Retailer Cutover

Retailer cutover is essentially the combined cutovers of all terminals belonging to the retailer. At retailer cutover, the Secondary Settlement process ensures that all of the retailer's terminals have been cut over to a new posting day and increases the retailer's POS Retailer Definition File (PRDF) posting dates. The Secondary Settlement process automatically cuts the retailer over at the end of the retailer's cutover window defined in the PRDF.

The BASE24-pos product provides the flexibility to set the retailer cutover window according to each retailer's needs and business practices. However, the BASE24-pos business day revolves around logical network and institution cutover. All institutions must be cut over before the logical network and all retailers must be cut over before institution cutover.

All retailers that process on the same day as the logical network must have cutover times that precede financial institution cutover by at least the number of minutes specified in the POS-SETL-FI-CUTOVER-LEAD-TIME param in the LCONF. If the cutover window defined in the PRDF does not allow for the time indicated in this param, the Secondary Settlement process automatically cuts the retailer over this number of minutes prior to financial institution cutover.

All retailers that process one day ahead of the network must wait for a specified interval after logical network cutover before they can begin cutting over. This time interval is specified in the POS-SETL-TERM-CUTOVER-DELAY-TIME param. If the cutover window defined in the PRDF does not allow for the time indicated in this param, the Secondary Settlement process automatically cuts the retailer over this number of minutes after logical network cutover.

Institution Cutover

Institution cutover consists of a number of institution-level actions. BASE24-pos retailer sponsors and card issuers are defined at the institution level. Actions directly related to BASE24-pos settlement include increasing the processing dates in the institution's Institution Definition File (IDF) record and verifying that all of the institution's retailers have been cut over.

Each institution controls its cutover time in the IDF. Like a retailer, an institution sets a window during which it can cut over to a new business day. At the end of this window, the Main Settlement process increases the institution's posting dates to the next day. If any of the institution's retailers have not been cut over by institution cutover, the BASE24-pos system performs force cutover actions to automatically cut the retailers over. In this way, institution cutover ensures that all retailers have been cut over to the new reporting day.

The BASE24-pos reporting day revolves around logical network and institution cutover. All institutions must be cut over before the logical network cuts over and all retailers must be cut over before the institution cuts over.

If the institution's cutover time is not set properly in the IDF, the Main Settlement process uses the value in the POS-SETL-LN-CUTOVER-LEAD-TIME param in the LCONF to determine the minimum number of minutes before logical network cutover that all institutions must be cut over. If an IDF record contains an institution cutover time that is less than the minimum number of minutes specified by this param, the Main Settlement process automatically cuts the institution over this number of minutes prior to logical network cutover.

Logical Network Cutover

At logical network cutover the entire logical network moves to a new business day. The BASE24-pos system reinitializes using the new processing dates established at institution cutover. The BASE24-pos system closes the POS Transaction Log File (PTLF) for the day just ended and this file then is available for extracting and reporting. The BASE24-pos system also creates a new PTLF. The Main Settlement process creates the report obey file used to start the BASE24-pos report programs. If configured to do so, the Main Settlement process also starts the BASE24-pos reports. Reports can either be produced immediately after logical network cutover or they can be set to start at a later time.

The logical network cutover time and the time at which reports must be run are defined in the Logical Network Configuration File (LCONF).

Interchange Cutover

If an interchange is associated with the logical network, interchange cutover is also a part of the overall BASE24-pos settlement and reporting picture. However, each participating interchange establishes its own cutover procedures. The BASE24 system's Interchange Interface processes are responsible for the initiation and completion of BASE24 cutover procedures for their respective interchanges (this

includes starting the appropriate report programs for the generation of reports).
The settlement process takes no responsibility for any interchange cutover activity
or report generation.

BASE24-pos Cutover and Reporting Database

The BASE24-pos product uses several files for cutover and reporting purposes. Transactions that have been processed for the day are logged to the POS Transaction Log File (PTLF). The BASE24-pos product uses the Interchange Log File (ILF) to log transactions according to the interchange business day. Other files are used in daily settlement and reporting processing.

POS Transaction Log File (PTLF)

The POS Transaction Log File (PTLF) is at the center of the BASE24-pos settlement and reporting effort. The BASE24-pos system logs daily transaction activity in the PTLF, and this file forms the database for BASE24-pos settlement and reporting. The PTLF contains one record for each transaction, approved or denied, processed during a single BASE24-pos reporting day. The Main Settlement process creates the PTLF at logical network cutover for use by the Authorization process. The Main Settlement process also purges old PTLFs at logical network cutover based on the POS-SETL-PTLF-FILE-RETENTION param in the LCONF. This param identifies the number of days PTLF records are to be kept on the system.

The BASE24-pos system bases its business day on its various cutovers and each PTLF is representative of those transactions the BASE24-pos product recognizes for the defined business day. By extracting the PTLF, institutions can process the transactions against their retailer and cardholder accounts, affecting the day's settlement with these accounts. Transactions in the PTLF can be extracted for use by both BASE24-pos retailer sponsors and card issuers as required for posting transactions to their files, creating their own on-host reports, and clearing daily transaction totals with national bank card settlement networks.

Institutions can gain access to information in the PTLF by perusing the file online, running reports, or extracting the file for host reporting. Operators with appropriate access can peruse the PTLF at any time. However, to run reports or extract the file, the PTLF must be closed by a daily network cutover. Refer to the ***BASE24-pos Files Maintenance Manual*** for more information concerning PTLF perusal.

Multiple Currency Impacts

When the BASE24-pos Multiple Currency add-on product is installed, transactions are logged to the PTLF and extracted from the PTLF with amounts in the Transaction currency. The Transaction currency is established for each terminal in the CURRENCY CODE field on BASE24-pos Terminal Data file (PTD) screen 2.

The Multiple Currency, POS Balances, and Original Currency 6.0 tokens can also be extracted from the PTLF, according to configuration and availability. Therefore, these tokens can be used to convey the transaction amounts in all applicable currencies to host systems, if required for host processing.

PTLF Extract files may contain records in different currencies, as the BASE24-pos system can process transactions (and settlement data) that are acquired in multiple currencies.

During cutover from one business day to the next, the Secondary Settlement process writes terminal totals records from the PTD to the PTLF. The MC TALLING TYPE field on PTD screen 2 indicates whether these totals are written to the PTD as hash totals (i.e., as is with no currency conversion) or in the primary terminal currency.

Note: Currency conversion is not supported in BASE24-pos reports. Therefore, ACI recommends that BASE24-pos reports not be used in a multiple currency environment since they may not reflect accurate amounts.

For more information on the BASE24-pos Multiple Currency add-on product, refer to the ***BASE24-pos Multiple Currency Support Manual***.

POS Institution Reporting File (PIRF)

The POS Institution Reporting File (PIRF) contains records of the approved and declined transaction totals that appeared on the Cardholder Activity Report (B24POS02). Each time the Cardholder Activity Report is produced, a record containing the approved and declined transaction totals from that report is created and placed in the POS Institution Reporting File (PIRF).

The BASE24-pos product uses the records in the PIRF to produce periodic reports.

POS Retailer Reporting File (PRRF)

The POS Retailer Reporting File (PRRF) contains summary records of retailer and terminal activity that appeared on the Retailer Activity Report (B24POS12). Each time the Retailer Activity Report is produced, a record containing the activity totals from that report is created and placed in the POS Retailer Reporting File (PRRF).

The BASE24-pos product uses the records in the PRRF to produce periodic reports.

Interchange Log File (ILF)

The Interchange Log File (ILF) contains a record of each transaction processed by an interchange. These records are then used to produce the settlement and statistical reports for that interchange. A new ILF is created by the Interchange Interface process each business day when the interchange cuts over. Interchange cutover is based on the requirements of the interchange. The business day reflected is that of the interchange, rather than the BASE24 system. The Interchange Interface process creates and writes a record once for each transaction.

Other Database Files

The following database files are also used by the Main and Secondary Settlement processes when performing settlement, cutover, and reporting functions.

- POS Retailer Definition File (PRDF)

The Secondary Settlement process reads the POS Retailer Definition File (PRDF) to determine the retailer's balancing window, during which the terminals connected to the retailer and the retailer itself are cut over. The PRDF is a partitioned file so that it can be accessed by several Secondary Settlement processes simultaneously, thus reducing the amount of time required to complete retailer cutover.

- BASE24-pos Terminal Data files (PTD)

The Secondary Settlement process reads the BASE24-pos Terminal Data files at terminal cutover to obtain processing parameters, reset terminal totals, and reset the posting date.

For Standard POS Device Handler (SPDH) and Hypercom terminal types, the Secondary Settlement process reads the BASE24-pos Terminal Data Dynamic File—general data (PTDD1), which contains dynamic data such as terminal totals that changes during transaction processing. For all other terminal types, the Secondary Settlement process reads the POS Terminal Data File (PTDF).

- Institution Definition File (IDF)

The Main and Secondary Settlement processes read and update the processing dates in the Institution Definition File (IDF). If the institution is processing transactions using the Negative Authorization with Usage Accumulation method, the Main Settlement process also retrieves the name of the Usage Accumulation File (UAF) from the IDF to clear usage accumulation totals.

- Log Message Configuration File (LMCF)

The Log Message Configuration File (LMCF) contains event message information, such as text, routing code, and severity, that has been modified by the user. At initialization, the Main and Secondary Settlement processes build a table in memory from the LMCF records. When an event message is generated, the Main or Secondary Settlement process accesses the table to verify whether that event message has been modified. If the event message has been modified, the Main or Secondary Settlement process logs the modified information instead of the standard event message.

- Logical Network Configuration File (LCONF)

The Main and Secondary Settlement processes read the Logical Network Configuration File (LCONF) at initialization and during a warmboot to retrieve the assigns and params they need for processing.

- Usage Accumulation File (UAF)

If the Main Settlement process is set up to clear the usage accumulation for an institution, it purges data in the institution's Usage Accumulation File (UAF) at institution cutover or at midnight. The time that the Main Settlement process purges data depends on a field in the IDF.

- POS Referral File (PRF)

The Main Settlement process creates the POS Referral File (PRF), which is used by the Authorization module of the Device Handler/Router/Authorization process. The Main Settlement process also purges old PRFs.

BASE24-pos Settlement Processes

Settlement processing is divided among a Main Settlement process and a configurable number of Secondary Settlement processes. The Main Settlement process is responsible for certain aspects of settlement processing while the Secondary Settlement process is responsible for other aspects of settlement processing.

Each logical network must have one Main Settlement process. The logical network can have up to 16 Secondary Settlement processes. Each Secondary Settlement process is responsible for cutting over its own range of retailers. Thus, multiple Secondary Settlement processes enable the system to cut retailers over more quickly than if one process was responsible for cutting over all retailers. Logical networks that are issuer-only do not require any Secondary Settlement processes.

The Main and Secondary Settlement processes communicate directly with one another except in the case of startup and startup-complete messages, warmboot and warmboot-complete messages, and checkmaster messages. In the case of these messages, the Main and Secondary Settlement processes communicate using the XPNET process. When the Main or Secondary Settlement process sends messages to another process, the messages are sent using the XPNET process.

Main Settlement Process

One Main Settlement process must be defined for each logical network. A logical network cannot have more than one Main Settlement process.

The Main Settlement process is responsible for the following tasks:

- Opening the Secondary Settlement processes for communication, if Secondary Settlement processes are defined in the system
- Creating the report obey files for running reports, if reports are configured to be run
- Creating and purging the POS Transaction Log Files (PTLFs) and POS Referral Files (PRFs)
- Cutting over the financial institutions (FIID cutover)
- Cutting over the logical network

Secondary Settlement Process

Up to 16 Secondary Settlement processes can be defined for each logical network. An issuer-only logical network does not require any Secondary Settlement processes as there are no terminals or retailers.

If one or more Secondary Settlement processes are defined in the logical network, one of the Secondary Settlement processes can be configured as the Senior Secondary Settlement process, although it is not required. The Senior Secondary Settlement process is identified using the POS-SENIOR-SLAVE-PROCESS param in the LCONF. The Senior Secondary Settlement process is responsible for periodically polling the Main Settlement process and verifying that the Main Settlement process is running. The Senior Secondary Settlement process is also responsible for all the tasks of a Secondary Settlement process.

A Secondary Settlement process is responsible for the following tasks:

- Opening and closing the PTLF
- Cutting over terminals
- Cutting over retailers

Each Secondary Settlement process controls the settlement of one or more retailers in the system. Retailers are identified by retailer ID numbers. Each Secondary Settlement process must be assigned a range of retailer IDs, which indicates the retailers for which the Secondary Settlement process is responsible. For example, Secondary A might be responsible for cutting over retailers 1 through 15, while Secondary B might be responsible for retailers 16 through 30.

Note: The number of retailers for which each Secondary Settlement process can be responsible depends on the configuration and performance issues of the specific customer site.

Timers

The settlement actions of the Main and Secondary Settlement processes are controlled and triggered by timers. These timers are listed and briefly described below. For more detailed information on each of these timers, refer to appendix A.

Retailer Cutover Timer

The Secondary Settlement process is responsible for setting the retailer cutover timer. This timer indicates when a retailer must cut over. When the retailer cutover timer expires, the Secondary Settlement process ensures that all terminals belonging to the retailer whose timer expired are cut over. If any of the terminals are not cut over, the Secondary Settlement process forces them to cut over. The Secondary Settlement process then cuts over the retailer.

Financial Institution Cutover Timer

The Main Settlement process is responsible for setting the financial institution cutover timer. This timer indicates when a financial institution within the logical network must cut over. When the financial institution cutover timer expires, the Main Settlement process verifies that each of the Secondary Settlement processes have finished cutting over all the retailers for the financial institution. When all the retailers have been cut over, the Main Settlement process cuts over the financial institution.

Logical Network Cutover Timer

The Main Settlement process is responsible for setting the logical network cutover timer. This timer indicates when the logical network must cut over. When the logical network cutover timer expires, the Main Settlement process cuts over the entire logical network to a new business day.

Midnight Timer

The Main Settlement process is responsible for setting the midnight timer when the IDF record is configured for the Main Settlement process to perform midnight settlement processing. This timer indicates that the data in the Usage Accumulation File (UAF) must be purged at midnight. When the midnight timer

expires, the Main Settlement process clears the UAF totals and updates the business date in the Institution Definition File (IDF). The midnight timer is optional.

Start Reports Timer

The Main Settlement process is responsible for setting the start reports timer at logical network cutover. When this timer expires, the Main Settlement process runs the daily BASE24-pos settlement and statistical reports. The start reports timer is optional.

Alive Timer

The Senior Secondary Settlement process is responsible for setting the alive timer at initialization and during warmbooting. When this timer expires, the Senior Secondary Settlement process opens the Main Settlement process if it is not open already and sends an *alive* message to the Main Settlement process to verify that it is running. The Senior Secondary Settlement process then sets another alive timer. If the Main Settlement process is not running, an event message is generated.

Note: If a Senior Secondary Settlement process is not defined in the POS-SETL-SENIOR-SLAVE param in the Logical Network Configuration File (LCONF), then no Secondary Settlement process verifies that the Main Settlement process is open and running.

Secondary Settlement Process Cutover Timer

The Main Settlement process is responsible for setting the Secondary Settlement process cutover timer. This timer indicates that all Secondary Settlement processes must be cut over. When this timer expires, the Main Settlement process checks the slave table to verify that all Secondary Settlement processes have cut over. If a Secondary Settlement process has not cut over, the Main Settlement process sends the Secondary Settlement process a cutover message and resets the Secondary Settlement process cutover timer.

BASE24-pos Report Overview

The BASE24-pos product provides reports that include both settlement information and statistical information. The statistical reports are an accumulation of totals for approved and declined transactions categorized by type of transaction, terminal ID, retailer ID, institution, and logical network. The settlement reports show the net position of each institution to the logical network, thus enabling participating retailers and institutions to settle funds amongst themselves.

The BASE24-pos system generates reports using transactions accumulated during each retailer's and each institution's business day. Thus, daily reports contain transactions accumulated during a single business day. Periodic reports are printed for an historical period (i.e., the previous week or month) and contain summary data accumulated from all daily activity in the reporting period.

The BASE24 system also provides interchange reports using transactions accumulated during each interchange's business day. These reports show each interchange's position to the logical network.

Note: Currency conversion is not supported in BASE24-pos reports. Therefore, ACI recommends that BASE24-pos reports not be used in a multiple currency environment since they may not reflect accurate amounts.

Daily Reports

Daily reports contain information from the POS Transaction Log File (PTLF). The PTLF is configured with up to five keys. The BASE24-pos system produces daily reports only when the appropriate PTLF keys have been configured.

Refer to the "PTLF Alternate Keys" topic in section 5 for more information about alternate key requirements.

Refer to the *BASE24-pos Files Maintenance Manual* for more information about the PTLF and PTLF keys.

The daily reports are listed and briefly described below. Refer to section 4 for details on each report.

Daily Network Settlement Report (B24POS01) — Reflects the debit, credit, and net position of each institution in the logical network.

Daily Cardholder Activity Report (B24POS02) — Provides a detailed listing and a summary of each institution's cardholder activity. It contains approved and denied cardholder transactions, chargebacks, representations, and sales drafts.

Daily Cardholder Exception Report (B24POS02-EXPT) — Provides a detailed listing of each institution's cardholder activity that resulted in exception processing. The transactions included on this report have been approved and completed. However, for some reason the BASE24-pos system was unable to post the transactions to all the appropriate BASE24 files. As a result, the institution must review these transactions for possible manual posting.

Daily Clerk Totals Report (B24POS04) — Reflects each clerk's transaction activity at a particular terminal during the business day.

Daily Disputed Transaction Report (B24POS05) — Provides a listing of each chargeback and representation transaction entered into the BASE24-pos system during the business day. This report is produced only if the institution has the BASE24-pos CRT Administration module.

Daily Retailer Summary Report (B24POS11) — Reflects the total number and amount of transactions that occurred at a particular retailer during the business day.

Daily Retailer Activity Report (B24POS12) — Provides a detailed listing by terminal of all activity that occurred at the retailer during the business day. The report also provides a summary of the activity by shift, terminal, retailer, region, group, and institution.

Daily Retailer Exception Detail Report (B24POS12-EXPT) — Provides a detailed listing by terminal of all exception transaction activity that occurred at the retailer during the business day. The transactions included on this report have been approved and completed. However, for some reason the BASE24-pos system was unable to post the transactions to all the appropriate BASE24 files. As a result, the institution must review these transactions for possible manual posting.

Network Referral Statistical Report (B24POS30) — Provides a listing and summary of records in the POS Referral File (PRF). These records represent transactions that have been referred elsewhere for authorization. The report is produced only if the institution has the BASE24-pos CRT Authorization module.

Retailer Reconciliation Report (B24POS35) — Provides reconciliation figures for the retailer's terminals. This reports includes only draft capture transactions.

Periodic Reports

The periodic reports are listed and briefly described below. Refer to section 4 for details on each report.

Periodic Institution Summary Report (B24POS21) — Summarizes approved and declined cardholder transaction totals for variable periods of time.

Periodic Retailer Summary Report (B24POS22) — Summarizes retailer transaction activity for variable periods of time.

Interchange Reports

The interchange reports are listed and briefly described below. Refer to section 4 for details on each report.

Clearings Statement Report (SETL01) — Reflects the debit, credit, and net position of the logical network to the interchange.

Transaction Detail Report (STAT09) — Provides a complete listing of all interchange activity based on transactions logged to the Interchange Log File (ILF).

ACI Commerce Gateway Support

The ACI Commerce Gateway product does not send the posting date in messages sent to the ISO Host Interface process. When the ACI Commerce Gateway product is used with the BASE24-pos product, a POS-SETL-NOTIFY*identifier* param must be configured in the Logical Network Configuration File (LCONF) for the ISO Host Interface process.

Each POS-SETL-NOTIFY*identifier* param contains the symbolic name of an ISO Host Interface process that the Main Settlement process must notify at the time specified in the POS-LN-CUTOVER param.

When the POS-SETL-NOTIFY*identifier* param is configured for the ISO Host Interface process, the Main Settlement process sends a 9502 cutover command to the ISO Host Interface process. When the ISO Host Interface process receives this command, it updates a global posting date value that is used to set the posting date for a transaction when the posting date is not included in the transaction.

Broadcast Notification

Broadcast notification enables BASE24 users to effectively minimize Settlement Initiator process notify lists in the LCONF, replacing multiple params with a single service name. Without broadcast notification, BASE24 users must configure a separate param for each process to be notified.

Services can be configured in the following LCONF params used by the Settlement Initiator process for the BASE24-pos product: POS-SETL-EAF-NOTIFY, POS-SETL-MIDNIGHT-NOTIFY, and POS-SETL-NOTIFY. When broadcast notification is not in use, these params contain an individual process to be notified instead of a service.

For broadcast notification, the BROADCAST-NOTIFY param in the LCONF must be set to a value of 2 and all processes requiring a settlement notify message must have a service configured that matches the entry in the appropriate settlement notification param. A second LCONF param associated with broadcast notification, BROADCAST-DELAY, is optional.

If the BROADCAST-NOTIFY param is missing or contains an invalid value, the Settlement Initiator process for the BASE24-pos product treats information defined in the POS-SETL-EAF-NOTIFY, POS-SETL-MIDNIGHT-NOTIFY, and POS-SETL-NOTIFY params as individual process names instead of service names.

For more information about service-based routing, including broadcast messages, see the *NET24-XPNET Network Control Operations Guide*.

2: Retailer Terminal Support

This section describes the retailer terminal support that the BASE24-pos product provides. The topics discussed in this section are BASE24-pos Terminal Data files service and terminal totals, BASE24-pos Terminal Data files clerk totals, terminal balancing, terminal cutover, terminal reconciliation, and BASE24-pos draft capture.

Terminal Totals

The BASE24-pos product provides significant support for retailer terminals in terms of balancing, cutover, and reconciliation. This support is facilitated by a comprehensive set of totals that the BASE24-pos system maintains for each terminal.

The BASE24-pos system keeps track of individual terminal activity using the BASE24-pos Terminal Data files. For Standard POS Device Handler (SPDH) and Hypercom devices, terminal activity is tracked in the BASE24-pos Terminal Data Dynamic File—general data (PTDD1). For all other types of terminals, terminal activity is tracked in the POS Terminal Data File (PTDF). As transactions are processed, they are added to totals in the PTDD1 or PTDF. These totals are then used to report and reconcile terminal activity.

The BASE24-pos system keeps three sets of totals in the PTDD1 or PTDF: service totals, terminal totals, and clerk totals.

Service Totals

Service totals are totals by card type and can be accessed using BASE24-pos Terminal Data files (PTD) screens 11, 12, 13, 14, 15, and 16. One set of totals exists for each card type supported by the terminal. For example, if a terminal supports MasterCard and Visa, the BASE24-pos Terminal Data files would be set up to maintain service totals individually for each card type. If other cards are supported, these cards would have their own service totals as well.

Service totals include counts and amounts for approved debit, credit, and adjustment transactions entered at the terminal. These totals are used to assist in reconciling terminal activity. Each time a terminal batch is closed, the system creates a service totals record, writes it to the PTLF if configured to do so in the BASE24-pos Terminal Data files, and clears the service totals. These records in the PTLF are then used to create the B24POS35, Retailer Reconciliation Report, which is used for reconciling terminal activity at the batch level.

Service totals include only draft capture transaction activity unless a card type is set up for authorization-only processing, in which case the service totals contain authorization-only totals. However, the system uses only draft capture totals for terminal reconciliation. Authorization-only activity is not included in terminal reconciliation.

Note: Throughout this manual, service totals are described as being logged to the PTLF in a single record. This is true when 17 services or less are defined in the BASE24-pos Terminal Data files. However, if more than 17 services are defined in the BASE24-pos Terminal Data files, the BASE24 system logs an additional service totals record to the PTLF.

Terminal Totals

Terminal totals, accessed using BASE24-pos Terminal Data files (PTD) screen 4, track overall terminal activity. The system uses these totals to support terminal balancing and reconciliation. Terminal totals are written to the PTLF each time the terminal is cut over or, if configured to do so in the BASE24-pos Terminal Data files, each time a batch, shift, or day is closed. The Main Settlement process then uses these PTLF records to create the B24POS35, Retailer Reconciliation Report, for reconciling terminal activity.

The PTD terminal totals include activity for all card types and include separate draft capture figures. The draft capture figures are the only figures used for terminal reconciliation on the B24POS35 report.

Batch totals contain BASE24-pos totals for the current batch in progress at the terminal.

Shift totals contain BASE24-pos totals for the current shift in progress at the terminal. If a terminal does not balance by shift, these totals will match the batch totals.

Daily totals contain BASE24-pos totals for the day in progress at the terminal. If a terminal does not balance by day, these totals will match the shift totals.

Current network totals contain totals for the current BASE24-pos reporting day. These totals may, in some cases, differ from the daily terminal totals if a terminal is forced to cut over.

Previous network totals contain totals for the previous BASE24-pos reporting day.

Clerk Totals

Clerk totals are totals accumulated for each terminal operator who has logged on to the terminal during the usage accumulation period. Only the BASE24 Standard POS Device Handler (SPDH), the Merchant Host Interface (MHI) Device Handler, and special device terminals support the clerk totals option.

The CLERK TOTALS FLAG on PTD screen 5 indicates whether the BASE24-pos system maintains clerk totals for a particular terminal. If the value in this field is 1, 2, or 3, clerk totals are maintained. If the value in this field is 0, no clerk totals are maintained for the terminal. The valid options for this field are:

- 0 = No clerk totals are maintained
- 3 = All totals (debits, credits, adjustments, cash out, and checks)

Clerk totals, accessed using PTD screen 5, include counts and amounts for approved debit, credit, adjustment, cash out, and check transactions the terminal operator entered at the terminal. The retailer can use these totals as an audit trail for each clerk's transaction activity during the business day. Clerk totals are maintained only for approved transactions.

One set of totals exists for each clerk who has logged on to the terminal. For example, if clerk 1 logged on to the terminal during the first work shift of the day and clerk 2 logged on to the terminal for the second work shift of the day, the system would maintain separate clerk totals for each clerk. If other terminal operators log on to the same terminal, these operators would also have their own individual clerk totals. Clerk totals are tallied in the BASE24-pos Terminal Data files for the clerk working at the terminal. These totals are then written to the PTLF and cleared when a new clerk takes over.

In addition, each time a terminal batch, shift, or day is closed, or the Device Handler cuts over a terminal (normal or forced), the system creates a clerk totals record, writes it to the PTLF, and clears the clerk totals. This ensures that terminal totals match all clerks working at the terminal for a given business day.

Note: The Main and Secondary Settlement processes do not log clerk totals; only the Device Handler module logs clerk totals.

The Main Settlement process then uses the clerk totals records in the PTLF to create the B24POS04, Daily Clerk Totals Report, which tracks each clerk's activity during the business day. This report is broken out by clerk within retailer.

Retailer sponsors can peruse clerk totals in the PTLF from the BASE24-pos PTLF file maintenance screens. Refer to the ***BASE24-pos Files Maintenance Manual*** for more information on perusing the PTLF.

Updating Service/Terminal/Clerk Totals

The BASE24-pos Terminal Data files service, terminal, and clerk totals are divided into three basic categories: debits, credits, and adjustments. Clerk totals has a fourth and fifth category for cash out and check amounts. The BASE24-pos system updates these totals for each approved transaction initiated at the terminal. The table below illustrates, by transaction, how the debit, credit, adjustment, cash, and check totals are increased in the BASE24-pos Terminal Data files.

Clearance of these BASE24-pos Terminal Data files service, terminal, and clerk totals depends on how the terminal is balanced and cut over. Additional information on clearing totals is presented later in this section. Appendix C includes several examples demonstrating how the BASE24-pos Terminal Data files totals are cleared based on different methods of terminal balancing.

Note: The cash out totals in the following table are used only with clerk totals. Cash out transactions are cash advances, purchases with cash back, and adjustment purchases with cash back transactions.

Updating BASE24-pos Terminal Data Files Totals					
Transaction	Debit	Credit	Adjust	Cash Out	Check
Normal purchase	+T				
Purchase preauthorization	no impact				
Preauthorized purchase completion	+T				
Mail or telephone order	+T				
Merchandise return		+T			
Cash advance	+T			+C	
Card verification	no impact				

Updating BASE24-pos Terminal Data Files Totals					
Transaction	Debit	Credit	Adjust	Cash Out	Check
Balance inquiry	no impact				
Purchase with cash back	+T			+C	
Purchase adjustment	+N		+O		
Merchandise return adjustment		+N	– O		
Cash advance adjustment	+N		+O		
Purchase with cash back adjustment				+\\-AC	
Check verification					+T
Check guarantee					+T

Key

-
- + = the amount is added
 - = the amount is subtracted
 - T = the total amount of the transaction
 - N = the new amount of the adjustment
 - O = the old amount of the adjustment
 - C = the amount of cash dispersed
 - AC = the adjusted cash amount

Terminal Balancing

Terminal balancing is closely linked with BASE24-pos settlement because by recognizing the monetary positions of its retailer terminals, a retailer sponsor is, essentially, recognizing its settlement position with the retailer. Terminal balancing, therefore, is an opportunity to ensure that BASE24-pos totals agree with actual terminal totals.

The BASE24-pos product supports terminal balancing according to the type of terminal being used. *Intelligent* (or second- or third-generation) terminals can perform interactive balancing functions such as batch and shift balancing. *Nonintelligent* (or first-generation dial) terminals cannot perform interactive balancing functions.

For terminals that support them, the BASE24-pos product provides a series of interactive transactions for balancing terminals by batch, shift, and day. For terminals that cannot enter balancing transactions to balance the terminal with the system, either the Secondary Settlement process or the Device Handler module opens and closes batch totals for the terminal. In this case, all transactions from the terminal for the business day are a part of a single batch, shift, and day.

A batch, which is the lowest processing level, consists of a small number of transactions that are grouped together and sent to the system throughout the business day. Batches are used to reconcile between BASE24-pos totals and the device totals.

A shift, which is the next processing level, consists of all the transactions occurring during a specific work period during the retailer's business day. The retailer defines the number of hours in each shift.

A day, which is the highest processing level, represents the retailer's business day. The retailer's business day consists of all transactions occurring during the number of hours the retailer considers to be a day as opposed to the number of hours the logical network considers to be a day. In this way, the retailer can accurately track transaction activity.

Balancing Transactions

The BASE24-pos product supports a number of balancing transactions to assist in terminal balancing. The availability of these transactions is dependent on the functions supported by the retailer's terminals. Some terminals support all of these balancing transactions; some terminals support only one or two of them; and some

terminals do not support any, in which case the terminals cannot balance online with the system. The balancing transactions the BASE24-pos product provides are described below.

Subtotal Transaction

The subtotal transaction requests BASE24-pos Terminal Data files totals. It can be used when balancing a terminal to determine whether the system totals are in agreement with the actual totals at the terminal. This transaction can be entered at any time during the business day and is not logged. Also, it does not affect the BASE24-pos Terminal Data files totals. The BASE24 system supports this transaction for batch, shift, end-of-day, and clerk totals.

Close Batch Transaction

The close batch transaction closes the current batch and opens a new batch. When the BASE24-pos system receives a close batch transaction, the system creates a batch totals record from the terminal totals in the BASE24-pos Terminal Data files and writes it to the PTLF. The BASE24-pos system then clears the batch totals in the BASE24-pos Terminal Data files.

If the terminal can send its own batch totals, the system also compares the terminal totals to the BASE24-pos Terminal Data files totals. If the debits, credits, and adjustments do not match, the BASE24-pos system writes the terminal totals to the PTLF immediately following the BASE24-pos Terminal Data files totals.

When a batch is closed, the BASE24-pos system automatically writes a service totals record to the PTLF prior to the batch totals record (if configured to do so in the BASE24-pos Terminal Data files) and clears the service totals in the BASE24-pos Terminal Data files.

Close Shift Transaction

The close shift transaction ends the current shift and starts a new shift. When the BASE24-pos system receives a close shift transaction, the system creates a shift totals record from the terminal totals in the BASE24-pos Terminal Data files and writes it to the PTLF. The BASE24-pos system then clears the shift totals in the BASE24-pos Terminal Data files.

If the terminal can send its own shift totals, the BASE24-pos system also compares the terminal totals to the BASE24-pos Terminal Data files totals. If the debits, credits, and adjustments do not match, the BASE24-pos system writes the terminal totals to the PTLF immediately following the BASE24-pos Terminal Data files totals. Close shift transactions, when entered, must be immediately preceded by a close batch transaction. If not, the system denies the close shift transaction.

Close Day Transaction

The close day transaction ends the current terminal day and starts a new day. When the BASE24-pos system receives a close day transaction, the system creates a daily totals record from the terminal totals in the BASE24-pos Terminal Data files and writes it to the PTLF. The BASE24-pos system then clears the daily totals in the BASE24-pos Terminal Data files.

If the terminal can send its own daily totals, the BASE24-pos system also compares the terminal totals to the BASE24-pos Terminal Data files totals. If the debits, credits, and adjustments do not match, the BASE24-pos system writes the terminal totals to the PTLF immediately following the BASE24-pos Terminal Data files totals.

Close day transactions, when entered, must be immediately preceded by a close shift transaction. If not, the system denies the close day transaction. Close day transactions should be entered only once during a reporting day to maintain reporting integrity.

Balancing Scenarios

The functionality of the BASE24-pos product focuses on two aspects of terminal balancing. First, the BASE24-pos system verifies that the physical terminal totals agree with the totals the BASE24-pos system keeps for the terminal. Second, the BASE24-pos system closes the current batch, shift, or daily totals and opens the new ones.

BASE24-pos terminal balancing is tied to the capabilities of the terminal. Therefore, some terminals use more BASE24-pos functions than others.

Balancing Nonintelligent Terminals

Since nonintelligent terminals do not support balancing transactions, there is no balancing interaction between the terminal operator and the BASE24-pos system. Therefore, totals cannot be verified online between the terminal and the system. Nor can an operator close out the terminal using a close batch, close shift, or close day transaction. Instead, the system closes terminal totals when the terminal cuts over.

Balancing Intelligent Terminals

From certain intelligent terminals that support balancing transactions, terminal operators can enter a balancing transaction to get terminal subtotals from the BASE24-pos system, compare the subtotals to those at the terminal, and, if necessary, enter transactions to bring the BASE24-pos totals and terminal totals into agreement. Then, once the totals are in agreement between the terminal and system, the operator can initiate a balancing transaction to close the batch, shift, or day.

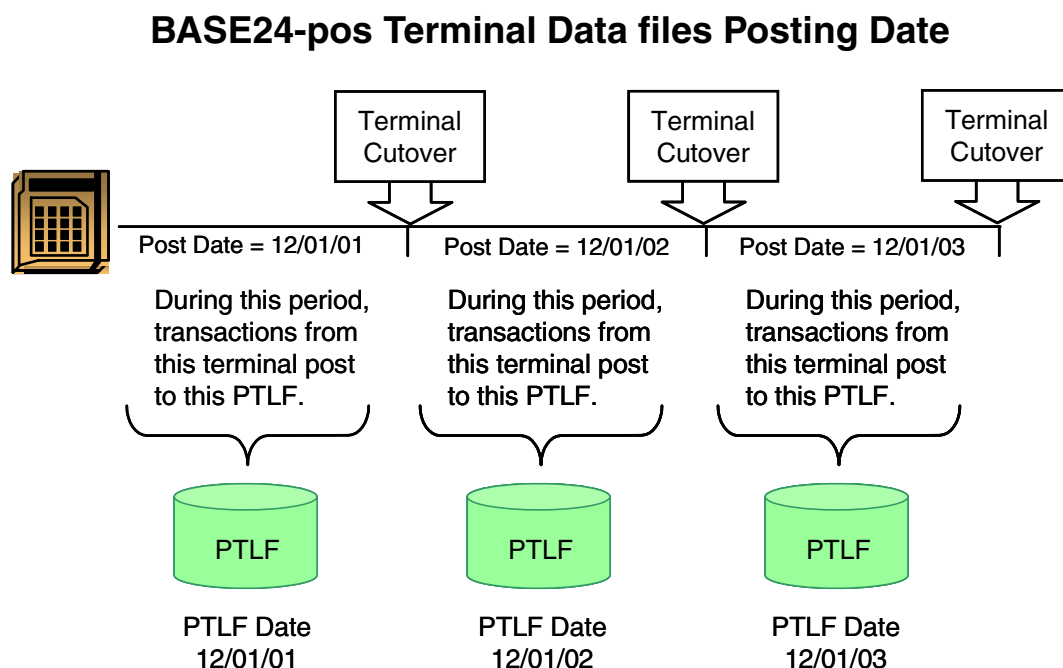
If an operator fails to bring the terminal and system totals into agreement, the system records both the system and terminal totals for reconciliation on the B24POS35, Retailer Reconciliation Report.

Terminal Cutover

The BASE24-pos product requires that every retailer terminal cut over each day. Terminal cutover closes the terminal's business day. Transactions prior to cutover are included in the current day's business; transactions after cutover are included in the next day's business.

At cutover, the posting date in the BASE24-pos Terminal Data files is increased by one calendar day. It is this date that indicates the PTLF to which a transaction from the terminal will post and, thus, the BASE24-pos reporting day in which it will be included.

The following diagram illustrates how the BASE24-pos Terminal Data files posting date is increased by one calendar day at terminal cutover.



The posting date in the BASE24-pos Terminal Data files controls the PTLF to which transactions from the terminal post. The BASE24-pos Terminal Data files posting date is increased by one calendar day at terminal cutover.

Retailer Cutover Window

Terminal cutover is enforced using a retailer cutover window configured in the RETAILER CUTOVER/BALANCE START and RETAILER CUTOVER/BALANCE END fields on POS Retailer Definition File (PRDF) screen 1. Terminal cutover is also enforced using a Device Handler cutover window, depending on the method by which the terminal is cut over. The Device Handler cutover window is defined by the RETAILER CUTOVER/BALANCE END field on PRDF screen 1 and the POS-DH-CUTOVER-LEAD-TIME param in the Logical Network Configuration File (LCONF). Refer to section 1 for more information on these cutover windows.

The cutover of intelligent terminals can be tied to the BASE24-pos balancing functionality. The cutover of nonintelligent terminals is *forced* each day by the BASE24-pos system and is not tied to terminal balancing. Tying cutover to the balancing function is a convenient way to close the terminal reporting day at the same time for both the terminal and the BASE24-pos system.

Terminals can be cut over by the following methods:

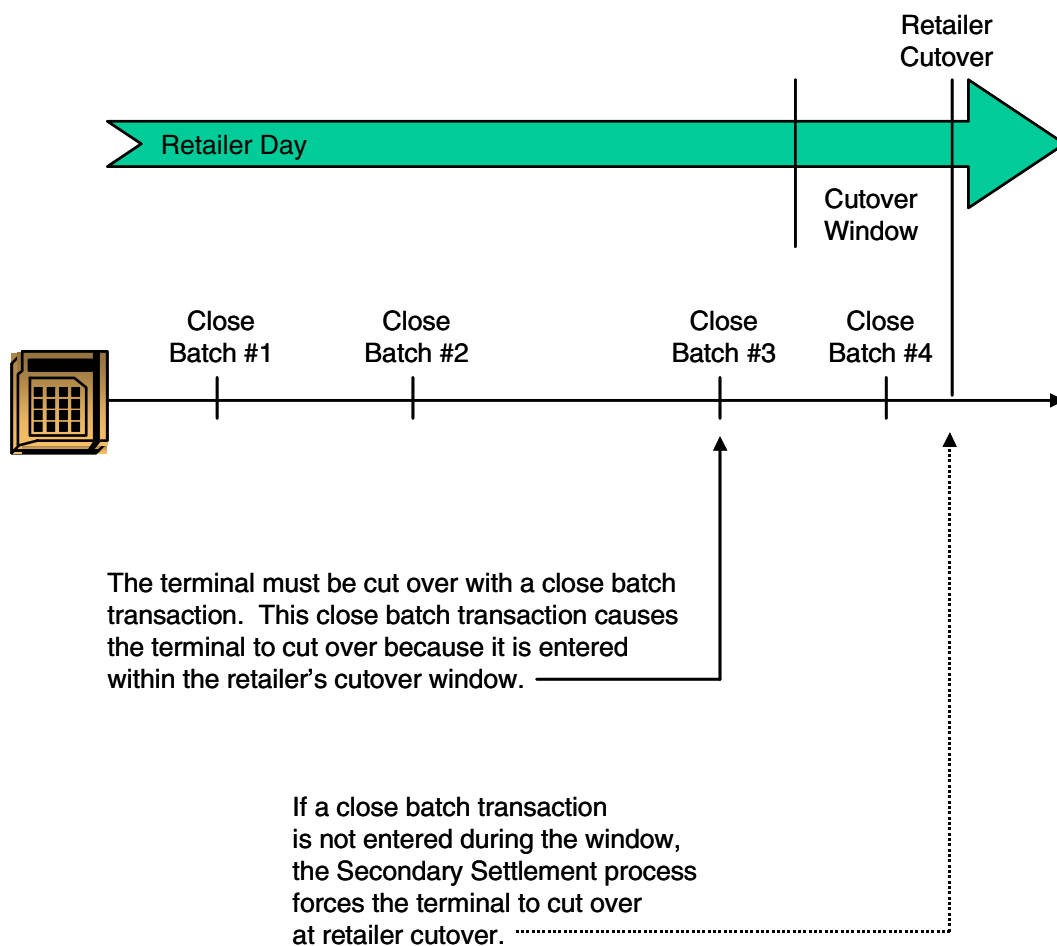
- A balancing and cutover transaction (close batch, close shift, close day) within the retailer cutover window.
- Device Handler module force cutover with the first financial transaction received within the Device Handler cutover window.
- Secondary Settlement process force cutover at the end of the retailer cutover window

Intelligent terminals are capable of cutting over under any of the methods listed above. Nonintelligent terminals must always be forced to cut over because they do not support balancing transactions.

At the time indicated in the RETAILER CUTOVER/BALANCE END field, the Secondary Settlement process verifies that all terminals for a retailer have cut over. If any terminals have not cut over, the Secondary Settlement process forces them to do so at this time.

The following diagram illustrates the link between terminal balancing and cutover.

Linking Terminal Balancing and Cutover



Terminal Cutover Options

The value in the **TERMINAL CUTOVER** field on PTD screen 3 indicates how each terminal is cut over. The following options can be set in this field:

Close Day Transaction (0)

The terminal cuts over when a clerk initiates a close day transaction during the retailer cutover window. Only the first close day transaction within the retailer cutover window cuts the terminal over. Subsequent close day transactions within

the retailer cutover window will not cause the terminal to cut over. Close day transactions should be entered only once during a reporting day to maintain reporting integrity.

If a clerk does not initiate a close day transaction during the retailer cutover window and the TERMINAL CUTOVER field contains the value 0, the Secondary Settlement process forces the terminal to cut over at the time indicated in the RETAILER CUTOVER/BALANCE END field on PRDF screen 1.

Close Shift Transaction (1)

The terminal cuts over when a clerk initiates a close shift transaction during the retailer cutover window. Only the first close shift transaction within the retailer cutover window cuts the terminal over. Subsequent close shift transactions within the retailer cutover window will not cause the terminal to cut over.

If a clerk does not initiate a close shift transaction during the retailer cutover window and the TERMINAL CUTOVER field contains the value 1, the Secondary Settlement process forces the terminal to cut over at the time indicated in the RETAILER CUTOVER/BALANCE END field on PRDF screen 1.

Close Batch Transaction (2)

The terminal cuts over when a clerk initiates a close batch transaction during the retailer cutover window. Only the first close batch transaction within the retailer cutover window cuts the terminal over. Subsequent close batch transactions within the retailer cutover window will not cause the terminal to cut over.

If a clerk does not initiate a close batch transaction during the retailer cutover window and the TERMINAL CUTOVER field contains the value 2, the Secondary Settlement process forces the terminal to cut over at the time indicated in the RETAILER CUTOVER/BALANCE END field on PRDF screen 1.

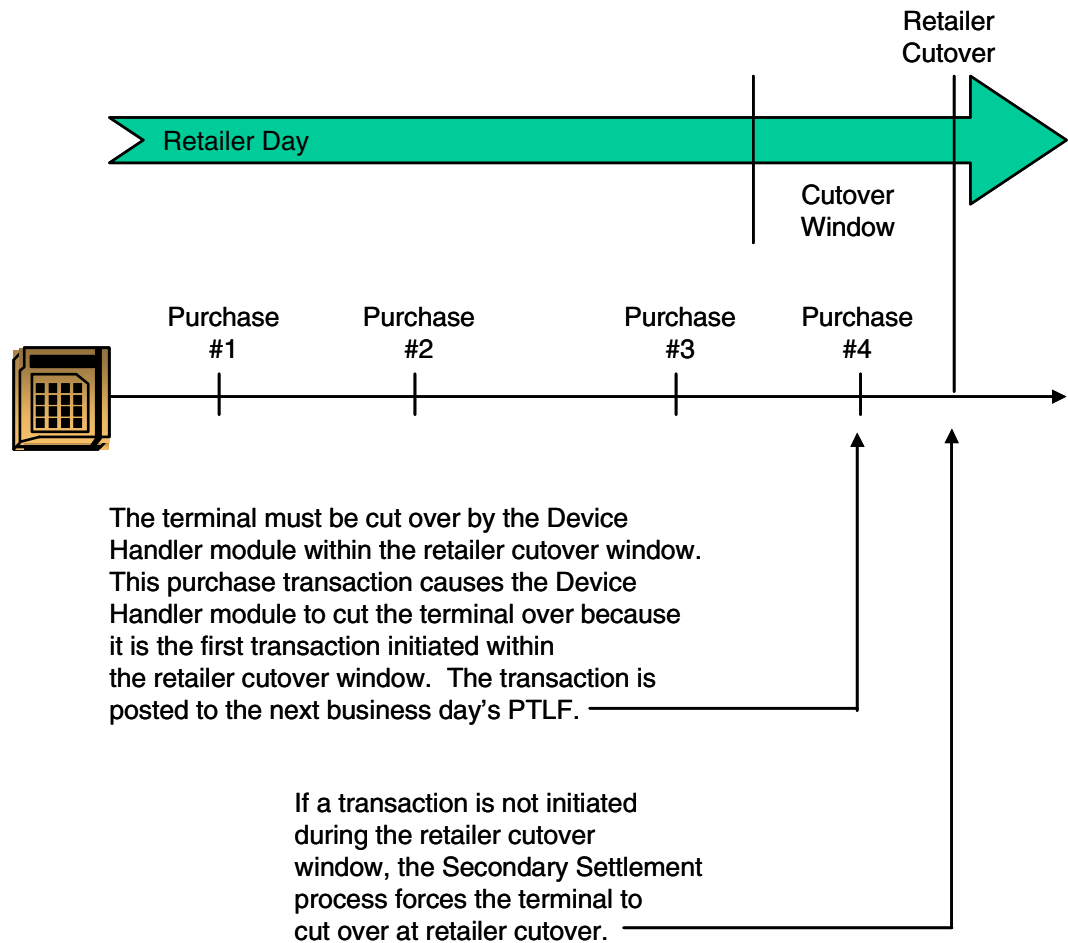
Secondary Settlement Process Force Cutover (3)

The Secondary Settlement process forces the terminal to cut over at the time indicated in the RETAILER CUTOVER/BALANCE END field on PRDF screen 1.

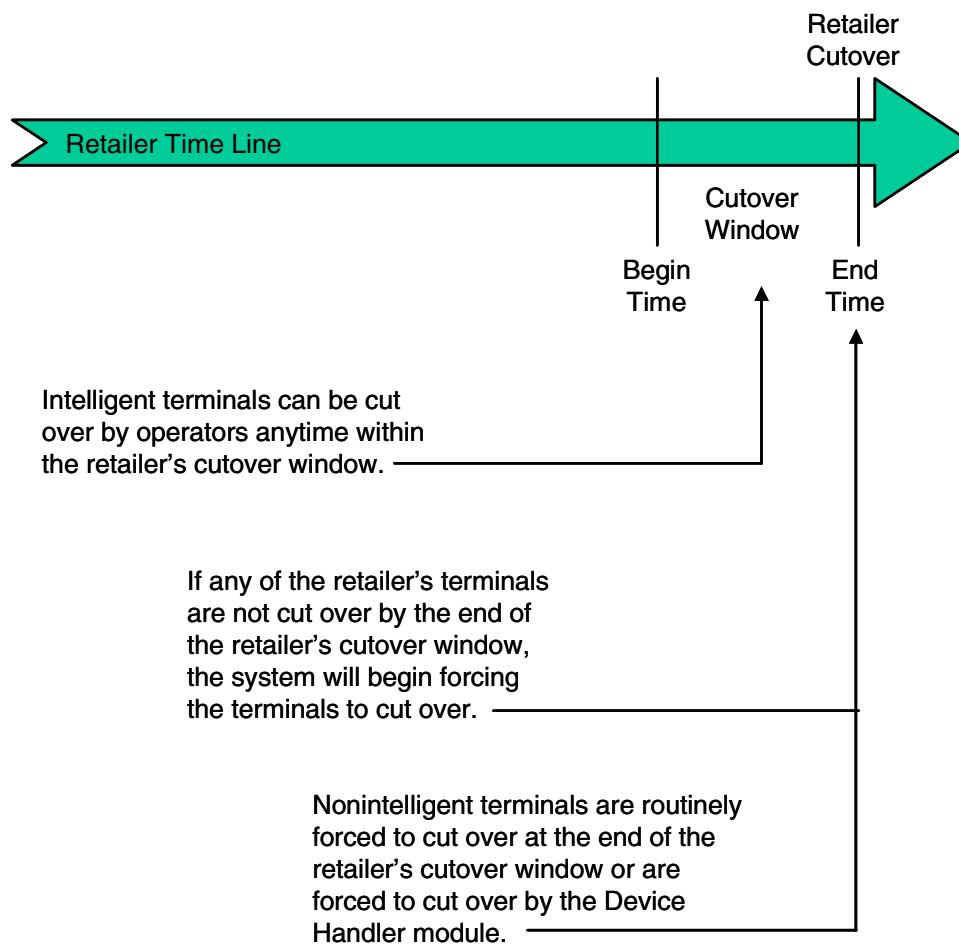
Device Handler Module Force Cutover (4)

The Device Handler module forces the terminal to cut over when it receives a financial transaction during the Device Handler cutover window. The transaction that causes the Device Handler module to cut over the terminal is then logged to the next day's PTLF. Only the first transaction within the Device Handler cutover window causes the Device Handler module to cut the terminal over. Subsequent transactions within the Device Handler cutover window will not cause the terminal to cut over.

The following diagrams illustrate how terminals are cut over during the retailer cutover window.



Retailer Cutover Window



Clearing BASE24-pos Terminal Data Files Totals at Cutover

The BASE24-pos system clears BASE24-pos Terminal Data files totals at cutover based on a value entered in the BALANCE SUPPORT field and TERMINAL CUTOVER field on PTD screen 3. The TERMINAL CUTOVER field is explained on previous pages. The setting in the BALANCE SUPPORT field identifies the types of terminal totals that the terminal can send to the BASE24 system.

The setting in the BALANCE SUPPORT field also indicates how BASE24-pos Terminal Data files totals are cleared based on the balancing transaction received, thus allowing the appropriate terminal's totals to continue accumulating through terminal cutover. For example, a terminal that sends batch and shift totals can cut over with a batch close while continuing to accumulate shift totals.

When the Device Handler module receives a close transaction within the retailer cutover window, it checks the TERMINAL CUTOVER and BALANCE SUPPORT fields. Totals in the BASE24-pos Terminal Data files are cleared based on the values entered in these fields. Network totals are always cleared at network cutover and service totals are always cleared when batch totals are cleared.

In order to close a shift, batch totals must have already been cleared. In order to close a day, both shift and batch totals must have already been cleared.

The options listed below can be set in the BALANCE SUPPORT field.

- 0 = The terminal can send batch totals to the BASE24-pos system.
- 1 = The terminal can send batch and shift totals to the BASE24-pos system.
- 2 = The terminal can send batch and day totals to the BASE24-pos system.
- 3 = The terminal can send batch, shift, and day totals to the BASE24-pos system.
- 4 = The terminal can send shift and day totals to the BASE24-pos system.
- 5 = The terminal can send day totals to the BASE24-pos system.
- 9 = The terminal cannot send totals to the BASE24-pos system.

Visa I Devices

Visa I devices cannot send totals or balancing and cutover transactions to the BASE24-pos system. Therefore, the Visa I Device Handler module clears BASE24-pos Terminal Data files totals as if the BALANCE SUPPORT field on PTD screen 3 is set to a value of 9. The Visa I Device Handler module does not actually check the BALANCE SUPPORT field. Visa I devices are always forced to cut over by the Secondary Settlement process or the Device Handler module. The BASE24-pos Terminal Data files batch, shift, and day totals are always cleared when the terminal is forced to cut over.

Visa II and Hypercom Devices

Visa II and Hypercom devices can send batch totals to the BASE24-pos system; they cannot send shift and day totals. These devices and their respective Device Handler modules also support close batch balancing and cutover transactions.

When a close batch balancing and cutover transaction is received within the retailer cutover window, the batch, shift, and day totals in the BASE24-pos Terminal Data files are cleared. When a close batch balancing transaction is received outside the retailer cutover window, the batch totals are cleared.

NCR/NDP Devices

NCR/NDP devices can send batch totals to the BASE24-pos system; they cannot send shift and day totals. These devices and the NCR/NDP Device Handler module also support close batch balancing and cutover transactions.

When a close batch balancing and cutover transaction is received within the retailer cutover window, the batch, shift, and day totals in the BASE24-pos Terminal Data files are cleared. When a close batch balancing transaction is received outside the retailer cutover window, batch totals are not cleared.

APACS 30/40 Devices

The APACS 40 terminal can send batch totals in reconciliation transactions to the BASE24-pos system. The APACS 40 terminal does not support shift and day totals. The APACS 30/40 Device Handler module performs close batch balancing and cutover when a reconciliation transaction request is received.

When a reconciliation transaction is received within the retailer cutover window, the batch totals in the BASE24-pos Terminal Data files are cleared. When a reconciliation transaction is received outside the retailer cutover window, the batch totals are not cleared.

APACS 30 devices cannot send totals or balancing and cutover transactions to the BASE24-pos system. Therefore, the APACS 30/40 Device Handler module clears BASE24-pos Terminal Data files totals as if the BALANCE SUPPORT field on PTD screen 3 is set to a value of 9. The APACS 30/40 Device Handler module does not actually check the BALANCE SUPPORT field. APACS 30 devices are always forced to cut over by the Secondary Settlement process or the Device Handler module. The BASE24-pos Terminal Data files batch totals are always cleared when the terminal is forced to cut over.

SPDH and MHI Devices

The Standard POS Device Handler (SPDH) and Merchant Host Interface (MHI) devices can send batch, shift, and day totals to the BASE24-pos system in any combination dictated by the value in the BALANCE SUPPORT field. These devices and their respective Device Handler modules also support close batch, close shift, and close day balancing and cutover transactions.

The table below illustrates which BASE24-pos Terminal Data files totals (batch, shift, or day) are cleared depending on the values in the BALANCE SUPPORT field and the TERMINAL CUTOVER field. This table assumes that the balancing transaction indicated in the second column is initiated within the retailer cutover window.

BALANCE SUPPORT field	TERMINAL CUTOVER field	Totals Cleared
Close batch (0)	Close batch	batch, shift, day
	Close shift	shift, day
	Close day	day
Close batch and shift (1)	Close batch	batch, day
	Close shift	shift, day
	Close day	day
Close batch and day (2)	Close batch	batch, shift
	Close shift	shift
	Close day	day
Close batch, shift, and day (3)	Close batch	batch
	Close shift	shift
	Close day	day
Close shift and day (4)	Close batch	batch
	Close shift	shift
	Close day	day

BALANCE SUPPORT field	TERMINAL CUTOVER field	Totals Cleared
Close day (5)	Close batch	batch, shift
	Close shift	shift
	Close day	day
None supported (9)	Close batch	batch, shift, day
	Close shift	shift, day
	Close day	day

Note: When the terminal is forced to cut over by the Device Handler, the value in the BALANCE SUPPORT field does not determine which totals are cleared. In this case, batch, shift, daily, network, and service totals are all cleared.

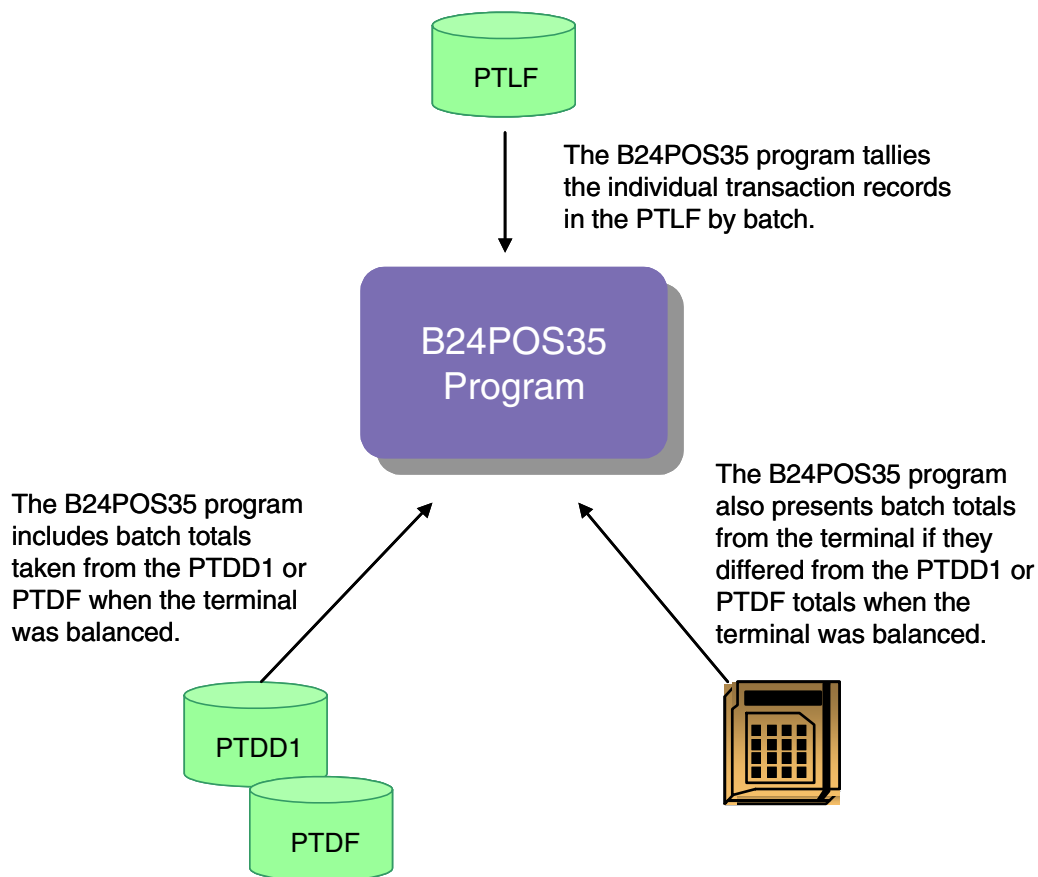
Terminal Reconciliation

The BASE24-pos product supports terminal reconciliation by comparing the retailer terminal totals in the BASE24-pos Terminal Data files and those sent from the terminal (if the terminal can provide totals) with the actual transaction totals in the PTLF. This comparison is provided using the B24POS35, Retailer Reconciliation Report and allows retailers and retailer sponsors to reconcile individual terminal activity with the transaction activity the BASE24-pos system has on file for the reporting day.

The Retailer Reconciliation Report presents terminal transaction activity, batch-by-batch, from three points of view. These points of view are derived from PTLF transaction records, BASE24-pos Terminal Data files totals, and actual totals submitted by the terminals. The report allows retailers and retailer sponsors to observe, at the batch level, that the transactions in the PTLF match those initiated at retailer terminals.

Batches are the fundamental units of terminal balancing. The BASE24-pos product always reconciles terminal activity at the batch level. Because shift and daily totals represent grouped batches, the BASE24-pos product also provides reconciliation tools at these levels for terminals that support them.

The following diagram illustrates how information provided from the PTLF and BASE24-pos Terminal Data files are reported on the Retailer Reconciliation Report.



The B24POS35 Report

The primary tool of terminal reconciliation is the Retailer Reconciliation Report (B24POS35). A sample of this report is shown below along with a numbered overlay indicating how the report information is grouped. For a description of each numbered segment refer to the report key on the following page. For descriptions of fields that appear on this report, additional report samples, and the spooler location for this report, refer to section 4.

The sample report and overlay illustrate the basic design of the Retailer Reconciliation Report and show how terminal figures are juxtaposed on the report for comparison. For additional information on this report, including field descriptions, refer to section 4.

		1	3		
REPORT NO: B24POS35 RETAILER RECONCILIATION REPORT		NETWORK: LLLL		JAN 20 YY PAGE 1	
INSTITUTION: BK01 GROUP: 001 REGION: 0001		RETAILER ID: RETAILER03		TERMINAL ID: 330000000 104 BATCH: 001	
BATCH TOTALS					
		-----DEPOSIT TOTALS-----		-----PREVIOUS/CARRYOVER-----	
---CARD TYPE---	---STATUS---	--- NETWORK (D) ---	--- NETWORK (T) ---	--- TERMINAL (T) ---	--- NETWORK (T) ---
(VISA CREDIT)	DB CNT:	15	15		
	DB AMT:	450.00	436.00		
	CR CNT:	2	2		
	CR AMT:	50.00	50.00		
	AD CNT:	0	0		
	AJ AMT:	0.00	0.00		
	TOTAL CNT:	17	17		
	TOTAL AMT:	400.00	400.00		
(MASTERCARD CREDIT)	DB CNT:	18	18		
	DB AMT:	700.00	700.00		
	CR CNT:	3	3		
	CR AMT:	75.00	75.00		
	AD CNT:	0	0		
	AJ AMT:	0.00	0.00		
	TOTAL CNT:	21	21		
	TOTAL AMT:	625.00	625.00		
*BATCH 001 TOTAL	DB CNT:	33	33	33	
	DB AMT:	1150.00	1150.00	1150.00	
	CR CNT:	5	5	5	
	CR AMT:	125.00	125.00	125.00	
	AD CNT:	0	0	0	
	AJ AMT:	0.00	0.00	0.00	
NET DEPOSIT	TOTAL CNT:	38	38	38	
	TOTAL AMT:	1025.00	1025.00	1025.00	
7	2	4	5	6	

Report Key

- 1 = Totals, by card type, the report program computed using PTLF transaction records.
- 2 = Batch totals the program computed by adding the card type totals for the batch computed in 1.
- 3 = Card type totals taken directly from the PTLF service record for the current batch. These totals were written to the PTLF from the BASE24-pos Terminal Data files when the batch was closed.
- 4 = Batch totals taken directly from the PTLF batch record for this batch. These totals were written to the PTLF from the BASE24-pos Terminal Data files when the batch was closed.

Report Key

- 5 = Batch totals the terminal sent when the batch was closed. If a terminal does not send batch totals, or if the totals match those in the BASE24-pos Terminal Data files, these totals are duplicates of the totals in 4.
- 6 = The portion of the batch that was a part of the previous day's settlement. Figures in this area are for incomplete batches only.
- 7 = The statement "Out of Balance" is printed in this column if the batch is incomplete or out-of-balance.

Incomplete and Out-of-Balance Batches

Incomplete batches or batches that do not balance are flagged with an asterisk on the Retailer Reconciliation Report so that institutions can research any out-of-balance amounts.

Incomplete Batches

Incomplete batches are batches that were in progress when the Secondary Settlement process forced the terminal to cut over. This situation only occurs when a terminal is set up to be balanced and cut over using an operator-initiated balancing and cutover transaction, but the appropriate transaction was not received.

When the Secondary Settlement process forces a terminal to cut over, the system closes the terminal's business day in the middle of a batch. Part of the batch is settled today and part tomorrow. While additional transactions may be included in the physical terminal's batch, only those transactions initiated prior to the forced cutover are included in today's BASE24-pos reporting.

The BASE24-pos product marks incomplete batches with messages on the current day's Retailer Reconciliation Report and presents the amount of the difference as carryover on the following day's report. The sample report on the following page shows this relationship. In this example, \$250 of the batch is settled the first day, and \$350 is settled the second day. The second day's report shows the amount of the batch that was settled on the previous day.

Terminal Reconciliation

REPORT NO: B24POS35 RETAILER RECONCILIATION REPORT - NETWORK: LLLL						JAN 26 YY	PAGE	1
INSTITUTION: BK01		GROUP: 001	REGION: 0001	RETAILER ID: RETAILER03	TERMINAL ID: 330000000	104	BATCH: 001	
BATCH TOTALS								
-----DEPOSIT TOTALS-----						PREVIOUS/CARRYOVER-		
---CARD TYPE---	---STATUS---	--- NETWORK (D) ---		--- NETWORK (T) ---		--- TERMINAL (T) ---		--- NETWORK (T) ---
(VISA CREDIT)	DB CNT:	5		5				
	DB AMT:	100.00		100.00				
	CR CNT:	0		0				
	CR AMT:	0.00		0.00				
	AD CNT:	0		0				
	AJ AMT:	0.00		0.00				
	TOTAL CNT:	5		5				
	TOTAL AMT:	100.00		100.00				
(MASTERCARD CREDIT)	DB CNT:	6		6				
	DB AMT:	150.00		150.00				
	CR CNT:	0		0				
	CR AMT:	0.00		0.00				
	AD CNT:	0		0				
	AJ AMT:	0.00		0.00				
	TOTAL CNT:	6		6				
	TOTAL AMT:	150.00		150.00				
*BATCH 001 TOTAL	INCOMPLETE BATCH							
	DB CNT:	11		11		11		
	DB AMT:	250.00		250.00		250.00		
	CR CNT:	0		0		0		
	CR AMT:	0.00		0.00		0.00		
	AD CNT:	0		0		0		
	AJ AMT:	0.00		0.00		0.00		
NET DEPOSIT	TOTAL CNT:	11		11		11		
	TOTAL AMT:	250.00		250.00		250.00		

Consecutive Days

REPORT NO: B24POS35 RETAILER RECONCILIATION REPORT - NETWORK: LLLL						JAN 27 YY	PAGE	1
INSTITUTION: BK01		GROUP: 001	REGION: 0001	RETAILER ID: RETAILER03	TERMINAL ID: 330000000	104	BATCH: 001	
BATCH TOTALS								
-----DEPOSIT TOTALS-----						PREVIOUS/CARRYOVER-		
---CARD TYPE---	---STATUS---	--- NETWORK (D) ---		--- NETWORK (T) ---		--- TERMINAL (T) ---		--- NETWORK (T) ---
(VISA CREDIT)	DB CNT:	3		8				
	DB AMT:	250.00		350.00				
	CR CNT:	1		1				
	CR AMT:	50.00		50.00				
	AD CNT:	0		0				
	AJ AMT:	0.00		0.00				
	TOTAL CNT:	4		9				
	TOTAL AMT:	200.00		300.00				
(MASTERCARD CREDIT)	DB CNT:	4		10				
	DB AMT:	150.00		300.00				
	CR CNT:	0		0				
	CR AMT:	0.00		0.00				
	AD CNT:	0		0				
	AJ AMT:	0.00		0.00				
	TOTAL CNT:	4		10				
	TOTAL AMT:	150.00		300.00				
*BATCH 001 TOTAL	DB CNT:	7		18		18		18
	DB AMT:	400.00		650.00		650.00		650.00
	CR CNT:	1		1		1		0
	CR AMT:	50.00		50.00		50.00		0.00
	AD CNT:	0		0		0		0
	AJ AMT:	0.00		0.00		0.00		0.00
NET DEPOSIT	TOTAL CNT:	8		19		19		11
	TOTAL AMT:	350.00		600.00		600.00		250.00

Settled Today

Total

Settled Yesterday

Out-of-balance Batches

Out-of-balance batches occur in two situations. The first situation, shown on page 5 of the sample report below, arises when the PTLF transaction records do not match the BASE24-pos Terminal Data files transaction totals. On the sample report, for example, the NETWORK(D) totals do not match the NETWORK(T) totals. This out-of-balance condition may occur because transactions have been double-posted to the PTLF as a result of a process failure and restart.

REPORT NO: B24POS35 RETAILER RECONCILIATION REPORT - NETWORK: XXXX				24-JUL-YY		PAGE	5
INSTITUTION: BK01		GROUP: 001	REGION: 0001	RETAILER ID: RETAILER03		TERMINAL ID: 330000000	104
"BATCH TOTALS"				BATCH: 005			
				-----DEPOSIT TOTALS-----		-----PREVIOUS/CARRYOVER-----	
----CARD TYPE----	---STATUS----	--- NETWORK (D) ---	--- NETWORK (T) ---	--- TERMINAL (T) ---	--- NETWORK (T) ---		
(VISA CREDIT)	DB CNT:	16	15				
	DB AMT:	450.00	410.00				
	CR CNT:	2	2				
	CR AMT:	50.00	50.00				
	AD CNT:	0	0				
	AJ AMT:	0.00	0.00				
	TOTAL CNT:	18	17				
	TOTAL AMT:	400.00	360.00				
(MASTERCARD CREDIT)	DB CNT:	18	18				
	DB AMT:	700.00	700.00				
	CR CNT:	3	3				
	CR AMT:	75.00	75.00				
	AD CNT:	0	0				
	AJ AMT:	0.00	0.00				
	TOTAL CNT:	21	21				
	TOTAL AMT:	625.00	625.00				
*BATCH 005 TOTAL	DB CNT:	34	33	33			
	DB AMT:	1150.00	1110.00	1110.00			
	CR CNT:	5	5	5			
	CR AMT:	125.00	125.00	125.00			
	AD CNT:	0	0	0			
	AJ AMT:	0.00	0.00	0.00			
NET DEPOSIT	TOTAL CNT:	39	38	38			
OUT OF BALANCE	TOTAL AMT:	1025.00	985.00	985.00			

Out of balance condition noted

Totals do not match

The second situation, shown on page 3 of the sample report on the following page, arises when a terminal submits its own batch totals and these totals do not match the totals in the BASE24-pos Terminal Data files. On the sample report, for example, the NETWORK(T) totals do not match the TERMINAL(T) totals. This condition occurs when a terminal does not balance its batches with the system.

REPORT NO: B24POS35 RETAILER RECONCILIATION REPORT - NETWORK: XXXX				28-JUL-YY		PAGE	3	
INSTITUTION: BK01		GROUP: 001	REGION: 0001	RETAILER ID: RETAILER03		TERMINAL ID: 330000000	104	BATCH: 003
"BATCH TOTALS"								
			-----DEPOSIT TOTALS-----			-----PREVIOUS/CARRYOVER-----		
---CARD TYPE---	---STATUS---	--- NETWORK (D) ---		--- NETWORK (T) ---		--- TERMINAL (T) ---		--- NETWORK (T) ---
(VISA CREDIT)	DB CNT:	15		15				
	DB AMT:	450.00		450.00				
	CR CNT:	2		2				
	CR AMT:	50.00		50.00				
	AD CNT:	0		0				
	AJ AMT:	0.00		0.00				
	TOTAL CNT:	17		17				
	TOTAL AMT:	400.00		400.00				
(MASTERCARD CREDIT)	DB CNT:	18		18				
	DB AMT:	700.00		700.00				
	CR CNT:	3		3				
	CR AMT:	75.00		75.00				
	AD CNT:	0		0				
	AJ AMT:	0.00		0.00				
	TOTAL CNT:	21		21				
	TOTAL AMT:	625.00		625.00				
*BATCH 003 TOTAL								
	DB CNT:	33		33		32		
	DB AMT:	1150.00		1150.00		1150.00		
	CR CNT:	5		5		5		
	CR AMT:	125.00		125.00		125.00		
	AD CNT:	0		0		0		
	AJ AMT:	0.00		0.00		0.00		
NET DEPOSIT		TOTAL CNT:	38	38		37		
OUT OF BALANCE		TOTAL AMT:	1025.00	1025.00		1025.00		

Out of balance condition noted

Totals do not match

The Retailer Reconciliation Report notes where these figures do not agree so that the institution can research and resolve discrepancies.

BASE24-pos Draft Capture

The BASE24-pos product enables institutions to process retailer transactions as draft capture or non-draft capture. Non-draft capture transactions are transactions for which the paper ticket represents the actual settlement instrument. The paper ticket must be physically presented for deposit before the transaction can be settled. While the BASE24-pos system may have authorized the transaction, the BASE24-pos record cannot be used to settle the transaction.

Draft capture transactions are electronically captured at the time of the transaction and are sufficient in themselves to enact a funds transfer for the transaction amount. The BASE24-pos product identifies draft capture transactions in its files and includes draft capture totals on its retailer reports so that the retailer sponsor can settle with the retailer.

Draft capture is a settlement tool that has no impact on how transactions are authorized. Cardholders are still checked against the same authorization criteria without regard for whether a transaction is draft capture or non-draft capture. Thus, the retailer is not incurring additional risk of cardholder fraud by using or not using draft capture.

The following draft capture options are available for each card type in the transaction profile (TP) fields on BASE24-pos Terminal Data files (PTD) screens 8, 9, and 10. Each option is explained in more detail on this and the following pages. For more information about the TP fields, refer to the ***BASE24-pos Files Maintenance Manual*** or the ***BASE24-pos Transaction Processing Manual***.

- Authorization only
- Authorization and draft capture
- Authorization with electronic follow-up
- Terminal determines which of the above options are used for the transaction

Authorization Only

The authorization-only option enables retailers to obtain an online authorization for the amount of a customer transaction. In an authorization-only environment, the card issuer waits for paper follow-up before posting the transaction and settling with the retailer. All POS devices support this option.

If a card type is set up with this option, all of the transactions for this card type from the specified terminal are non-draft capture. The BASE24-pos system logs transactions to the POS Transaction Log File (PTLF) with a draft capture flag of 0, indicating the transaction is an authorization-only transaction. Authorization-only transactions do not update the draft capture totals in the BASE24-pos Terminal Data files.

Authorization and Draft Capture

The authorization and draft capture option enables retailers to obtain an authorization on a transaction and enables institutions to automatically capture the draft electronically at the same time. With electronic draft capture, the card issuer does not require paper follow-up to post the transaction and settle with the retailer. The APACS 40, BASE24 Standard POS Device Handler (SPDH), Merchant Host Interface, NCR/NDP, and Hypercom devices support this option.

The BASE24-pos product logs transactions to the POS Transaction Log File (PTLF) with a draft capture flag of 1, indicating that the transaction is draft capture. If a terminal is set up with this option, the draft capture totals in the BASE24-pos Terminal Data files are updated.

Authorization with Electronic Follow-up

The authorization with electronic follow-up option provides for a two-step transaction that first obtains online authorization for the amount of a customer transaction and then, in a second step, transmits the information needed to perform settlement to the card issuer. The APACS 30, Visa II, and Hypercom devices support this option. The two steps are described below.

1. Obtain an online authorization for the amount of a customer transaction. This step is similar to the authorization-only option. If the transaction is authorized, the second step in this option is performed. If the customer transaction request is declined, the second step is not taken.

The BASE24-pos product logs these initial transactions to the POS Transaction Log File (PTLF) with a draft capture flag of 2, indicating the transaction has been authorized but requires an electronic follow-up transaction.

2. If the transaction amount has been authorized in step 1, collect the ticket information electronically and transmit it to the card issuer for settlement.

The BASE24-pos product logs the second transaction to the PTLF with a draft capture flag of 3, indicating that this is the electronic follow-up to a previously approved transaction. Electronic follow-up transactions update the draft capture totals in the BASE24-pos Terminal Data files.

Terminal-Defined Draft Capture

The terminal-defined draft capture option enables the terminal to determine the draft capture option for each transaction based on the card type.

If a card type is set up with this option, the terminal must select one of the draft capture options previously described (authorization and draft capture, authorization with electronic follow-up, or authorization only) for each transaction involving this card type. The BASE24 Standard POS Device Handler (SPDH) and the Merchant Host Interface (MHI) devices support this option.

The BASE24-pos product logs transactions to the POS Transaction Log File (PTLF) with a draft capture flag of 0 or 1, indicating the draft capture option selected by the terminal. If a terminal selects a *draft capture* option, the draft capture totals in the BASE24-pos Terminal Data files are updated.

Transaction Identification

The following table summarizes the relationship between the transaction profile (TP) field on BASE24-pos Terminal Data files (PTD) screens 8, 9, and 10 and how the draft capture flag is set in the PTLF.

Transaction Profile—PTD	Draft Capture Flag—PTLF
0 = Authorization only	0 = Authorization only
1 = Authorization and draft capture	1 = Authorization and draft capture
2 = Authorization with electronic follow-up (two step transaction)	2 = Authorization-only with electronic follow-up expected 3 = Electronic follow-up
3 = Terminal determines the draft capture option by transaction	0 = Authorization only 1 = Authorization and draft capture

3: Settlement Support

This section describes BASE24-pos settlement support. The BASE24-pos product supports a number of settlement relationships between network participants. These relationships include retailer settlement, card issuer settlement, logical network settlement, and interchange settlement.

Retailer Settlement

Retailer settlement is the settlement between network retailers and retailer sponsors for the day's transaction activity. It is a two-step process that consists of crediting retailer accounts for the day's draft capture transactions and then clearing these items through the appropriate settlement channels. The BASE24-pos product provides the support data to facilitate both aspects of retailer settlement.

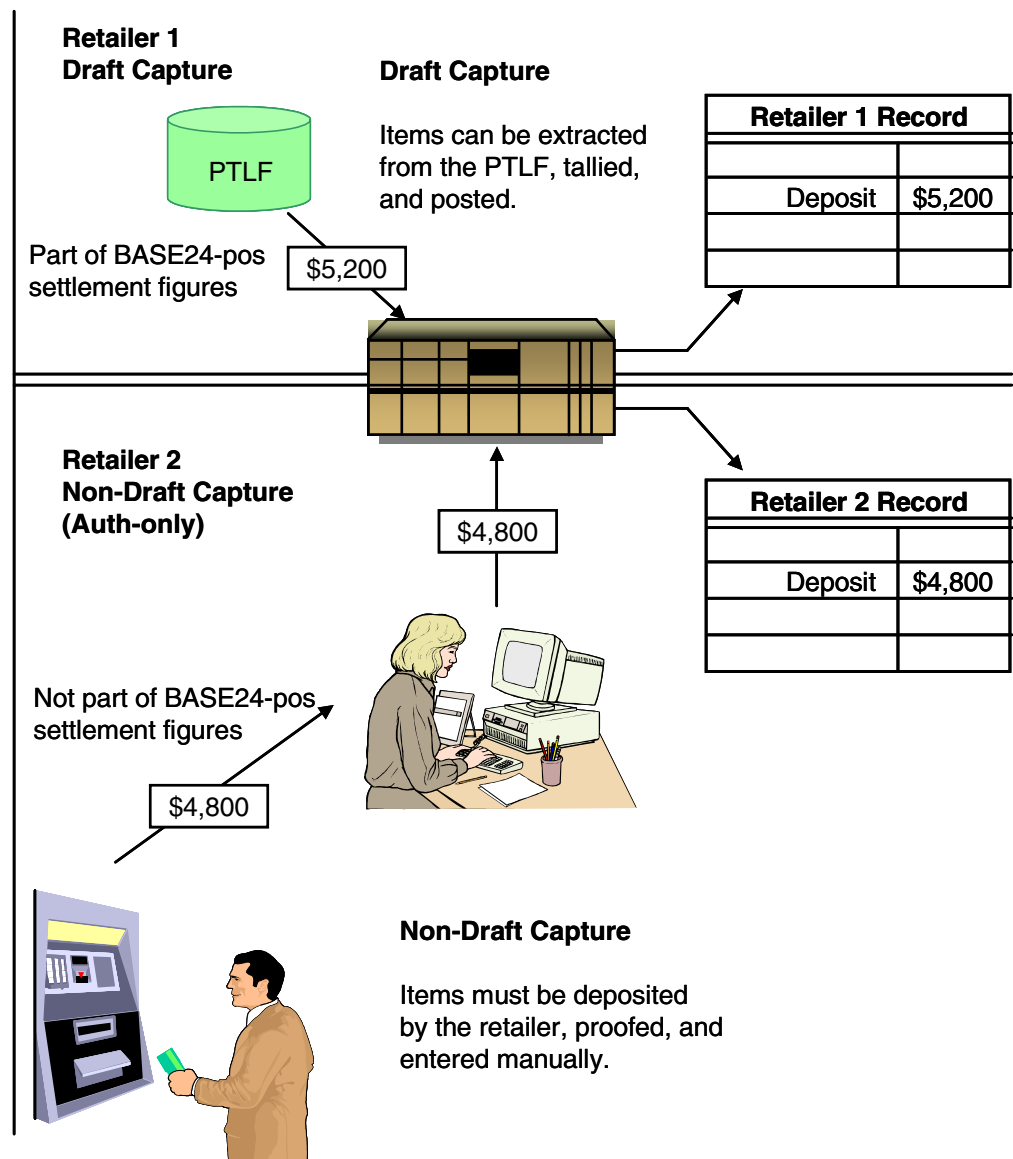
Crediting Retailers

From the retailer's point of view, there are two methods of being credited for debit and credit card activity. The first method involves draft capture transactions. Draft capture transactions are electronically-captured transactions. The electronic record is sufficient to enact settlement for the item. The retailer need not submit actual sales tickets in order to be credited for the transactions (see Retailer 1 on the next page).

The second method involves non-draft capture transactions for which the actual sales tickets must be bundled and deposited. In this case, the sales ticket itself acts as the settlement instrument (see Retailer 2 on the next page).

Although all transactions processed through the BASE24-pos system are included in the PTLF and on the BASE24-pos reports, only draft capture transactions are included in BASE24-pos retailer settlement totals. The settlement totals represent the transactions for which the retailer must be credited *without* deposit of a sales ticket.

Retailer sponsors can credit their retailer accounts by extracting customer transactions from the PTLF for their own processing and reporting (see Retailer 1 on the next page), or they can credit retailer accounts directly from the BASE24-pos reports. The total amount due each retailer for draft capture activity is summarized on the Retailer Summary Report (B24POS11) and Retailer Activity Report (B24POS12). Draft capture activity is then reconciled with retailer terminal totals using the Retailer Reconciliation Report (B24POS35). Disputed transactions are listed in the Daily Disputed Transactions Report (B24POS05).



Clearing Retailer Items

Crediting a retailer's account for the day's transactions leaves these transactions in an intermediate stage. The retailer has been paid for them, but the retailer sponsor has yet to collect for them. The retailer sponsor must clear these transactions in order to recover the amount paid to the retailer.

A retailer sponsor can clear the transactions for which its retailers have been credited through several channels. Transactions can be cleared:

- To the retailer sponsor's own cardholder accounts

These are transactions initiated using cards issued by the retailer sponsor. In this case, the retailer sponsor controls both the retailer and cardholder accounts. These transactions can be split out from the PTLF and posted directly to the retailer sponsor's cardholder accounts.

- To other institutions in the BASE24-pos network

These are transactions initiated using cards issued by other institutions in the network.

Debit and credit card transactions are included in a net clearings figure on the Network Settlement Report (B24POS01). Retailer sponsors cannot clear these transactions individually because other institutions have posted the transactions from the PTLF.

Retailer sponsors can extract credit card transactions from the PTLF to generate files used to settle with other BASE24-pos card issuers.

- To interchanges connected to the BASE24-pos network

These are transactions initiated using cards serviced by interchanges. For settlement purposes, there are two types of interchanges.

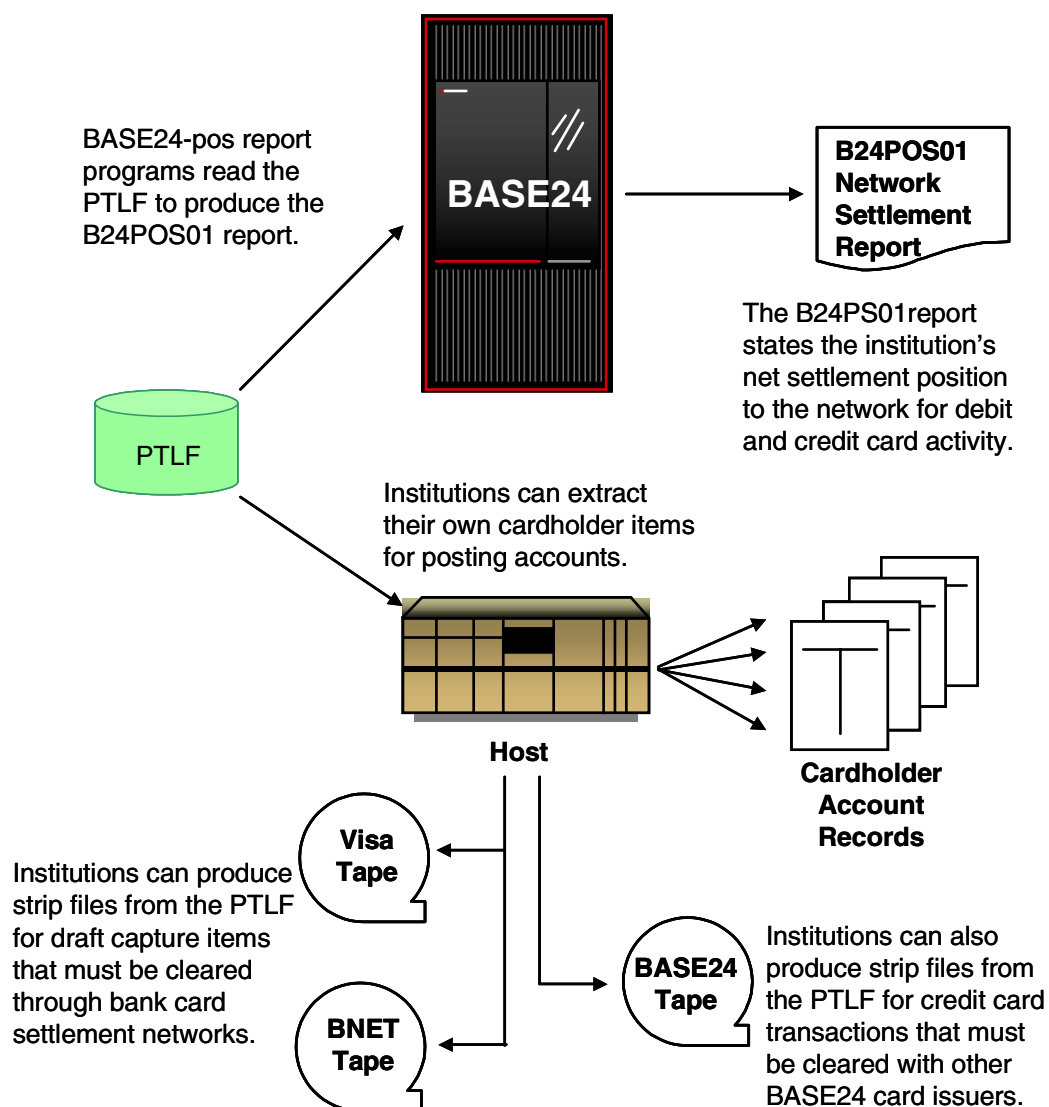
The first type of interchange authorizes and captures transactions at the same time. Typically debit card interchanges, the amounts due from these interchanges for their cardholder activity are included in the BASE24-pos network settlement totals on the B24POS01 report.

The second type of interchange authorizes and settles transactions separately. These interchanges settle transactions through their own national bank card settlement networks (i.e., Visa, BNET, and AMEX). The BASE24-pos product does not include amounts due from these interchanges in its network settlement totals. To clear these transactions, the retailer sponsor must settle the transactions through the appropriate national bank card settlement network. The BASE24-pos product enables retailer sponsors to create

transaction files, called strip files, from the PTLF to assist in this process. If the interchange allows for paper truncation, retailer sponsors can strip off those draft capture transactions to be sold to the interchange.

The following diagram shows how retailer items are cleared through BASE24-pos settlement processing.

Clearing Retailer Items



Card Issuer Settlement

Card issuer settlement is the settlement between card issuers and the various acquiring organizations that have accepted their cards.

Card issuer settlement first involves posting transactions to cardholder accounts. Secondly, it involves settling transaction activity between the card issuer and the various acquiring organizations.

Posting Cardholder Accounts

All transactions the BASE24-pos system processes are recorded in the PTLF. BASE24-pos card-issuing institutions can then extract the PTLF to post transactions individually to their cardholder accounts.

The BASE24-pos product checks each PTLF record to determine how the transaction must be posted to the appropriate cardholder account. Some transactions affect cardholder account balances, while others affect only the available credit and float.

The BASE24-pos product provides a detail listing and summary of transactions in the PTLF by cardholder for card issuers to use as an audit trail. This report, the Daily Cardholder Activity Report (B24POS02), can be used for researching cardholder transactions. Also produced with the B24POS02 report is the Daily Cardholder Exception Report, which can be used to research transactions that the BASE24-pos system was unable to post correctly. The Daily Disputed Transactions Report (B24POS05), lists transactions that have been disputed by the cardholder.

Settling with Acquirers

Card issuer transactions are settled based on the type of acquirer, or retailer, initiating the transaction. Transactions can originate from:

- The card issuer's own retailers
Transactions are initiated at the card issuer's own retailers. In such a case, the card issuer is also the retailer sponsor. The retailer is credited directly for draft capture transactions as a part of retailer settlement.

- Retailers belonging to other institutions in the BASE24-pos network

Settlement of debit and credit card transactions is included in the network settlement totals on the Daily Network Settlement Report (B24POS01).

- From interchanges connected to the BASE24-pos network

Transactions are initiated through interchange networks. For settlement purposes, there are two types of interchanges.

The first type of interchange allows card issuers to authorize and capture transactions at the same time. Typically debit card interchanges, the amounts due to these interchanges are included in the network settlement totals on the Daily Network Settlement Report (B24POS01).

The second type of interchange requires that transactions be authorized and settled separately. These interchanges settle transactions through their own national bank card settlement networks (i.e., VisaNet, Banknet, American Express). These transactions must be presented to the card issuer by other means, and they are not included in the BASE24-pos network settlement totals on the Daily Network Settlement Report (B24POS01).

Logical Network Settlement

Logical network settlement is the settlement between BASE24-pos institutions. The Daily Network Settlement Report (B24POS01) provides a net position to the network for each institution in the network.

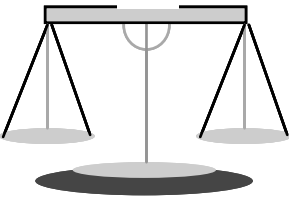
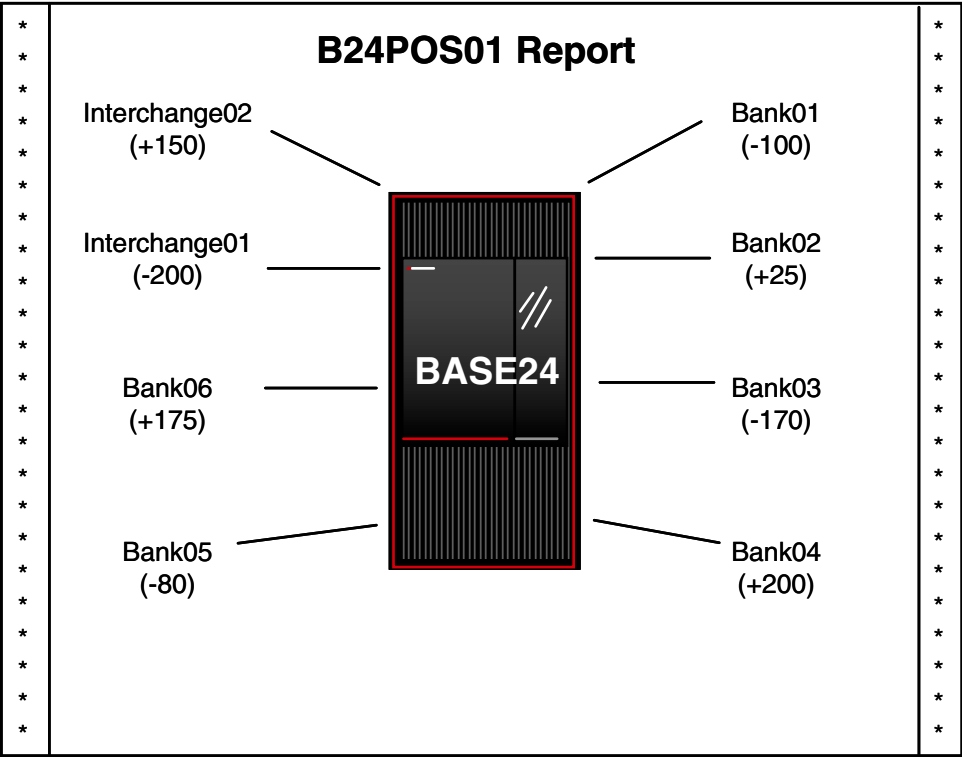
BASE24-pos retailer sponsors and card issuers can settle their debit and credit card activity using the B24POS01 report. Retailer sponsors are reimbursed for those amounts credited to their retailers, and card issuers pay for the transactions their cardholders have initiated at other institutions' retailers.

Included on the B24POS01 report are net settlement figures for those interchanges the BASE24-pos system considers to be settlement entities. The BASE24-pos system does not consider interchanges that require transaction settlement through national bank card settlement networks to be settlement entities.

The B24POS01 report brings the BASE24 network into balance with the amounts owed to the network and by the network netting to zero. In the following diagram, the amount owed to the network is \$550 and the amount owed by the network is \$550. These figures net to zero.

The following diagram illustrates the typical settlement positions on the B24POS01 report.

- Card issuers are typically in a negative position to the network. They owe the network for their cardholders' transactions.
- Retailer sponsors are typically in a positive position to the network. The network owes them for the amounts they have credited their retailers.
- If an institution is both a card issuer and retailer sponsor, the amount they owe is subtracted from the amount they are owed to arrive at a net position to the network, positive or negative.

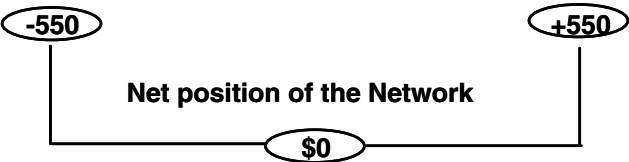


Owed to the Network

Bank 01 -100
Bank 03 -170
Bank 05 -80
Interchange 01 -200

Owed by the Network

Bank 02 +25
Bank 04 +200
Bank 06 +175
Interchange 02 +150

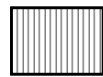
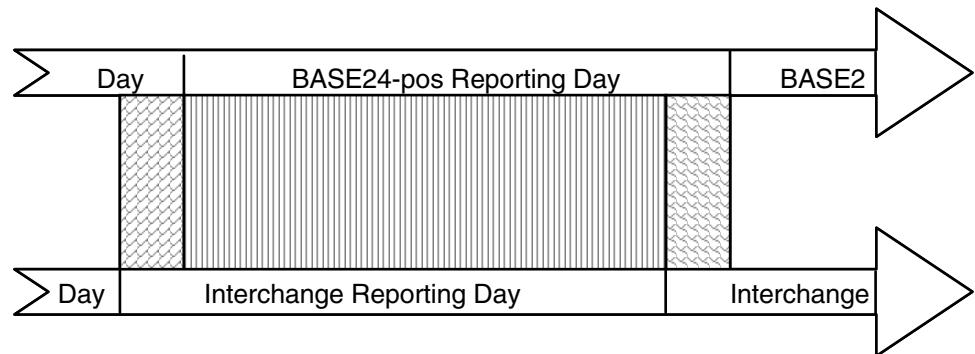


Interchange Settlement

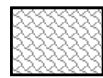
The BASE24-pos product recognizes a settlement position with only those interchanges that do not settle transactions through separate bank card settlement networks. This settlement position is provided on the Daily Network Settlement Report (B24POS01), which contains a net clearing amount for each BASE24-pos settlement entity.

Interchanges settle transactions based on their own business days. Since interchanges rarely cut over at the same time as the BASE24-pos network, the business day and the transactions to be settled differ between the two. The BASE24 system produces a set of interchange reports to reconcile these points of view. Interchange reports vary by interchange; however, they do allow the BASE24 participants to reconcile the settlement positions of the interchange and the BASE24 system. Where the B24POS01 report contains BASE24's settlement viewpoint, the BASE24 interchange reports contain both points of view along with a reconciliation of the two. The following diagrams illustrate how differing BASE24 and interchange business days are reconciled.

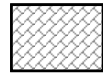
Interchanges Cutting Over Before The BASE24 System



Transactions during this time period are considered part of the day's settlement figures by BOTH the BASE24-pos system and the interchange.



Transactions during this time period are included in BASE24-pos settlement figures, but are NOT settled by the interchange until the next day.



Transactions during this time period are NOT included in BASE24-pos settlement figures, but are considered part of settlement by the interchange.

The B24POS01 report includes transactions in these periods:



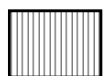
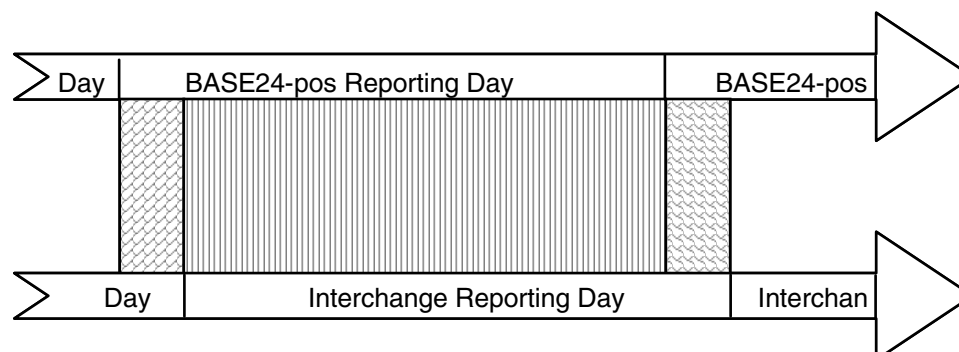
Interchange settlement figures include transactions in these time periods:



The BASE24 interchange reports reconcile BASE24 figures with interchange figures basically as follows:

$$\text{B24POS01 figures} + \text{[Cross-hatched box]} - \text{[Hatched box]}$$

Interchanges Cutting Over After The BASE24 System



Transactions during this time period are considered part of the day's settlement figures by BOTH the BASE24-pos system and the interchange.



Transactions during this time period are included in BASE24-pos settlement figures, but are NOT settled by the interchange until the next day.



Transactions during this time period are NOT included in BASE24-pos settlement figures, but are considered part of settlement by the interchange.

The B24POS01 report includes transactions in these periods:



Interchange settlement figures include transactions in these time periods:



The BASE24 interchange reports reconcile BASE24 figures with interchange figures basically as follows:

$$\text{B24POS01 figures} + \text{hatched} - \text{hatched}$$

4: BASE24-pos Reporting

This section describes the reports the BASE24-pos product provides to meet the reporting needs of its institutions and retailers. The information contained in these reports enables institutions to settle with the network; settle with their retailers; settle with interchanges; track cardholder, retailer, clerk, and referral activity; reconcile retailer terminals; and track cardholder disputed transactions. Retailers can also use the information in these reports to assess their transaction volumes and keep track of transaction activity at the shift, terminal, and clerk levels.

The BASE24-pos reports discussed in this section are divided into three categories—daily reports, periodic reports, and daily interchange reports. The reports shown and described in this section are listed below under the appropriate category.

Daily Reports

- Daily Network Settlement Report (B24POS01)
- Daily Cardholder Activity Report (B24POS02)
- Daily Cardholder Exception Report (B24POS02-EXPT)
- Daily Clerk Totals Report (B24POS04)
- Daily Disputed Transaction Report (B24POS05)
- Daily Retailer Summary Report (B24POS11)
- Daily Retailer Activity Report (B24POS12)
- Daily Retailer Exception Detail Report (B24POS12-EXPT)
- Network Referral Statistical Report (B24POS30)
- Retailer Reconciliation Report (B24POS35)

Periodic Reports

- Periodic Institution Summary Report (B24POS21)
- Periodic Retailer Summary Report (B24POS22)

Daily Interchange Reports

- Clearings Statement Report (SETL01)
- Transaction Detail Report (STAT09)

Daily Network Settlement Report (B24POS01)

The Daily Network Settlement Report (B24POS01) illustrates the perspective of the BASE24 logical network. Network participants can use this report to settle funds amongst themselves.

This report indicates the position each network participant holds in respect to the logical network. The B24POS01 lists each participant FIID and shows that participant's credit, debit, and net position to the network. Based on its net position, each network participant can determine the amount it owes or is owed by the network clearing account for today's transactions.

Both debit and credit card transactions are reflected on this report. However, only those transactions the BASE24-pos product classifies as settlement items are included on this report; transactions that must be settled through a national credit card settlement network, such as Visa, are not included in these totals. Adjustments to purchases with cash back are not included on this report because the settlement amounts for these transactions are already included in the corresponding purchase with cash back transactions. When an interchange associated with the network is not listed on the report, the BASE24-pos system did not process draft capture items for that interchange.

The BASE24-pos system generates one Daily Network Settlement Report (B24POS01) per logical network each day. The BASE24-pos system obtains the information for this report from the POS Transaction Log File (PTLF). The report is printed at the location specified in the POS-SETL-RPT-1 assign in the Logical Network Configuration File (LCONF). If an invalid spooler location is specified in the LCONF, the report program uses a default spooler location of \$\$.#ISR.

A sample Daily Network Settlement Report (B24POS01) is shown on the following page, followed by a description of each field.

REPORT NO: B24POS01 NETWORK SETTLEMENT REPORT				LOGICAL NETWORK: LLLL		JAN 20 YY		PAGE	1
INSTITUTION IDENTIFICATION		---INSTITUTION NAME---		-----CREDITS-----		-----DEBITS-----		NET	
				COUNT	AMOUNT	COUNT	AMOUNT	AMOUNT	AMOUNT
BK01	FIRST NATIONAL BANK	146,872	2,203,080.17	23,486	352,290.57-		1,850,789.60		
BK02	NATIONAL BANK OF TYLER	20,019	301,419.17	69,684	1,045,309.29-		743,890.12-		
BK03	U.S. NATIONAL BANK	18,476	277,140.32	76,482	1,158,364.36-		881,224.04-		
BK04	REDWOOD COUNTY BANK	61,364	1,173,685.62	23,573	353,595.58-		820,090.04		
BK05	GREENFIELD NATIONAL BANK	27,341	410,115.26	16,562	248,430.15-		161,685.11		
BK06	FIRST BANK OF OMAHA	83,691	1,731,667.41	28,637	383,398.67-		1,348,268.74		
BK07	FIRST BANK OF MELBOURNE	67,832	952,552.00	22,911	246,022.63-		706,639.37-		
BK08	BANK OF LONDON	32,622	527,202.67	65,958	976,886.22-		449,683.55-		
BK09	SECOND BANK OF PARIS	80,002	998,323.47	86,217	1,678,956.69-		690,633.22		
BK10	MALIBU BANK	23,676	250,981.55	18,622	215,638.20-		35,343.35		
BK11	SOUTHSIDE BANK	57,862	969,934.69	68,499	1,939,869.38-		969,934.69-		
FIG		26,029	360,435.68	104,529	1,567,886.27-		1,207,450.59-		
	NETWORK TOTALS	645,786	10,166,648.01	605,160	10,166,648.01		0,000,000.00-		
End of report									
****End of run****									

Format Description

The report header contains the report number, the report name, the logical network name, the reporting date, and the page number.

INSTITUTION IDENTIFICATION — The FIID that identifies this institution or the logical network of this interchange.

INSTITUTION NAME — The name of this institution or interchange. Institution names are taken from the Institution Definition File (IDF) and interchange names are taken from the Interchange Configuration File (ICF) or Enhanced Interchange Configuration File (ICFE).

CREDITS — Credits represent a positive position to the network. Credits include the transactions for which other network participants owe funds to this institution or interchange.

COUNT — The number of transactions for which the other network participants collectively owe this institution or interchange.

AMOUNT — The amount the other network participants collectively owe to this institution or interchange.

DEBITS — Debits represent a negative position to the network. Debits include the transactions for which this institution or interchange owes other network participants.

COUNT — The number of transactions for which this institution or interchange owes other network participants.

AMOUNT — The amount this institution or interchange owes other network participants.

NET AMOUNT — The net position of this institution or interchange to the logical network (i.e., credits minus debits). A minus sign indicates a negative position in which this institution or interchange owes funds to the network clearing account. No sign indicates a positive position in which this institution or interchange is owed funds from the network clearing account.

NETWORK TOTALS — The total debits and credits of all the network participants that the BASE24-pos product considers to be settlement entities, and the net amount of the total debits and credits. Total debits must always equal total credits with a net amount equal to zero.

END MESSAGES — The following messages appear at the end of the Network Settlement Report (B24POS01):

```
*** End of report ***  
**** End of run ****
```

When the BASE24-pos system cannot locate any network records and, therefore, cannot report network record information, the following message appears on the report format:

```
*** No reportable records ***
```

Daily Cardholder Activity Report (B24POS02)

The Daily Cardholder Activity Report (B24POS02) contains a listing and summary of each institution's cardholder activity for the day. The report lists both approved and declined cardholder transactions, chargebacks, representations, and sales drafts.

Institutions can use the Daily Cardholder Activity Report (B24POS02) to track cardholder activity, resolve customer service problems, and determine proper authorization on disputed transactions. This report is an audit report only; it contains no settlement information.

The B24POS02 report is produced in two parts. Part One provides a detailed listing of the transactions performed by the institution's cardholders in ascending cardholder number order. Part Two recaps the cardholder activity provided in Part One.

This report is an optional report. The BASE24-pos system produces it based on the REPORT SET field on Institution Definition File (IDF) screen 21. See section 5 or the *BASE24 Base Files Maintenance Manual* for more information on configuring this field for producing reports.

The masking of cardholder primary account numbers (PANs) and administrative PANs on this report depends on the values of the four REPORT DATA MASKING PARAMETER fields on Institution Definition File (IDF) screen 4.

The BASE24-pos system generates one Daily Cardholder Activity Report (B24POS02) for each institution in the logical network each day, if configured to do so in the REPORT SET field on IDF screen 21. The BASE24-pos product obtains the information for this report from the POS Transaction Log File (PTLF). The report is printed at the location specified in the CARDHOLDER ACTIVITY REPORT PRINT LOCATION field on IDF screen 21. If no spooler location is specified in the IDF, the report program uses a default spooler location of \$\$. #CARD.

A sample Daily Cardholder Activity Report (B24POS02) is shown on the following pages, followed by a description of each field.

Nov-2007 R6.0v8 PS-AE000-06
ACI Worldwide, Inc.

Report No: B24POS02 Cardholder Activity Report										BK01 - FIRST NATIONAL BANK		DD MM YY		Page	
-----Cardholder Number-----		Mbr		-----Account Number-----		Type		--Merchant Number--		--Terminal Id--		--Tran Description--			
-----Tran Amount-----		-Zip Code-		-----Cash Out-----		Rsp		Tran Date		Tran time		----Seq No----		Src	
-Cardholder Address-				Addr Stat										Auth Cde	
*****7115		* 332615		5.59		0.00		01 RETAILERDR		12:03:41		1212121212121212		PURCHASE	
								01 JUN 93				000000000000		B24	
														49627631	
*****7226		* 341726		6.69		0.00		01 RETAILERDR		12:05:36		1212121212121212		PURCHASE	
								01 JUN 93				000000000000		B24	
														82436511	
*****7271		* 343629		1.19		0.00		01 RETAILERDR		12:32:15		1212121212121212		PURCHASE	
								01 JUN 93				000000000000		B24	
														968099 B	
*****7313		* 348896		5.55		0.00		01 RETAILERDA		12:43:15		1212121212121212		PURCHASE	
								01 JUN 93				000000000000		B24	
														802673 B	
*****7350		* 349762		5.55		0.00		01 RETAILERDA		12:44:25		1212121212121212		PURCHASE	
								01 JUN 93				000000000000		B24 *RVSAL*	
														802673 B	
*****7367		* 355671		1.19		0.00		01 RETAILERDA		12:44:25		1212121212121212		PURCHASE	
								01 JUN 93				000000000000		B24	
														351826 B	
*****7402		* 359932		1.19		0.00		01 RETAILERDA		12:48:38		1212121212121212		PURCHASE	
								01 JUN 93				000000000000		B24 *RVSAL*	
														351826 B	
*****7488		* 362145		8.89		0.00		01 RETAILERDA		12:51:17		1212121212121212		PURCHASE	
								01 JUN 93				000000000000		B24	
														450245 B	
*****7491		* 363127		8.89		0.00		01 RETAILERDA		12:52:29		1212121212121212		PURCHASE	
								01 JUN 93				000000000000		B24 *RVSAL*	
														450245 B	

Part One Format Description

Most of the report fields are taken directly from corresponding fields in the PTLF records. Where this is true, the documentation parenthetically includes the name of the PTLF record field from which the information is taken.

The report header contains the report number, the report name, the institution FIID, the institution name (taken from the IDF), the reporting date, and the page number.

CARDHOLDER NUMBER — (HEAD.CRD.CARD.CRD-NUM) The card number used in the transaction. Values of the four REPORT DATA MASKING PARAMETER fields on Institution Definition File (IDF) screen 4 control masking of values in this field.

MBR — (HEAD.CRD.CARD.MBR-NUM) The member number associated with the card used in the transaction. The member number distinguishes between cards when multiple cards are issued with the same card number. If an institution does not support member numbers, this field defaults to 000.

ACCOUNT NUMBER — (AUTH.ACCT) The number of the application account the transaction impacted.

TYPE — (AUTH.TRAN-CDE.AA) A code identifying the type of application account the transaction impacted. Valid codes are:

00 = None
01–09 = DDA
11–19 = Savings
31–39 = Credit

MERCHANT NUMBER — (HEAD.RETL.KY.RDFKEY.ID) The identification number for the retailer associated with this transaction.

TERMINAL ID — (HEAD.RETL.TERM-ID) The identification number for the terminal at which the transaction was performed.

TRAN DESCRIPTION — A literal description of the type of transaction (e.g., PURCHASE, PRE-AUTH PUR, CASH ADVANCE, CHARGEBACK, SALES DRAFT, etc.).

TRAN AMOUNT — (AUTH.AMT-1) The requested transaction amount. In the case of an adjustment transaction, this field contains the amount originally requested. For a preauthorization request, this field contains the amount to be held. For a preauthorization completion, this field contains the completed amount. For a cash back transaction, this field contains the total amount of the transaction (the actual cash back amount of the transaction is contained in the CASH OUT field).

CASH OUT — (AUTH.AMT-2) For a purchase with cash back transaction, this field contains the actual cash back amount (the TRAN AMOUNT field contains the total amount). For an adjustment, this field contains the new amount.

RSP — (AUTH.RESP-CDE) The response code given to the transaction to indicate whether it was approved or declined. Refer to the ***BASE24-pos Transaction Processing Manual*** for response codes and their descriptions.

TRAN DATE — (AUTH.TRAN-DAT) The date (DD MMM YY) on which the transaction occurred.

TRAN TIME — (AUTH.TRAN-TIM) The time of day (hh:mm:ss) at which the transaction occurred.

SEQ NO — (AUTH.SEQ-NUM) The sequence number that either the originating terminal or the BASE24 system assigned to this transaction.

SRC — (AUTH.ORIG) A literal description of the transaction originator. Transactions can originate from the following sources:

- A terminal directly connected to the BASE24 system. In this case, the literal “B24” appears in this field.
- An acquirer terminal not controlled by the BASE24 system. In this case, the literal “HOST” appears in this field.
- An interchange. In this case, the literal indicates the originating interchange (e.g., BASE, ACDS, and AMEX).
- A retailer. In this case, the literal “RETL” appears in this field.

AUTH CDE — (AUTH.APPRV-CDE) The approval code assigned to the transaction. When transactions are approved, an approval code is assigned for reference.

<*RVRSAL*> — The names listed below are printed following the transaction source, if any of the corresponding transactions appear on this report.

RVRSAL = Reversal
RV ADJ = Reversal Adjustment
X RVL = Exception Reversal
X RVLA = Exception Reversal Adjustment
F POST = Force Post
FP ADJ = Force Post Adjustment
X FP = Exception Force Post
XFPA = Exception Force Post Adjustment
ADJUST = Adjustment
X Adj = Exception Adjustment

CARDHOLDER ADDRESS — (AUTH.ADDR) The cardholder billing address received with the transaction when performing address verification.

ZIP CODE — (AUTH.ZIP-CDE) The cardholder billing ZIP code received with the transaction when performing address verification.

ADDR STAT — (AUTH.ADDR-VRFY-STAT) A code identifying the result of comparing address verification information received in the transaction and address verification information contained in the processor's database.

The following values can be set by BASE24-pos through its address verification processing. If a transaction is sent to an interchange for processing, the response message may contain other interchange-specific values (BASE24 does not change values received from the interchanges).

A = ADDRESS—Addresses matched, but ZIP codes did not match.
E* = ERROR—The transaction was not eligible for address verification or an editing error occurred while attempting to process the message.
N = NO—Addresses did not match and ZIP codes did not match.
R* = RETRY—Primary and secondary authorizers were unavailable or declined the transaction and address verification was not performed on the BASE24-pos system.
S* = SERVICE NOT SUPPORTED—The BASE24-pos system authorized the transaction, but did not have the add-on Address Verification module.
U* = UNAVAILABLE—Address information was not available to the processor performing address verification.
W = WHOLE ZIP—Nine-digit ZIP codes matched, but addresses did not match.
X = EXACT—Addresses and nine-digit ZIP codes matched.

- Y = YES—Addresses and five-digit ZIP codes matched.
- Z = ZIP—Five-digit ZIP codes matched, but addresses did not match.
- b* = Address verification information was not included in this transaction (b denotes a blank character).
- 0* = Address verification information was included in this transaction, but was not verified. A transaction to be verified by a host or interchange carries this code. A transaction to be verified on the BASE24-pos system, but declined before address verification could be performed also carries this code.
- * This value identifies a reason that a comparison was not made. Each interchange specifies the value it uses to identify this reason. BASE24 Interchange Interfaces substitute the interchange-specific value before sending the message from the BASE24-pos system to the interchange. Refer to section 1 of the ***BASE24-pos Address Verification Manual*** for a listing of interchange-specific values.

REPORT NO: B24POS02		CARDHOLDER ACTIVITY REPORT		BK01 - FIRST NATIONAL BANK		JAN 20 YY		PAGE	
INSTITUTION TOTALS:								3	
--TRAN DESCRIPTION--		--ON-US RETAILER--		NOT ON-US RETAILER--		-----INTERCHANGE-----		-----TOTAL-----	
NORMAL PURCHASE	APP	TRAN COUNT	7		8		0	15	
		TRAN AMOUNT	140.00		0.00		0.00	140.00	
	DEC	TRAN COUNT	3		2		0	5	
		TRAN AMOUNT	47.79		12.28		0.00	60.07	
PRE-AUTH PURCHASE	APP	TRAN COUNT	1		0		0	1	
		TRAN AMOUNT	20.00		0.00		0.00	20.00	
	DEC	TRAN COUNT	0		0		0	0	
		TRAN AMOUNT	0.00		0.00		0.00	0.00	
PRE-AUTH PUR COMPL	APP	TRAN COUNT	1		0		0	1	
		TRAN AMOUNT	20.00		0.00		0.00	20.00	
	DEC	TRAN COUNT	0		0		0	0	
		TRAN AMOUNT	0.00		0.00		0.00	0.00	
MERCHANDISE RETURN	APP	TRAN COUNT	1		0		0	1	
		TRAN AMOUNT	20.00		0.00		0.00	20.00	
	DEC	TRAN COUNT	0		0		0	0	
		TRAN AMOUNT	0.00		0.00		0.00	0.00	
ADJUST PURCHASE	APP	TRAN COUNT	1		0		0	1	
		TRAN AMOUNT	20.00		0.00		0.00	20.00	
	DEC	TRAN COUNT	0		0		0	0	
		TRAN AMOUNT	0.00		0.00		0.00	0.00	
TOTAL	APP	TRAN COUNT	11		8		0	19	
		TRAN AMOUNT	220.00		0.00		0.00	220.00	
	DEC	TRAN COUNT	3		2		0	5	
		TRAN AMOUNT	47.79		12.28		0.00	60.07	
TOTAL TRANSACTIONS		TRAN COUNT	14		10		0	24	
		TRAN AMOUNT	267.79		12.28		0.00	280.07	
*** End of report ***									

Part Two Format Description

TRAN DESCRIPTION — Literal descriptions of the transactions for which the report totals are provided: APP identifies the transaction totals for approved transactions; DEC identifies the transaction totals for declined transactions.

ON-US RETAILER — Totals for transactions performed by the institution's cardholders at the institution's retailers. These transactions are considered on-us transactions.

TRAN COUNT — The total number of on-us transactions.

TRAN AMOUNT — The total amount, in fractional currency units, of on-us transactions.

NOT ON-US RETAILER — Totals for transactions this institution's cardholders performed at other institutions' retailers (within the same logical network). These transactions are considered not-on-us transactions.

TRAN COUNT — The total number of not-on-us transactions.

TRAN AMOUNT — The total amount, in fractional currency units, of not-on-us transactions.

INTERCHANGE — Totals for transactions the institution's cardholders performed at retailers belonging to interchanges.

TRAN COUNT — The total number of transactions the institution's cardholders performed at retailers belonging to interchanges.

TRAN AMOUNT — The total amount, in fractional currency units, of transactions the institution's cardholders performed at retailers belonging to interchanges.

TOTAL — The totals for each type of transaction listed on the report.

TRAN COUNT — The total number of each type of transaction listed on the report.

TRAN AMOUNT — The total amount, in fractional currency units, of each type of transaction listed on the report.

TOTAL APP — The totals for approved transactions in the ON-US RETAILER, NOT-ON-US RETAILER, INTERCHANGE, and TOTAL columns.

TRAN COUNT — The total number of approved transactions listed in the ON-US RETAILER, NOT-ON-US RETAILER, INTERCHANGE, and TOTAL columns.

TRAN AMOUNT — The total amount, in fractional currency units, of approved transactions listed in the ON-US RETAILER, NOT-ON-US RETAILER, INTERCHANGE, and TOTAL columns.

TOTAL DEC — Totals for declined transactions in the ON-US RETAILER, NOT ON-US RETAILER, INTERCHANGE, and TOTAL columns.

TRAN COUNT — The total number of declined transactions listed in the ON-US RETAILER, NOT-ON-US RETAILER, INTERCHANGE, and TOTAL columns.

TRAN AMOUNT — The total amount, in fractional currency units, of declined transactions listed in the ON-US RETAILER, NOT-ON-US RETAILER, INTERCHANGE, and TOTAL columns.

TOTAL TRANSACTIONS — The totals for all transactions in the ON-US RETAILER, NOT ON-US RETAILER, INTERCHANGE, and TOTAL columns.

TRAN COUNT — The total number of all transactions listed in the ON-US RETAILER, NOT-ON-US RETAILER, INTERCHANGE, and TOTAL columns.

TRAN AMOUNT — The total amount, in fractional currency units, of all transactions listed in the ON-US RETAILER, NOT-ON-US RETAILER, INTERCHANGE, and TOTAL columns.

END MESSAGES — The following messages appear at the end of the Cardholder Activity Report (B24POS02):

*** End of report ***

When the BASE24-pos system cannot locate any cardholder records and, therefore, cannot report cardholder record information, the following message appears on the report format:

*** No reportable records ***

Daily Cardholder Exception Report (B24POS02-EXPT)

The Daily Cardholder Exception Report (B24POS02-EXPT) contains a listing and summary of each institution's cardholder activity that resulted in exception processing that day. The report includes all transactions that have the following record types:

- 21 = Exception not posted to any BASE24 files
- 22 = Exception posted to some but not all applicable BASE24 files
- 23 = Exception not posted because of invalid data

The BASE24-pos product performs exception processing when an authorizer other than the BASE24 system has approved the transaction and the transaction contains data or is missing data that prevents the BASE24-pos system from posting the transaction correctly to BASE24 files. Transactions that may result in exception processing include force posts, reversals, and requests. This report does not include administrative or settlement records.

Institutions can use the Daily Cardholder Exception Report (B24POS02-EXPT) to research exception transactions and ensure that they have been appropriately posted to the BASE24 files. If the BASE24-pos system was unable to post the transaction appropriately, the institution must update the appropriate files manually to keep the BASE24 database accurate.

The B24POS02-EXPT is produced in the same format as Part One of the Daily Cardholder Activity Report (B24POS02). It provides a detailed listing of the transactions involved in exception processing in ascending cardholder number.

This report is an optional report. The BASE24-pos product produces it based on the POS-EXPT-RPT-FLG param in the LCONF. See section 5 or the ***BASE24 Logical Network Configuration File Manual*** for more information about this param.

The BASE24-pos product generates one Daily Cardholder Exception Report (B24POS02-EXPT) for each institution in the logical network each day, if configured to do so in the LCONF. The BASE24-pos system obtains the information for this report from the POS Transaction Log File (PTLF). If no exception records exist in the PTLF for this institution, the report contains the following comment:

```
*** NO EXCEPTION RECORDS FOR THIS INSTITUTION ***
*** END OF REPORT ***
```

The report is printed at the spooler location specified in the POS-SETL-RPT-1R param in the LCONF. If this param contains an invalid value, the report program uses a default spooler location of \$S.#EXPT2.

Because this report is identical in format to Part One of the Daily Cardholder Activity Report (B24POS02), a sample report is not provided in this manual. Refer to the sample of the B24POS02 report for a description of the Daily Cardholder Exception Report (B24POS02-EXPT).

Daily Clerk Totals Report (B24POS04)

The Daily Clerk Totals Report (B24POS04) enables retailers to track each clerk's transaction activity at a particular terminal during the business day. Clerk totals include counts and amounts for approved debit, credit, adjustment, check guarantee, check verification, and cash out transactions the clerk entered at the terminal. The report also provides a grand total for each clerk. This report includes only approved transactions.

The report also provides totals for all transactions performed at the terminal by all clerks for the particular retailer ID.

The BASE24-pos system can produce the Daily Clerk Totals Report (B24POS04) only if clerk totals are being written to the PTLF. Clerk totals are written to the PTLF for clerks at a specific terminal based on the CLERK TOTALS FLAG on BASE24-pos Terminal Data files (PTD) screen 5. If this field contains a value other than 0, the BASE24-pos system maintains clerk totals for the terminal. The BASE24-pos system maintains clerk totals in the BASE24-pos Terminal Data files until the operator logs off of the terminal or a new operator logs on. At this time, the BASE24-pos system creates a clerk totals record, writes the clerk totals record to the PTLF, and clears the BASE24-pos Terminal Data files clerk totals.

The BASE24-pos system generates one Daily Clerk Totals Report (B24POS04) for each terminal, retailer, and institution in the system every day. This report is separated according to clerk IDs within a retailer. With each new batch, a line item is displayed on the report. With each new clerk ID, a grand totals section is printed for the previous clerk ID and a new section begins on the report. With each new retailer ID, a new page begins. Retailer totals are provided for each retailer and institution.

The BASE24-pos system obtains information for the B24POS04 report from the PTLF. The report is printed at the location specified in the PRINT LOCATION field on POS Retailer Definition File (PRDF) screen 4. If this field contains an invalid spooler location, the report program uses a default spooler location of \$S.#RETL.

A sample Daily Clerk Totals Report (B24POS04) is shown on the following page, followed by a description of each field.

REPORT NO: B24POS04 CLERK TOTALS REPORT				BK01 - FIRST NATIONAL BANK		MM DD YY		PAGE 1	
CLERK WITHIN RETAILER						MM DD YY			
RETAILER ID: RETAILER 01						MM DD YY			
B24 TERM ID	SHIFT	---DEBITS---		---CREDITS---		B24 DATE		---CHECKS---	
TIME LOGGED	BATCH	CNT/AMT		CNT/AMT		CNT/AMT		CNT/AMT	
CLERK ID: CLRK01									
90000000000	002	2	4						
11:22:36	001	32.56-	26.37						
90000000000	002	10	6						
14:53:21	002	200.00-	47.62						
TOTALS FOR									
CLRK01		12	10						
		232.56-	73.99						
CLERK ID: CLRK02									
90000000000	003	11	17			1		1	
15:27:16	001	213.67-	462.98			15.00		20.00	
90000000000	003	8	6						
17:41:01	002	147.39-	153.51			10.00			
90000000000	003	1	2					2	
17:57:02	003	30.00-	50.00					53.90	
TOTALS FOR									
CLRK02		20	25			1		3	
		391.06-	666.49			15.00		73.90	
*** End of report ***									
*** End of run ***									

Format Description

Most of the report fields are taken directly from corresponding fields in the POS Transaction Log File (PTLF) records. Where this is true, the documentation parenthetically includes the name of the PTLF field from which the information is taken.

The report header contains the report number, the report name, the FIID and the name of the financial institution for which this report is being generated, the reporting date, and the page number.

RETAILER ID — (TKEY.RKEY.RETAILER-ID) The retailer ID of the retailer associated with the clerk performing these transactions.

B24 DATE — The date (MMM DD YY) for which the clerk totals are being reported.

B24 TERM ID — (TKEY.TERM-ID) The terminal ID of the terminal for which data is being reported.

SHIFT — (RETL.SHIFT-NUM) The number identifying the shift with which these clerk totals are associated.

DEBITS CNT/AMT — (CLERK-TOT.SET-REC5.DB-CNT and CLERK-TOT.SET-REC5.DB-AMT) The total number and amount, in fractional currency units, of approved debit transactions for this batch and shift.

CREDITS CNT/AMT — (CLERK-TOT.SET-REC5.CR-CNT and CLERK-TOT.SET-REC5.CR-AMT) The total number and amount, in fractional currency units, of approved credit transactions for this batch and shift.

ADJUSTMENTS CNT/AMT — (CLERK-TOT.SET-REC5.ADJ-CNT and CLERK-TOT.SET-REC5.ADJ-AMT) The total number and amount, in fractional currency units, of approved adjustment transactions for this batch and shift.

CASH OUT CNT/AMT — (CLERK-TOT.SET-REC5.CASH-CNT and CLERK-TOT.SET-REC5.CASH-AMT) The total number and amount, in fractional currency units, of approved cash out transactions for this batch and shift.

CHECKS CNT/AMT — (CLERK-TOT.SET-REC5.CHK-CNT and CLERK-TOT.SET-REC5.CHK-AMT) The total number and amount, in fractional currency units, of approved check guarantee and check verification transactions for this batch and shift.

TIME LOGGED — (TERM.TIM) The time (hh:mm:ss) the totals for this clerk were written to the PTLF.

BATCH — (RETL.BATCH-NUM) The number identifying the batch with which these clerk totals are associated.

CLERK ID — (TKEY.RKEY.CLERK-ID) The clerk ID of the clerk whose totals are being presented.

TOTALS FOR CLERK — The combined total number and amount, in fractional currency units, of all approved debit, credit, adjustment, cash out, check guarantee, and check verification transactions for this batch and shift.

END MESSAGES — The following messages appear at the end of the Daily Clerk Totals Report (B24POS04):

```
*** End of report ***  
*** End of run ***
```

When the BASE24-pos system cannot locate any clerk totals records and, therefore, cannot report clerk totals record information, the following message appears on the report format:

```
*** No reportable records ***
```

Daily Disputed Transactions Report (B24POS05)

The Daily Disputed Transactions Report (B24POS05) aids retailers in monitoring chargeback and representation transactions. This report lists each chargeback and representation transaction entered into the system during the current business day. This report also displays force balances. Retailers can use the B24POS05 report as a tool to research and resolve disputed transactions.

Note: The BASE24-pos system generates the Daily Disputed Transactions Report (B24POS05) only if the institution is using the BASE24-pos CRT Administration add-on product.

The report includes the type of transaction, original amount of the transaction, replacement amount of the transaction, and reason for the chargeback. The information on the report is organized according to retailer ID. Each new retailer ID begins a new page.

Chargebacks are monetary transactions that cardholders have disputed. The acquiring institution or retailer must research the disputed transaction to determine if, in fact, the cardholder performed the transaction and is required to pay the transaction amount. An example of a disputed transaction would be if a cardholder were charged twice for the same transaction.

Representations are monetary transactions that a cardholder has disputed (charged back to the acquiring institution or retailer), but the acquiring institution or retailer has researched the transaction and determined the transaction to be accurate as it stands. Thus, the acquiring institution or retailer is presenting the transaction for settlement again. In this case, the cardholder is responsible for paying the transaction amount.

The BASE24-pos system generates one B24POS05 report for each institution and acquirer retailer each day. The report is printed at the spooler location specified in the MERCHANT SETTLEMENT REPORT PRINT LOCATION field on Institution Definition File (IDF) screen 21. If this field contains an invalid spooler location, the report program uses a default spooler location of \$\$.#RETL.

The masking of cardholder primary account numbers (PANs) and administrative PANs on this report depends on the values of the four REPORT DATA MASKING PARAMETER fields on Institution Definition File (IDF) screen 4.

A sample the Daily Disputed Transactions Report (B24POS05) is shown on the following page, followed by a description of each field.

REPORT NO: B24POS05 DISPUTED TRANSACTIONS REPORT										PAGE	
RETAILER ID: 012391398										1	
-----CARDHOLDER NUMBER-----											
--ORIGINAL AMOUNT-- MBR -INST CODE- ---TERM ID--- CARD ACC ID CDE ORIG REF NUMBER ORIG TYPE ORIG DATE ORIG TIME											
---REPLACE AMOUNT-- APPR CODE AUTH RESP -----REASON FOR CHARGE BACK-----											
*****0000	*** 02101239139 SPDH2	10.00	15.00	00000000000	000000000202	0200	05/01	10:31:55			
*****0000	*** 00000000000 SPDH2	10.00	15.00	00000000000	960 03 - INVALID MERCHANT	0200	05/01	10:31:55			
*****0000	*** 00000000000 SPDH2	10.00	15.00	00000000000	001 03 - INVALID MERCHANT	0200	05/03	14:44:17			
*****0000	*** 00000000000 SPDH2	10.00	15.00	00000000000	960 59 - SUSPECTED FRAUD	0200	05/03	14:44:17			
*****0000	*** 00000000000 SPDH2	10.00	15.00	00000000000	001 59 - SUSPECTED FRAUD	0200	05/03	14:44:17			
TOTAL FOR RETAILER: --ORIGINAL AMOUNT-- ---REPLACE AMOUNT-- TRAN COUNT											
		40.00	60.00	4							
TOTAL TRANSACTION COUNT = 4											
TOTAL ORIGINAL AMOUNT = 40.00											
TOTAL REPLACEMENT AMOUNT = 60.00											
*** End of report ***											
*** End of run ***											

Format Description

The report header contains the report number, the report name, the FIID and name of the financial institution for which this report is being generated, the reporting date, and the page number.

RETAILER ID — The retailer ID of the retailer associated with these disputed transactions.

CARDHOLDER NUMBER — The account number identifying the card associated with the disputed transaction. All digits in this number, except the last four, are masked as per the application's default setting.

MBR — The member number of the card. The member number distinguishes between cards when multiple cards are issued with the same card number. This number is masked as per the application's default settings.

INST CODE — The Route Transit Number, transit/routing number, or issuer identification number of the acquiring institution.

TERM ID — The terminal ID of the POS terminal where the original transaction was performed.

CARD ACC ID CDE — A code identifying the host card acceptor when the transaction originated from an acquirer host. This field is used only when the acquirer is not the actual card acceptor. Otherwise, this field is left blank.

ORIG REF NUMBER — The sequence number the system assigned to the original transaction when the transaction was originally performed. The sequence number uniquely identifies a transaction.

ORIG TYPE — The original message type associated with the transaction being disputed.

ORIG DATE — The original date (MMDD) on which the transaction being disputed occurred.

ORIG TIME — The original time (hh:mm:ss) at which the transaction being disputed occurred.

ORIGINAL AMOUNT — The original amount of the transaction being disputed.

REPLACE AMOUNT — The new amount of the disputed transaction. This amount replaces the original amount of the transaction if it has changed due to a chargeback.

APPR CODE — The approval code if the transaction was sent to an interchange that returns approval codes to be printed on the receipt. Or, if the Authorization module and the host approve credit cards, they generate a value for this field. If the BASE24-pos system generates the approval code, a B appears in the last byte of this field.

AUTH RESP — The response code given to the original transaction to indicate whether it was approved or declined. Refer to the *BASE24-pos Transaction Processing Manual* for a list of response codes and their descriptions.

REASON FOR CHARGE BACK — The reason the transaction was charged back. Valid codes are:

- 03 = Invalid merchant
- 12 = Invalid transaction
- 18 = Customer dispute
- 59 = Suspected fraud
- 63 = Security violation
- 64 = Original transaction amount is incorrect
- 68 = Supporting documentation received too late
- 93 = Transaction in violation of the law
- 94 = Duplicate transaction
- 96 = System malfunction

TOTAL FOR RETAILER — The following fields contain the total of the original amounts, replacement amounts, and the number of disputed transactions occurring for the retailer.

ORIGINAL AMOUNT — The combined total amount, in fractional currency units, of original disputed transactions displayed on the report for the retailer.

REPLACE AMOUNT — The combined total replacement amount, in fractional currency units, of disputed transactions for the retailer.

TRAN COUNT — The total number of disputed transactions displayed on the report for the retailer.

TOTAL TRANSACTION COUNT — The combined number of disputed transactions displayed for all retailers on this report.

TOTAL ORIGINAL AMOUNT — The combined original amount, in fractional currency units, for all retailers on this report.

TOTAL REPLACEMENT AMOUNT — The combined total replacement amount, in fractional currency units, of disputed transactions for all retailers on this report.

END MESSAGES — The following messages appear at the end of the Daily Disputed Transactions Report (B24POS05):

```
*** End of report ***  
***   End of run   ***
```

When the BASE24-pos system cannot locate any retailer records and, therefore, cannot report disputed transaction record information, the following message appears on the report format:

```
*** No reportable records ***
```

Daily Retailer Summary Report (B24POS11)

Institutions can use the Daily Retailer Summary Report (B24POS11) to settle with their retailers. The figures on this report provide comprehensive totals to aid the retailer sponsor in reconciling with its retailers. The report reflects the total number and amount of transactions that occurred during the business day. These totals appear on the report according to the types of cards the retailer accepts.

This report also provides separate draft capture totals for the retailers. These draft capture totals specify the amount each retailer must be credited for the business day's transaction activity.

The BASE24-pos system generates the Daily Retailer Summary Report (B24POS11) each day using information in the POS Transaction Log File (PTLF). The report is printed at the spooler location specified in the POS-RPT-2 assign in the Logical Network Configuration File (LCONF). If the BASE24-pos system cannot find the POS-RPT-2 assign in the LCONF, it sends the report to the location specified in the PRINT LOCATION field on POS Retailer Definition File (PRDF) screen 4. If neither the PRDF or the LCONF contains a valid print location, the report program uses a default spooler location of \$S.#RETL.

Note: The BASE24-pos system produces the Daily Retailer Summary Report (B24POS11), the Daily Retailer Activity Report (B24POS12), and the Daily Retailer Exception Detail Report (B24POS12-EXPT) at the same time. The B24POS11 report is combined with the institution-level portion of the B24POS12 into a single spooler job. If the B24POS11 and B24POS12 must be combined for all retailers and printed at the same time, the BASE24-pos product must use the LCONF assign spooler location. The BASE24-pos product must use the PRDF field when the B24POS11 and B24POS12 reports are to be printed individually for each retailer. When the PRDF field is to be used to obtain the spooler location, the LCONF assign should not be configured.

A sample Daily Retailer Summary Report (B24POS11) is shown on the following pages, followed by descriptions of its fields.

REPORT NO: B24POS11 RETAILER SUMMARY REPORT		BK01 - FIRST NATIONAL BANK		26 OCT 93		PAGE 1	
RETAILER NUMBER: RETAILER10		RETAILER10		GROUP: 0001		REGION: 0001	
-----CARD TYPE-----		-----ON US-----		-----NOT ON US-----		-----TOTAL-----	
TRAN CNT		TRAN AMOUNT		TRAN CNT		TRAN AMOUNT	
CARD TYPE TOTAL (PROP DEBIT)		9		1		10	
CAPTURE TOTAL		9		1		10	
CARD TYPE TOTAL (MASTERCARD DEBIT)		137		20		157	
CAPTURE TOTAL		137		20		157	
CARD TYPE TOTAL (VISA DEBIT)		153		15		168	
CAPTURE TOTAL		153		15		168	
RETAILER TOTALS:							
CARD TYPE TOTAL **ALL CARD TYPES**		299		36		335	
CAPTURE TOTAL		299		36		335	

Format Description

Line 1 of the report header contains the report number, report name, institution FIID, institution name (taken from the IDF), reporting date, and page number.

Line 2 of the report header contains the retailer ID, retailer name, group, and region taken from the PRDF.

CARD TYPE — The card types the retailer accepts (e.g., MasterCard, Visa, Diners Club, etc.).

ON US — The count and amount totals for transactions performed at the retailer where the cardholder and the retailer belong to the same institution.

TRAN CNT — The transaction counts for on-us transactions.

TRAN AMOUNT — The transaction amounts for on-us transactions.

NOT ON US — The count and amount totals for transactions performed at the retailer where the cardholder and the retailer belong to different institutions.

TRAN CNT — The transaction counts for not-on-us transactions.

TRAN AMOUNT — The transaction amounts for not-on-us transactions.

TOTAL — The combined count and amount totals for all transactions performed at the retailer.

TRAN CNT — The combined number of all transactions performed at the retailer.

TRAN AMOUNT — The combined amount of all transactions performed at the retailer.

CARD TYPE TOTAL — The totals for all transactions initiated using cards of the type specified in the CARD TYPE field. Each card type has its own set of totals. These totals include all authorization-only transactions and all authorization and capture transactions.

CAPTURE TOTAL — The totals for the day's draft capture transactions from this retailer. These totals represent the amount the institution must credit the retailer for draft capture transactions.

RETAILER TOTALS — The retailer totals for all card types identified in the CARD TYPE field.

INSTITUTION TOTALS — The totals for all retailers.

END MESSAGES — The following message appears at the end of the Daily Retailer Summary Report (B24POS11):

*** End of report ***

Daily Retailer Activity Report (B24POS12)

Retailers use the Daily Retailer Activity Report (B24POS12) to track activity by shift for an individual terminal. The report provides the retailer with a record of the total activity of all terminals. Institutions use the report to track their retailer activity by group, region, and retailer.

The B24POS12 report is produced in two parts. Part One lists all transactions in chronological order. Part Two summarizes the information in Part One.

The Daily Retailer Activity Report (B24POS12) is an optional report. The BASE24-pos system generates this report based on the value in the REPORT SET field on PRDF screen 5 as follows:

- 0 = Full report; the report program generates the entire report.
- 1 = Separate totals; the report program generates the entire report, but produces Part One and Part Two as separate spooler jobs.
- 2 = Totals only; the report program generates Part Two of the report only.
- 3 = No report; the report program does not generate the report.

The BASE24-pos system generates the B24POS12 report each day using information in the POS Transaction Log File (PTLF). The report is printed at the location specified in the POS-RPT-2 assign in the Logical Network Configuration File (LCONF). If the BASE24-pos product cannot find the POS-RPT-2 assign in the LCONF, it sends the report to the location specified in the PRINT LOCATION field on PRDF screen 4. If neither the LCONF or the PRDF contains a valid print location, the report program uses a default spooler location of \$\$.#RETL.

Note: The BASE24-pos system generates the Daily Retailer Summary Report (B24POS11), the Daily Retailer Activity Report (B24POS12), and the Daily Retailer Exception Detail Report (B24POS12-EXPT) at the same time. The institution-level portions of the B24POS12 report are combined with the B24POS11 report into a single spooler location. If the B24POS11 and B24POS12 reports must be combined for all retailers and printed at the same time, the BASE24-pos system must use the spooler location specified by the LCONF assign. The BASE24-pos system must use the PRDF field when the B24POS11 and B24POS12 reports are to be printed individually for each retailer. If the PRDF field is to be used to obtain the spooler location, the LCONF assign should not be configured.

The masking of cardholder primary account numbers (PANs) and administrative PANs on this report depends on the values of the four REPORT DATA MASKING PARAMETER fields on Institution Definition File (IDF) screen 4.

A sample Daily Retailer Activity Report (B24POS12) is shown on the following pages, followed by descriptions of its fields.

REPORT NO: B24POS12 RETAILER ACTIVITY REPORT 1										26 OCT YY		PAGE	1
GRP: 0001 RGN: 0001 RETAILER INFO: RETAILER10										TERM: 011216 CFIDANTE		BCH: 001	001
-----CARDHOLDER NUMBER-----										BK01 - FIRST NATIONAL BANK			
-CARDHOLDER ADDRESS-										RETAILER 10			
MBR ---TRAN DESCRIPTION--- DC										---TRAN AMOUNT---		---CASH OUT---	
TERM OPR SEQUENCE NUM										TRAN DATE		RESP CODE	
-ZIP CODE- ADDR STAT										TRAN TIME		AUTH CODE	
												TRAN DEST	
*****4276	*** PURCHASE	0000000000332	*							20.00	24 OCT YY 001	974542 B	B24
*****4288	*** PURCHASE	0000000000333	*							20.00	24 OCT YY 001	223579 B	B24
*****4289	*** PURCHASE	0000000000334	*							20.00	24 OCT YY 001	282562 B	B24
*****4442	*** PURCHASE	0000000000335	*							20.00	24 OCT YY 001	933851 B	B24
*****4462	*** PRE AUTH PUR	0000000000336	*							20.00	24 OCT YY 001	979726 B	B24
*****4656	*** P-A PUR COMP	0000000000337	*							20.00	24 OCT YY 001	122562 B	B24
*****4761	*** MERCH RTN	0000000000338	*							20.00	24 OCT YY 001	181544 B	B24
*****5832	*** PURCHASE	0000000000339	*							20.00	24 OCT YY 001	240526 B	B24
*****5921	*** PURCHASE	0000000000340	*							20.00	24 OCT YY 001	999413 B	B24
*****5981	*** ADJ PURCHASE	0000000000341	*							20.00	24 OCT YY 001	155355 B	B24
*****6003	*** PURCHASE	0000000000342	*							20.00	24 OCT YY 001	388494 B	B24
*****6673	*** PURCHASE	0000000000343	*							20.00	24 OCT YY 054	00000000	B24

REPORT NO: B24POS12 RETAILER ACTIVITY REPORT 1										PAGE		2	
GRP: 0001 RGN: 0001 RETAILER INFO: RETAILER10										BCH: 001			
-----CARDHOLDER NUMBER-----										26 OCT YY			
--CARDHOLDER ADDRESS--										TERM: 011216			
*** MAIL/PHONE ORDER										TRAN DATE		TRAN DEST	
11835 CASS PLAZA #3										TRAN TIME		RESP CODE	
*****3365										AUTH CODE			
8108 HAUSER													
*****3716													
5601 OUTLOOK													
*****3728													
330 S 108TH AVE													
*****3740													
*****3771													
*****3829													
*****3851													
*****3922													
*****3948													
*****3976													
*****4217													
*****4223													
*****4256													

Part One Format Description

Most of the report fields are taken directly from corresponding fields in the PTLF records. Where this is true, the documentation parenthetically includes the name of the PTLF record field from which the information is taken.

Line 1 of the report header contains the report number, report name, the institution FIID, institution name (taken from the IDF), reporting date, and page number.

Line 2 of the report header contains the group number, region number, retailer ID, retailer name, terminal ID, and batch number identifying the transactions being listed.

CARDHOLDER NUMBER — (HEAD.CRD.CARD.CRD-NUM) The card number used in the reported transaction. All digits in this number, except the last four, are masked based on the application's default setting.

MBR — (HEAD.CRD.CARD.MBR-NUM) The member number of the card. When multiple cards are issued with the same card number, the member number distinguishes between the cards. This number is masked based on the application's default settings.

TRAN DESCRIPTION — A literal description of the type of transaction (e.g., PURCHASE, PRE AUTH PUR, MERCH RTN, P-A PUR COMP, ADJ PURCHASE, etc.).

DC — A code indicating whether or not the transaction was a draft capture transaction. Valid codes are:

* = Yes

b = No

TRAN AMOUNT — (AUTH.AMT-1) The requested transaction amount.

For an adjustment, this field contains the amount originally requested. For a preauthorization request, this field contains the amount to be held. For a preauthorization completion, this field contains the completed amount. For a purchase with cash back transaction, this field contains the total amount of the transaction (the actual cash back amount of the transaction is contained in the CASH OUT field).

CASH OUT — (AUTH.AMT-2) For a purchase with cash back transaction, this field contains the amount of cash back involved in the transaction. For an adjustment transaction, this field contains the new amount.

TRAN DATE — (AUTH.TRAN-DAT) The date (DD MMM YY) on which the transaction occurred.

RESP CODE — (AUTH.RESP-CDE) The response code given to the transaction to indicate whether it was approved or declined. For a list of response codes and their descriptions, refer to the ***BASE24-pos Transaction Processing Manual***.

TRAN DEST — (AUTH.DEST) A literal description of the transaction's authorizer. Transactions can be authorized by the following entities:

- The BASE24-pos system. In this case, the literal "B24" appears in this field.
- A back-end host connected to the BASE24-pos system. In this case, the literal "HOST" appears in this field.
- An interchange. In this case, the literal indicates the interchange (e.g., BASE, ACDS and AMEX).
- A retailer. In this case, the literal "RETL" appears in this field.

TERM OPR — (AUTH.CLERK-ID) An alphanumeric code identifying the terminal operator who entered the transaction.

SEQUENCE NUM — (AUTH.SEQ-NUM) The sequence number the terminal or Device Handler module assigned to the transaction.

TRAN TIME — (AUTH.TRAN-TIM) The time of day (hh:mm:ss) at which the transaction occurred.

AUTH CODE — (AUTH.APPRV-CDE) The approval code assigned to the transaction. When transactions are approved, they are assigned an approval code for reference.

<*RVRSAL*> — The names listed below are printed following the AUTH CODE field if any of the corresponding transactions appear on this report.

*RVRSAL = Reversal
*
X RVL = Exception Reversal
RV ADJ = Reversal Adjustment
X RVLA = Exception Reversal Adjustment
ADJUST = Adjustment
X ADJ = Exception Adjustment
F POST = Force Post
X FP = Exception Force Post
FP ADJ = Force Post Adjustment
X FPA = Exception Force Post Adjustment

CARDHOLDER ADDRESS — (AUTH.ADDR) The cardholder billing address received with the transaction when performing address verification.

ZIP CODE — (AUTH.ZIP-CDE) The cardholder billing ZIP code received with the transaction when performing address verification.

ADDR STAT — (AUTH.ADDR-VRFY-STAT) A code identifying the result of comparing address verification information received in the transaction and address verification information contained in the processor's database.

The following values can be set by BASE24-pos through its address verification processing. If a transaction is sent to an interchange for processing, the response message may contain other interchange-specific values (BASE24 does not change values received from the interchanges).

A = ADDRESS—Addresses matched, but ZIP codes did not match.
E* = ERROR—The transaction was not eligible for address verification or an editing error occurred while attempting to process the message.
N = NO—Addresses did not match and ZIP codes did not match.
R* = RETRY—Primary and secondary authorizers were unavailable or declined the transaction and address verification was not performed on the BASE24-pos system.
S* = SERVICE NOT SUPPORTED—The BASE24-pos system authorized the transaction, but did not have the add-on Address Verification module.
U* = UNAVAILABLE—Address information was not available to the processor performing address verification.

- W = WHOLE ZIP—Nine-digit ZIP codes matched, but addresses did not match.
- X = EXACT—Addresses and nine-digit ZIP codes matched.
- Y = YES—Addresses and five-digit ZIP codes matched.
- Z = ZIP—Five-digit ZIP codes matched, but addresses did not match.
- b* = Address verification information was not included in this transaction (b denotes a blank character).
- 0* = Address verification information was included in this transaction, but was not verified. A transaction to be verified by a host or interchange carries this code. A transaction to be verified on the BASE24-pos system, but declined before address verification could be performed also carries this code.
- * This value identifies a reason that a comparison was not made. Each interchange specifies the value it uses to identify this reason. BASE24 Interchange Interfaces substitute the interchange-specific value before sending the message from the BASE24-pos system to the interchange. Refer to section 1 of the ***BASE24-pos Address Verification Manual*** for a listing of interchange-specific values.

REPORT NO: B24POS12 RETAILER ACTIVITY REPORT				BK01 - FIRST NATIONAL BANK		APR 09 YY	PAGE
*** INSTITUTION TOTALS - ALL GROUPS:							3
---TRAN DESCRIPTION---				TRAN COUNT	-----TRAN AMOUNT-----	TRAN COUNT	-----TRAN AMOUNT-----
NORMAL PURCHASE	APPROVED	42,858		642,870.65	DECLINED	2,543	38,145.28
MAIL/PHONE ORDER	APPROVED	3,108		52,836.97	DECLINED	109	1,635.12
MERCHANIDISE RETURN	APPROVED	2,856		42,840.34	DECLINED	0	0.00
PURCHASE CASH BACK	APPROVED	7,122		391,710.17	DECLINED	1,623	89,265.61
CHECK VERIFICATION	APPROVED	2,526		0.00	DECLINED	236	0.00
ADJUST - PURCHASE	APPROVED	447		6,705.06	DECLINED	0	0.00
ADJUST - MERC RETURN	APPROVED	390		5,850.81	DECLINED	0	0.00
TOTAL APPROVED				59,307	1,142,814.00	DECLINED	4,511
TOTAL TRANSACTIONS				63,818	1,271,860.01		129,046.01
SALES CAPTURE TOTALS				59,307	1,142,814.00		
				*** End of report ***			

Part Two Format Description

These summary totals include approved and declined transaction counts and amounts by batch, terminal, retailer, region, group, and institution. The summaries are identified as follows:

- TERMINAL TOTALS FOR BATCH XXX
- TERMINAL TOTALS - ALL BATCHES
- RETAILER TOTALS - ALL TERMINALS
- REGION TOTALS - ALL RETAILERS
- GROUP TOTALS - ALL REGIONS
- INSTITUTION TOTALS - ALL GROUPS

TRAN DESCRIPTION — A literal description of the transaction type (e.g., NORMAL PURCHASE, PURCHASE COMPLETION, MERCHANDISE RETURN, etc.)

APPROVED — The following fields indicate the counts and amounts for approved transactions.

TRAN COUNT — The total number of approved transactions.

TRAN AMOUNT — The total amount, in fractional currency units, of approved transactions.

DECLINED — The following fields indicate the counts and amounts for declined transactions.

TRAN COUNT — The total number of declined transactions.

TRAN AMOUNT — The total amount, in fractional currency units, of declined transactions.

TOTAL APPROVED — The total count and amount, in fractional currency units, of approved transactions for this summary.

TOTAL DECLINED — The total count and amount, in fractional currency units, of declined transactions for this summary.

TOTAL TRANSACTIONS — The total count and amount, in fractional currency units, of transactions for this summary without regard for whether they were approved or declined.

SALES CAPTURE TOTALS — The total count and amount, in fractional currency units, of draft capture transactions included in this summary.

END MESSAGES — The following message appears at the end of the Daily Retailer Activity Report (B24POS12):

*** End of report ***

Daily Retailer Exception Detail Report (B24POS12-EXPT)

The Daily Retailer Exception Detail Report (B24POS12-EXPT) contains a listing and summary of each retailer's activity that resulted in exception processing that day. The report includes all transactions that have the following record types:

- 21 = Exception not posted to any BASE24 files
- 22 = Exception posted to some but not all applicable BASE24 files
- 23 = Exception not posted because of invalid data

The BASE24-pos product performs exception processing when an authorizer other than the BASE24 system has approved the transaction and the transaction contains data or is missing data that prevents the BASE24-pos system from posting the transaction correctly to BASE24 files. Transactions that may result in exception processing include force posts, reversals, and requests. This report does not include administrative or settlement records.

Retailers can use the Daily Retailer Exception Detail Report (B24POS12-EXPT) to research exception transactions by individual terminal. The report provides the retailer with a record of the total exception activity of all terminals. Institutions use the report to track their retailer exception activity by group, region, and retailer.

The B24POS12-EXPT is produced in the same format as Part One of the Daily Retailer Activity Report (B24POS12). It lists all exception transactions in chronological order.

This report is an optional report. The BASE24-pos system produces it based on the POS-EXPT-RPT-FLG param in the LCONF. See section 5 or the ***BASE24 Logical Network Configuration File Manual*** for more information about this param.

The BASE24-pos system generates one Daily Retailer Exception Detail Report (B24POS12-EXPT) for each institution and retailer in the logical network each day, if configured to do so in the LCONF. The BASE24-pos product obtains the information for this report from the POS Transaction Log File (PTLF). If no exception records exist in the PTLF for this retailer, the report contains the following comment:

*** NO EXCEPTION RECORDS EXIST FOR THIS RETAILER ***

The report is printed at the spooler location specified in the POS-RPT-2R param in the LCONF. If this param is not configured or contains an invalid value, the report program uses a default spooler location of \$S.#EXPT12.

Because this report is identical in format to Part One of the Daily Retailer Activity Report (B24POS12), a sample report is not provided in this manual. Refer to the sample of the B24POS12 report for a description of the Daily Retailer Exception Detail Report (B24POS12-EXPT).

Network Referral Statistical Report (B24POS30)

The Network Referral Statistical Report (B24POS30) provides a listing and summary of records in the POS Referral File (PRF). These records represent transactions which, for some reason, have been identified for referral by a BASE24-pos authorizing institution.

Note: The BASE24-pos product generates the Daily Disputed Transactions Report (B24POS05) only if the institution is using the BASE24-pos CRT Authorization add-on product.

Report data are grouped according to authorizing institution (authorizer). Transaction details are listed first, followed by totals for the authorizer. An additional set of totals is also provided at the network level.

The figures on the Network Referral Statistical Report (B24POS30) provide institutions with a summary of actions taken on their referral items. The BASE24-pos product computes percentages for referrals that were subsequently approved or denied and for referrals on which no subsequent action was taken. Institutions can use this report as an audit trail to review and analyze their referral activity for fine-tuning their authorization efforts. For example, if a certain type of referral is being routinely approved, the institution might want to reset their authorization parameters so that the transaction would be automatically approved rather than being referred.

The BASE24-pos system generates one B24POS30 report for the logical network each day. The BASE24-pos system uses information in the POS Referral File (PRF) to produce this report. The report is printed at the spooler location specified in the POS-RPT-30 assign in the Logical Network Configuration File (LCONF). If the assign does not exist or the specified spooler location in the assign is invalid, the report program uses a default spooler location of \$\$.#RPT30.

The masking of cardholder primary account numbers (PANs) and administrative PANs on this report depends on the values of the four REPORT DATA MASKING PARAMETER fields on Institution Definition File (IDF) screen 4.

A sample Network Referral Statistical Report (B24POS30) is shown on the following page, followed by descriptions of its fields.

Report No: B24POS30 Network Referral Statistical Report for Authorizer BK01 10 JAN YY Page 1

--CARDHOLDER SEQ NO	NUMBER-- OVERRIDE	--MEMBER NBR--	--TRAN DESCRIPTION-- TRAN DATE	TRAN TIME	---TRAN AMOUNT--- CRT OPER	RETAILER ID	RSP CODE	AUTH CODE
*****0221		***	NORMAL PURCHASE		1000.00	STORE01	050	571622
00007900001	DEC		20 JAN YY 12:57:15	99,99				
*****2253		***	NORMAL PURCHASE		600.00	STORE04	107	
00062400011	NOACT		20 JAN YY 10:32:57	79,99				
*****4270		***	NORMAL PURCHASE		1000.00	STORE01	000	657134
00100212543	APP		20 JAN YY 14:12:32	102,102				
*****7076		***	NORMAL PURCHASE		900.00	STORE03	000	456673
00166667891	APP		20 JAN YY 16:43:55	50,50				
*****8087		***	NORMAL PURCHASE		365.45	STORE05	000	476892
00166688911	APP		20 JAN YY 16:45:20	50,50				

Format Description

Most of the detail report fields are taken directly from corresponding fields in the POS Referral File (PRF). Where this is true, the documentation parenthetically includes the name of the PRF field from which the information is taken.

The report header contains the report number, report name, authorizer for which the transactions are being reported, reporting date, and the page number.

CARDHOLDER NUMBER — (PSTM.TRAN.TRACK2) The card number of the card used in the transaction. All digits in this number, except the last four, are masked based on the application's default setting. If no IDF record exists, the entire primary account number is masked.

MEMBER NBR — (PSTM.TRAN.MBR-NUM) The member number associated with the card used in the transaction. The member number distinguishes between cards when multiple cards are issued with the same card number. If an institution does not support member numbers, this field defaults to 000. This number is masked based on the application's default settings.

TRAN DESCRIPTION — (PSTM.TRAN.TRAN-CDE) A literal description of the transaction type (e.g., NORMAL PURCHASE, PRE-AUTH COMPLETION, MERCHANDISE RETURN, ADJUST - PURCHASE, etc.).

TRAN AMOUNT — (PSTM.TRAN.AMT-1) The requested transaction amount, in fractional currency units.

RETAILER ID — (RFRL-KEY.RETAILER-ID) The ID of the retailer originating the transaction.

RSP CODE — (PSTM.TRAN.RESP-CDE) The response code given to the transaction to indicate whether it was approved or declined. For a list of response codes and their descriptions, refer to the ***BASE24-pos Transaction Processing Manual***.

AUTH CODE — (PSTM.TRAN.APPRV-CDE) The approval code assigned to the transaction. When transactions are approved, they are assigned an approval code for reference.

SEQ NO — (PSTM.SEQ-NUM) The sequence number the terminal or BASE24-pos system assigned to the transaction.

OVERRIDE — A literal description of the action taken. Values are:

APP = Indicates the referral was approved. This literal is used for a response code of 000.

DEC = Indicates the referral was declined. This literal is used for a response code of 050.

NOACT = Indicates no action was taken. This literal is used for response codes other than 000 or 050.

TRAN DATE — (PSTM.TRAN-DAT) The date (DD MMM YY) on which the transaction occurred.

TRAN TIME — (PSTM.TRAN-TIM) The time (hh:mm:ss) at which the transaction occurred.

CRT OPER — (CRT-OPR.GRP and CRT-OPR.USER-ID) The group and user number (separated by a comma) of the CRT operator who handled the referral. The group and user number in this field reflect the operator's group and user number in the Security File (SEC).

Report No: B24POS30 Network Referral Statistical Report for Authorizer BK01										20 JAN YY	Page	2
-TOTALS BY TRAN TYPE-												
--TRAN DESCRIPTION---												
			APPROVED			DECLINED			NO ACTION			TOTAL
			COUNT	PERCENTAGE		COUNT	PERCENTAGE		COUNT	PERCENTAGE		COUNT
			-----AMOUNT/COUNT/PERCENTAGE-----									
NORMAL PURCHASE			3	2265.45	60%	1	1000.00	20%	1	600.00	20%	5
												3865.45
Total Transactions			3	2265.45	60%	1	1000.00	20%	1	600.00	20%	5
												3865.45

Report No: B24POS30 Network Referral Statistical Report for All Authorizers

-TOTALS BY TRAN TYPE-		APPROVED		DECLINED		NO ACTION		TOTAL	
--TRAN DESCRIPTION---	COUNT	PERCENTAGE	COUNT	PERCENTAGE	COUNT	PERCENTAGE	COUNT	PERCENTAGE	COUNT
NORMAL PURCHASE	3	2265.45	1	1000.00	1	600.00	5	3865.45	
CASH ADVANCE	3	2265.45	1	1000.00	1	600.00	5	3865.45	
TOTAL TRANSACTIONS	6	4530.90	2	2000.00	2	1200.00	10	7730.90	

*** End of report ***
*** End of run ***

Format Description

The report header contains the report number, report name, authorizer for which the totals are being reported, reporting date, and page number.

TOTALS BY TRAN TYPE

TRAN DESCRIPTION — A literal description of the transaction type (e.g., NORMAL PURCHASE, PRE-AUTH COMPLETION, MERCHANDISE RETURN, ADJUST-PURCHASE, etc.).

AMOUNT/COUNT/PERCENTAGE — The total amount, in fractional currency units, of referred transactions (TOTAL), broken down according to the amount of the transactions that were approved and declined and for which no action was taken.

The total number of referred transactions (TOTAL), broken down according to the number of transactions that were approved and declined and for which no action was taken.

The percentage of referred transactions that were approved and declined and for which no action was taken. This percentage is based on the value in the COUNTS field.

TOTAL TRANSACTIONS — Cumulative totals for all transaction types.

END MESSAGES — The following messages appear at the end of the Network Referral Statistical Report (B24POS30):

```
*** End of report ***  
***   End of run   ***
```

Retailer Reconciliation Report (B24POS35)

The Retailer Reconciliation Report (B24POS35) provides reconciliation figures for the retailer's terminals. To reconcile terminals, the BASE24-pos system compares transaction figures from several system sources to verify that terminal transactions are in balance in the system. Institutions can then look at this report to identify out-of-balance conditions and work to research and resolve them.

The B24POS35 report reconciles transaction activity at the terminal batch level. The report also provides additional totals by shift and day when the terminal supports these balancing levels. The report shows cumulative totals by retailer, region, group, institution, and network. The report is divided into the following totals sections:

****BATCH TOTALS****
****SHIFT TOTALS****
****END OF DAY TOTALS****
****NETWORK DAY TOTALS****
****RETAILER TOTALS - ALL TERMINALS****
****REGIONS TOTALS - ALL RETAILERS****
****GROUP TOTALS - ALL REGIONS****
****INSTITUTION TOTALS - ALL GROUPS****
****NETWORK TOTALS - ALL INSTITUTIONS****

Only draft capture transactions appear on the Retailer Reconciliation Report (B24POS35). Draft capture transactions are the only transactions for which the retailer can be credited without the deposit of supporting paper.

The BASE24-pos product generates one B24POS35 report each day for each retailer, terminal, region, group, institution, and network. The BASE24-pos system uses information in the POS Transaction Log File (PTLF) to produce this report. The B24POS35 report is printed at the spooler location specified in the POS-RPT-35 param in the Logical Network Configuration File (LCONF). If an invalid spooler location is indicated in the LCONF, the report program uses a default spooler location of \$S.#RPT35.

For additional information on terminal reconciliation and the Retailer Reconciliation Report (B24POS35), refer to section 2.

A sample Retailer Reconciliation Report (B24POS35) is shown on the following page, followed by descriptions of its fields.

REPORT NO: B24POS35 RETAILER RECONCILIATION REPORT - NETWORK: LLLL									
INSTITUTION: BK01		GROUP: 001	REGION: 0001	RETAILER ID: RETAILER03	TERMINAL ID: 330000000	104	BATCH: 001	JAN 20	PAGE 2
BATCH TOTALS									
-----DEPOSIT TOTALS-----PREVIOUS/CARRYOVER-----									
--- CARD TYPE --- STATUS --- NETWORK (D) --- NETWORK (T) --- TERMINAL (T) --- NETWORK (T) ---									
(VISA CREDIT)									
	DB CNT:	15	450.00	15	436.00				
	DB AMT:								
	CR CNT:	2	50.00	2	36.00				
	CR AMT:								
	AD CNT:	0	0.00	0	0.00				
	AJ AMT:								
	TOTAL CNT:	17	400.00	17	400.00				
	TOTAL AMT:								
(MASTERCARD CREDIT)									
	DB CNT:	18	700.00	43	1650.00				
	DB AMT:								
	CR CNT:	3	75.00	5	125.00				
	CR AMT:								
	AD CNT:	0	0.00	0	0.00				
	AJ AMT:								
	TOTAL CNT:	21	625.00	48	1525.00				
	TOTAL AMT:								
(* USER DEFINED *)									
	DB CNT:	10	500.00	10	500.00				
	DB AMT:								
	CR CNT:	0	0.00	0	0.00				
	CR AMT:								
	AD CNT:	0	0.00	0	0.00				
	AJ AMT:								
	TOTAL CNT:	10	500.00	10	500.00				
	TOTAL AMT:								
*BATCH 001 TOTAL									
	DB CNT:	43	1650.00	43	1650.00				
	DB AMT:								
	CR CNT:	5	125.00	5	125.00				
	CR AMT:								
	AD CNT:	0	0.00	0	0.00				
	AJ AMT:								
	TOTAL CNT:	48	1525.00	48	1525.00				
	TOTAL AMT:								
NET DEPOSIT									
	TOTAL CNT:	48	1525.00	48	1525.00				
	TOTAL AMT:								

REPORT NO: B24POS35 RETAILER RECONCILIATION REPORT - NETWORK: LLLL										JAN 20 YY		PAGE	3
INSTITUTION: BK01				GROUP: 001	REGION: 0001	RETAILER ID: RETAILER03				TERMINAL ID: 330000000 104 BATCH: 001			
SHIFT TOTALS													
-----DEPOSIT TOTALS-----PREVIOUS/CARRYOVER-----													
-----CARD TYPE-----		-----STATUS-----		NETWORK (D)		NETWORK (T)		TERMINAL (T)		NETWORK (T) --			
*SHIFT 001 TOTAL		DB CNT:	63	2500.00	63	2500.00	63	2500.00					
		DB AMT:	6	275.00	6	275.00	6	275.00					
		CR CNT:	0	0.00	0	0.00	0	0.00					
		CR AMT:	69	2225.00	69	2225.00	69	2225.00					
		AD CNT:											
		AJ AMT:											
NET DEPOSIT		TOTAL CNT:											
		TOTAL AMT:											

REPORT NO: B24POS35 RETAILER RECONCILIATION REPORT - NETWORK: LLLL				JAN 20 YY	PAGE	4
INSTITUTION: BK01 GROUP: 001		REGION: 0001	RETAILER ID: RETAILER03	TERMINAL ID: 330000000 104		
END OF DAY TOTALS						
-----DEPOSIT TOTALS-----PREVIOUS/CARRYOVER-----						
--- NETWORK (D) --- NETWORK (T) --- TERMINAL (T) --- NETWORK (T) ---						
DB CNT:		63	2500.00	63	2500.00	
DB AMT:		6	275.00	6	275.00	
CR CNT:		0	0.00	0	0.00	
CR AMT:		69	2225.00	69	2225.00	
AD CNT:						
AJ AMT:						
TOTAL CNT:						
TOTAL AMT:						
End of report						
End of run						

REPORT NO: B24POS35 RETAILER RECONCILIATION REPORT - NETWORK: LLLL									
INSTITUTION: BK01		GROUP: 001	REGION: 0001	RETAILER ID: RETAILER03	TERMINAL ID: 33000000	104	BATCH: 001	JAN 20 YY	PAGE 1
BATCH TOTALS									
-----DEPOSIT TOTALS-----PREVIOUS/CARRYOVER-----									
--- NETWORK (D) --- NETWORK (T) --- TERMINAL (T) --- NETWORK (T) ---									
---CARD TYPE-----STATUS-----									
(VISA CREDIT)									
	DB CNT:	15	450.00	15	436.00				
	DB AMT:								
	CR CNT:	2	50.00	2	36.00				
	CR AMT:								
	AD CNT:	0	0.00	0	0.00				
	AJ AMT:								
	TOTAL CNT:	17	400.00	17	400.00				
	TOTAL AMT:								
(MASTERCARD CREDIT)									
	DB CNT:	18	700.00	43	1650.00				
	DB AMT:								
	CR CNT:	3	75.00	5	125.00				
	CR AMT:								
	AD CNT:	0	0.00	0	0.00				
	AJ AMT:								
	TOTAL CNT:	21	625.00	48	1525.00				
	TOTAL AMT:								
(* USER DEFINED *)									
	DB CNT:	10	500.00	10	500.00				
	DB AMT:								
	CR CNT:	0	0.00	0	0.00				
	CR AMT:								
	AD CNT:	0	0.00	0	0.00				
	AJ AMT:								
	TOTAL CNT:	10	500.00	10	500.00				
	TOTAL AMT:								
*BATCH 001 TOTAL									
	DB CNT:	43	1650.00	43	1650.00				
	DB AMT:								
	CR CNT:	5	125.00	5	125.00				
	CR AMT:								
	AD CNT:	0	0.00	0	0.00				
	AJ AMT:								
	TOTAL CNT:	48	1525.00	48	1525.00				
	TOTAL AMT:								
NET DEPOSIT									
	TOTAL CNT:	48	1525.00	48	1525.00				
	TOTAL AMT:								

Format Description

Line 1 of the report header contains the report number, report name, logical network name, reporting date, and page number.

Line 2 of the report header contains the institution FIID, group, region, retailer ID, terminal ID, and batch number.

CARD TYPE — The card types the retailer accepts (e.g., MasterCard, Visa, Diners Club, etc.). If the retailer accepts user-defined cards, these are noted as * USER DEFINED *.

STATUS — A status notation that flags certain conditions. The following status notations can appear on this report.

OUT OF BALANCE—Indicates the NETWORK(D) totals do not agree with the NETWORK(T) totals or that the NETWORK(T) totals do not agree with the TERMINAL(T) totals. In both cases, the report program compares the net figures to make this determination.

INCOMPLETE BATCH—Indicates that the system cut over the terminal in the middle of a batch and that a portion of the batch must be settled tomorrow.

DEPOSIT TOTALS — The following fields contain deposit totals information used for retailer reconciliation. Transactions are divided by card type and totaled by batch (and shift and day, if the terminal supports these balancing levels).

NETWORK (D) — The report program calculates the COUNT and AMOUNT totals in this column from individual transaction records in the PTLF.

DB CNT — The total number of NETWORK (D) debit transactions occurring for the card type.

DB AMT — The total amount, in fractional currency units, of NETWORK (D) debit transactions occurring for the card type.

CR CNT — The total number of NETWORK (D) credit transactions occurring for the card type.

CR AMT — The total amount, in fractional currency units, of NETWORK (D) credit transactions occurring for the card type.

AD CNT — The total number of NETWORK (D) adjustment transactions occurring for the card type.

AJ AMT — The total amount, in fractional currency units, of NETWORK (D) adjustment transactions occurring for the card type.

TOTAL CNT — The total number of NETWORK (D) debit, credit, and adjustment transactions occurring for the card type.

TOTAL AMT — The total amount, in fractional currency units, of NETWORK (D) debit, credit, and adjustment transactions occurring for the card type.

NETWORK (T) — The COUNT and AMOUNT totals in this column are taken directly from terminal total records in the PTLF. These records reflect transaction totals in the BASE24-pos Terminal Data files at the times the records were created.

CARD TYPE totals = taken from service total records
 BATCH totals = taken from the batch totals records
 SHIFT totals = taken from the shift totals records
 DAILY totals = taken from the daily totals records
 NETWORK totals = taken from the network totals records

DB CNT — The total number of NETWORK (T) debit transactions occurring for the card type.

DB AMT — The total amount, in fractional currency units, of NETWORK (T) debit transactions occurring for the card type.

CR CNT — The total number of NETWORK (T) credit transactions occurring for the card type.

CR AMT — The total amount, in fractional currency units, of NETWORK (T) credit transactions occurring for the card type.

AD CNT — The total number of NETWORK (T) adjustment transactions occurring for the card type.

AJ AMT — The total amount, in fractional currency units, of NETWORK (T) adjustment transactions occurring for the card type.

TOTAL CNT — The total number of NETWORK (T) debit, credit, and adjustment transactions occurring for the card type.

TOTAL AMT — The total amount, in fractional currency units, of NETWORK (T) debit, credit, and adjustment transactions occurring for the card type.

TERMINAL (T) — The COUNT and AMOUNT totals in this column reflect what the actual terminal indicates the transaction totals should be.

If a terminal can balance with the system, these totals represent the terminal's version of the totals. If a terminal cannot balance with the system, these totals are duplicated from the NETWORK (T) column. However, terminals do not send totals by card type. Therefore, no card type totals appear in this column.

DB CNT — The total number of TERMINAL (T) debit transactions occurring for the batch.

DB AMT — The total amount, in fractional currency units, of TERMINAL (T) debit transactions occurring for the batch.

CR CNT — The total number of TERMINAL (T) credit transactions occurring for the batch.

CR AMT — The total amount, in fractional currency units, of TERMINAL (T) credit transactions occurring for the batch.

AD CNT — The total number of TERMINAL (T) adjustment transactions occurring for the batch.

AJ AMT — The total amount, in fractional currency units, of TERMINAL (T) adjustment transactions occurring for the batch.

TOTAL CNT — The total number of TERMINAL (T) debit, credit, and adjustment transactions occurring for the batch.

TOTAL AMT — The total amount, in fractional currency units, of TERMINAL (T) debit, credit, and adjustment transactions occurring for the batch.

PREVIOUS/CARRYOVER NETWORK (T) — This column is only used when a portion of the batch was settled on the previous day. This situation occurs when the system forces the terminal to cut over in the middle of a batch. On the first day, the batch is shown as INCOMPLETE; on the second day, the amounts settled on the first day are listed in this column.

NET DEPOSIT — The computed net figure for each of the columns. Net amounts are computed as follows: debits, minus credits, minus adjustments. The term deposit is used because all transactions on the report are draft capture transactions, which indicates that the retailer is to be credited for these transactions without the subsequent deposit of tickets.

TOTAL CNT — The combined number of transactions contained in all of the columns.

TOTAL AMT — The combined net amount, in fractional currency units, contained in all of the columns.

END MESSAGES — The following messages appear at the end of the Retailer Reconciliation Report (B24POS35):

```
*** End of report ***
***   End of run   ***
```

Periodic Institution Summary Report (B24POS21)

The Periodic Institution Summary Report (B24POS21) allows the institution to review transaction totals from the Cardholder Activity Reports (B24POS02) for variable periods of time.

Each time the BASE24-pos system generates the B24POS02 report, the system creates a record containing the approved and declined transaction totals from that report and places it in the POS Institution Reporting File (PIRF). The BASE24-pos product then uses the records in the PIRF to produce the Periodic Institution Summary Report (B24POS21).

This report is optional. An operator must request the B24POS21 report from the POS Report Generator (PRG) screen. On the PRG screen, the operator specifies the institution and number of days for which totals are to be reported. The report program then reads the records in the PIRF for the specified institution and date range and produces the B24POS21 report totals.

The B24POS21 report provides a tool that institutions can use to analyze cardholder activity. With this report, institutions can summarize and compare cardholder activity for user-specified lengths of time (e.g., holidays, summer months, promotional periods, etc.).

When an operator requests the B24POS21 report, the BASE24-pos product generates one set of totals for the institution the operator specified on the PRG screen. The BASE24-pos product uses information in the POS Institution Reporting File (PIRF) to produce this report. The report is printed at the spooler location specified in the POS-RPT-21 assign in the Logical Network Configuration File (LCONF). If the specified spooler location is invalid, the report program uses a default spooler location of \$\$.#RPT21.

A sample Periodic Institution Summary Report (B24POS21) is shown on the following page, followed by descriptions of its fields.

REPORT NO: B24POS21

PERIODIC INSTITUTION SUMMARY REPORT

BK01 - FIRST NATIONAL BANK

JAN 20 YY PAGE 1

SELECTION CRITERIA: 28 DEC YY - 20 JAN YY PRINT LOCATION: #RPT21

*** INSTITUTION TOTALS:

--TRAN DESCRIPTION--	TRAN COUNT	--TRAN AMOUNT--	TRAN COUNT	--TRAN AMOUNT--
NORMAL PURCHASE	1,247,328	18,709,920.46	20,147	302,205.23
PRE-AUTH PURCHASE	20,178	302,670.13	1,136	17,040.54
PRE-AUTH COMPLETION	19,042	285,630.94	0	0.00
MAIL/PHONE ORDER	78,472	1,647,912.15	1,026	19,494.85
CASH ADVANCE	1,897	94,850.00	97	4,850.00
MERCHANDISE RETURN	647	9,705.84	0	0.00
CARD VERIFICATION	96,473	0.00	840	0.00
PURCHASE - CASH BACK	12,316	197,056.23	821	20,525.78
ADJUST - PURCHASE	1,465	21,975.16	0	0.00
ADJUST - MERC RETURN	2,062	30,930.42	0	0.00
ADJUST - CASH ADVANCE	395	19,750.00	0	0.00
TOTAL APPROVED	1,480,275	21,320,401.33	24,067	360,116.40
TOTAL TRANSACTIONS	1,504,342	21,684,517.73		
CAPTURE TOTALS	1,480,275	21,320,401.33		

*** End of report ***
 *** End of run ***

Format Description

Line 1 of the report header contains the report number, report name, institution FIID, institution name (taken from the IDF), reporting date, and page number.

Line 2 of the report header contains the report beginning and ending dates and the location at which this report was printed.

TRAN DESCRIPTION — A literal description for each transaction type (e.g., NORMAL PURCHASE, PRE-AUTH PURCHASE, PRE-AUTH COMPLETION, MERCHANDISE RETURN, etc.).

APPROVED — The following fields contain the counts and amounts for approved transactions. Approved transactions have response codes of 000 through 009. For a list of response codes and their descriptions, refer to the *BASE24-pos Transaction Processing Manual*.

TRAN COUNT — The total number of approved transactions.

TRAN AMOUNT — The total amount, in fractional currency units, of approved transactions.

TOTAL APPROVED — The total count and amount, in fractional currency units, of approved transactions for this summary.

DECLINED — The following fields contain the counts and amounts for declined transactions. Declined transactions have response codes of 050 or greater. For a list of response codes and their descriptions, refer to the *BASE24-pos Transaction Processing Manual*.

TRAN COUNT — The total number of declined transactions.

TRAN AMOUNT — The total amount, in fractional currency units, of declined transactions.

TOTAL DECLINED — The total count and amount, in fractional currency units, of declined transactions for this summary.

TOTAL TRANSACTIONS — The total count and amount, in fractional currency units, of transactions for this summary without regard for whether they were approved or declined.

CAPTURED TOTALS — The total count and amount, in fractional currency units, of draft capture transactions included in this summary.

END MESSAGES — The following messages appear at the end of the Periodic Institution Summary Report (B24POS21):

```
*** End of report ***  
***   End of run   ***
```

Periodic Retailer Summary Report (B24POS22)

The Periodic Retailer Summary Report (B24POS22) enables the institution and its retailers to review transaction totals from the Retailer Activity Report (B24POS12) for variable periods of time. Each time the BASE24-pos system generates the B24POS12 report, it creates a record containing the approved and declined transaction totals from the B24POS12 report and places the record in the POS Retailer Reporting File (PRRF). The BASE24-pos product then uses the records in the PRRF to produce the Periodic Retailer Summary Report (B24POS22).

This report is optional. An operator must request the B24POS22 report from the POS Report Generator (PRG) screen. On the PRG screen, the operator specifies the institution and number of days for which totals are to be reported. The report program then reads the records in the PRRF for the specified institution and date range specified and produces the B24POS22 report totals.

The B24POS22 report provides a tool that institutions and retailers can use to analyze retailer activity. With this report, institutions and retailers can summarize and compare retailer activity for user-specified lengths of time (e.g., holidays, summer months, promotional periods, etc.).

The report provides totals at the institution, group, region, retailer, and terminal level as specified by the operator.

When an operator requests the B24POS22 report, the BASE24-pos system generates one set of totals for the operator-specified institution, group, region, retailer, or terminal. The BASE24-pos product uses information in the POS Retailer Reporting File (PRRF) to produce this report. The report is printed at the spooler location specified in the POS-RPT-22 assign in the Logical Network Configuration File (LCONF). If the assign is not present or the specified spooler location is invalid, the report program uses a default spooler location of \$\$.#RPT22.

A sample Periodic Retailer Summary Report (B24POS22) is shown on the following page, followed by descriptions of its fields.

REPORT NO: B24POS22	PERIODIC RETAILER SUMMARY REPORT	BK01 - FIRST NATIONAL BANK	JAN 20 YY	PAGE 1
SELECTION CRITERIA: 28 DEC YY - 20 JAN YY	GRP: 0008 RGN: 002	RETAILER: MERCHANT01	LOC: #RPT22	
--TRAN DESCRIPTION--	TRAN COUNT	---TRAN AMOUNT---	TRAN COUNT	---TRAN AMOUNT---
NORMAL PURCHASE	2,463	36,945.18	263	3,945.21
MERCHANDISE RETURN	92	1,380.42	0	0.00
ADJUST - PURCHASE	50	750.13	0	0.00
ADJUST - MERC RETURN	6	90.43	0	0.00
TOTAL APPROVED	2,611	39,166.18	263	3,945.21
TOTAL TRANSACTIONS	2,874	43,111.39		
CAPTURE TOTALS	2,611	39,166.18		
*** End of report ***				
*** End of run ***				

Format Description

Line 1 of the report header contains the report number, report name, institution FIID, institution name (taken from the IDF), reporting date, and page number.

Line 2 of the report header contains the production criteria for the report, including the report beginning and ending dates; the institution, group, region, retailer, or terminal for which totals are being printed; and the spooler location at which the report was printed.

TRAN DESCRIPTION — A literal description for each transaction type (e.g., NORMAL PURCHASE, MERCHANDISE RETURN, ADJUST-PURCHASE, PRE-AUTH COMPLETION, etc.).

APPROVED — The following fields contain the counts and amounts for approved transactions. Approved transactions have response codes of 000 through 009. For a list of response codes and their descriptions, refer to the ***BASE24-pos Transaction Processing Manual***.

TRAN COUNT — The total number of approved transactions.

TRAN AMOUNT — The total amount, in fractional currency units, of approved transactions.

TOTAL APPROVED — The total count and amount, in fractional currency units, of approved transactions for this summary.

DECLINED — The following fields contain the counts and amounts for declined transactions. Declined transactions have response codes of 050 or greater. For a list of response codes and their descriptions, refer to the ***BASE24-pos Transaction Processing Manual***.

TRAN COUNT — The total number of declined transactions.

TRAN AMOUNT — The total amount, in fractional currency units, of declined transactions.

TOTAL DECLINED — The total count and amount, in fractional currency units, of declined transactions for this summary.

TOTAL TRANSACTIONS — The total count and amount, in fractional currency units, of transactions for this summary without regard for whether they were approved or declined.

CAPTURE TOTALS — The total count and amount of draft capture transactions included in this summary.

END MESSAGES — The following messages appear at the end of the Periodic Retailer Summary Report (B24POS22):

```
*** End of report ***  
***   End of run   ***
```

Clearings Statement Report (SETL01)

The Clearings Statement Report (SETL01) reflects the debit, credit, and net position of the logical network to the interchange. However, since it is from the interchange's point-of-view, the debit and credit amounts are inverted.

This report contains the totals of all transactions logged to the Interchange Log File (ILF) since the interchange cutover on the previous day. The report also shows the net position of each logical network to the interchange.

The central institution running the logical network can use this report to settle with participating interchanges.

The Interchange Interface process generates one SETL01 report each day for each interchange in the BASE24-pos network. The BASE24-pos product uses the information in the ILF to produce this report.

A sample Clearings Statement Report (SETL01) is shown on the following pages, followed by descriptions of its fields.

SETL01 - FIG		CLEARINGS STATEMENT REPORT FOR FIG		DATE YY/MM/DD
				PAGE 1
		TRAN DATE	DEBIT	CREDIT
		YY/MM/DD		NET
*****PRO1 TODAY*****				
MINUS INTERCHANGE NOT-TODAY				
INTERCHANGE		YY/MM/DD	1000.00	2000.00-
SETTLEMENT SUBTOTAL		YY/MM/DD	1000.00	500.00
		YY/MM/DD	9000.00	2500.00-
*****ADD INTERCHANGE TODAY*****				
INTERCHANGE		YY/MM/DD	2000.00	1000.00-
SETTLEMENT SUBTOTAL		YY/MM/DD	11000.00	3500.00-
		**** End of report ****		

SETL01 - FIG		CLEARINGS STATEMENT REPORT FOR FIG		DATE YY/MM/DD
				PAGE 1
		TRAN DATE	DEBIT	CREDIT
*****SPAN TODAY*****		YY/MM/DD		NET
		YY/MM/DD	1500.00	10000.00-
MINUS INTERCHANGE NOT-TODAY		YY/MM/DD	12000.00	10000.00-
INTERCHANGE		YY/MM/DD	3000.00	0.00
SETTLEMENT SUBTOTAL				
*****ADD INTERCHANGE TODAY*****		YY/MM/DD	2000.00	1000.00-
INTERCHANGE		YY/MM/DD	5000.00	1000.00-
SETTLEMENT SUBTOTAL				
		***** End of report *****		

Format Description

The report header contains the report number, report name, Interchange name, reporting date (YY/MM/DD), and page number.

******* PROx TODAY ******* — The settlement position of the logical network to the interchange for the day this report is produced.

TRAN DATE — The Interchange Log File (ILF) date (YY/MM/DD) to which these transactions belong.

DEBIT — The amount, in fractional currency units, that the logical network owes the interchange, according to the interchange. Interchange debits include:

- Merchandise returns by interchange cardholders at terminals owned by a BASE24 institution.
- Withdrawals, cash advances, and purchases by cardholders belonging to a BASE24 institution at interchange terminals.

CREDIT — The amount, in fractional currency units, that the interchange owes the logical network, according to the interchange. Interchange credits include:

- Merchandise returns by cardholders belonging to a BASE24 institution at interchange terminals.
- Withdrawals, cash advances, and purchases by interchange cardholders at terminals owned by a BASE24 institution.

NET — The net position of the logical network to the interchange, according to the interchange (credits minus debits). A minus sign (-) indicates a debit position or the logical network owes the interchange.

******* MINUS INTERCHANGE NOT-TODAY ******* — The following fields contain the debits, credits, and net amounts that are not considered part of the interchange business day, but are considered part of the BASE24-pos business day. This situation occurs when the interchange has a different cutover time than the BASE24-pos system, which allows some transactions to be posted on different days.

INTERCHANGE — The total amount, in fractional currency units, of debits, credits, and net amounts not considered part of the interchange business day.

SETTLEMENT SUBTOTAL — The debit, credit, and net figures not considered part of the interchange business day are subtracted from today's business. The result is a settlement subtotal for BASE24-pos activity that the interchange considers to be today's business.

******* ADD INTERCHANGE TODAY ******* — The following fields contain the debits, credits, and net amounts considered part of the interchange business day but not part of the BASE24 business day. This situation occurs when the interchange has a different cutover time than the BASE24-pos system, which allows some transactions to be posted on different days.

INTERCHANGE — The total amount, in fractional currency units, of debits, credits, and net amounts considered part of the interchange business day.

SETTLEMENT SUBTOTAL — The debit, credit, and net figures considered part of the interchange business day are added to today's business. The result is a settlement subtotal for BASE24-pos activity that the interchange considers to be today's business.

******* SPAN TODAY ******* — This section summarizes the net position of the logical network to the interchange. The fields on this portion of the report are similar to those described above.

END MESSAGES — The following message appears at the end of the Clearings Statement Report (SETL01):

*** End of report ***

Transaction Detail Report (STAT09)

The Transaction Detail Report (STAT09) provides a complete listing of all interchange activity. The BASE24-pos product produces this report using information in the current day's Interchange Log File (ILF). The Interchange Interface process generates this report each day after the interchange has cut over to the new reporting day.

The exact name of this report depends on the name of the interchange. The format of this report also varies depending on the interchange.

Some interchanges do not perform settlement reporting on weekends and holidays. Under these circumstances, the reports for those days do not contain detail records. Instead, the reports inform the institution that no transactions occurred on those dates. The detailed report received after the weekend or holiday contains all transaction activity on one report.

A sample Transaction Detail Report (STAT09) is shown on the following page, followed by descriptions of its fields.

4-78 Nov-2007 R6.0v8 PS-AE000-06
ACI Worldwide, Inc.

Format Description

The report header contains the report number, report name, interchange name, reporting date (YY/MM/DD), and page number.

CARDHOLDER NUMBER — The number identifying the cardholder who initiated the transaction.

TRAN DESC — A brief description of the transaction (e.g., cash advance, purchase, etc.).

PST DAT — The BASE24 posting date.

B24 DAT-TIM — The date (YYMMDD) and time (hhmmss) this transaction occurred. This value represents the terminal date and time, if available.

B24 RESP — The response code given to the original transaction to indicate whether it was approved or declined. For a list of response codes and their descriptions, refer to the *BASE24-pos Transaction Processing Manual*.

CRD LN — The logical network to which this cardholder belongs.

ATM FROM ACCT #/POS ACCT # — The account accessed by this transaction. This field applies to BASE24-pos transactions and to the *from* account for BASE24-atm transactions.

TRAN/BAL/ORIG AMOUNT — If this is a POS transaction, this field contains the original amount. If this is an ATM transaction, this field contains the transaction amount. If this is a balance inquiry, this field contains the balance amount.

TRM LN — The logical network to which this terminal belongs.

TRM ID — The terminal identification number of the terminal that originated this transaction.

SEQUENCE # — The sequence number the terminal assigned to this transaction.

TRACE # — An additional identification number the interchange assigned to the transaction.

SWI DAT-TIM — The date (YYMMDD) and time (hhmmss) this transaction occurred, according to the interchange's external message. This value represents the date and time the interchange processed the transaction, if available. This field may contain the same value as the B24 DAT-TIM field, depending on the interchange.

SWI RESP — The response code, assigned by the interchange, that indicates the action taken for the transaction.

RV CDE — The identifier that indicates the reversal code, if the transaction was reversed.

ATM TO ACCOUNT #/POS APPR CODE — If this is a POS transaction, this field represents the approval code assigned to this transaction. If this is an ATM transaction, this field represents the *to* account.

COMPLETED AMOUNT — The total amount, in fractional currency units, of the completed transaction.

ATM TRM OWNER/POS RETLR — If this is a POS transaction, the value in this field indicates the retailer who owns this terminal. If this is an ATM transaction, the value in this field indicates the terminal-owning institution.

TRM NAME/LOCATION — The terminal name, as assigned by the terminal-owning institution, and the terminal location.

TRM CITY — The city in which the terminal is located.

TRM ST — The state in which the terminal is located.

PRD — The product identifier indicating the type of product that initiated the transaction (e.g., POS, ATM, etc.).

B24 TYP — The message type the BASE24 system assigned to the transaction.

SWI TYP — The message type the interchange assigned to the transaction.

POS TRANSACTION SUMMARY — A summary page, by transaction type, of all approved and denied POS transactions (e.g., purchases, cash advances, withdrawals, etc.).

END MESSAGES — The following message appears at the end of the Transaction Detail Report (STAT09):

*** End of report ***

ACI Worldwide, Inc.

5: Configuring Cutover and Reporting

This section provides an overview of the configuration tasks that a user must perform to establish BASE24-pos cutover and reporting. These tasks involve setting assigns and params in the Logical Network Configuration File (LCONF) and setting fields in POS, Base, and interchange files.

Refer to the *BASE24 Logical Network Configuration File Manual* for information on accessing and configuring the LCONF and for detailed descriptions and valid values for each assign and param.

Refer to the *BASE24-pos Files Maintenance Manual* for more information on BASE24-pos Terminal Data files and POS Retailer Definition File (PRDF) fields and field settings. Refer to the *BASE24 Base Files Maintenance Manual* for more information on the Institution Definition File (IDF) fields and field settings and for more information on the Interchange Configuration File (ICF) and Enhanced Interchange Configuration File (ICFE) fields and field settings.

Configuration Options

The Main and Secondary Settlement processes can be configured several ways within the BASE24-pos system, depending on the requirements of the logical network and the functions of the institutions within the logical network. Each logical network must have one Main Settlement process; the logical network can have up to 16 Secondary Settlement processes.

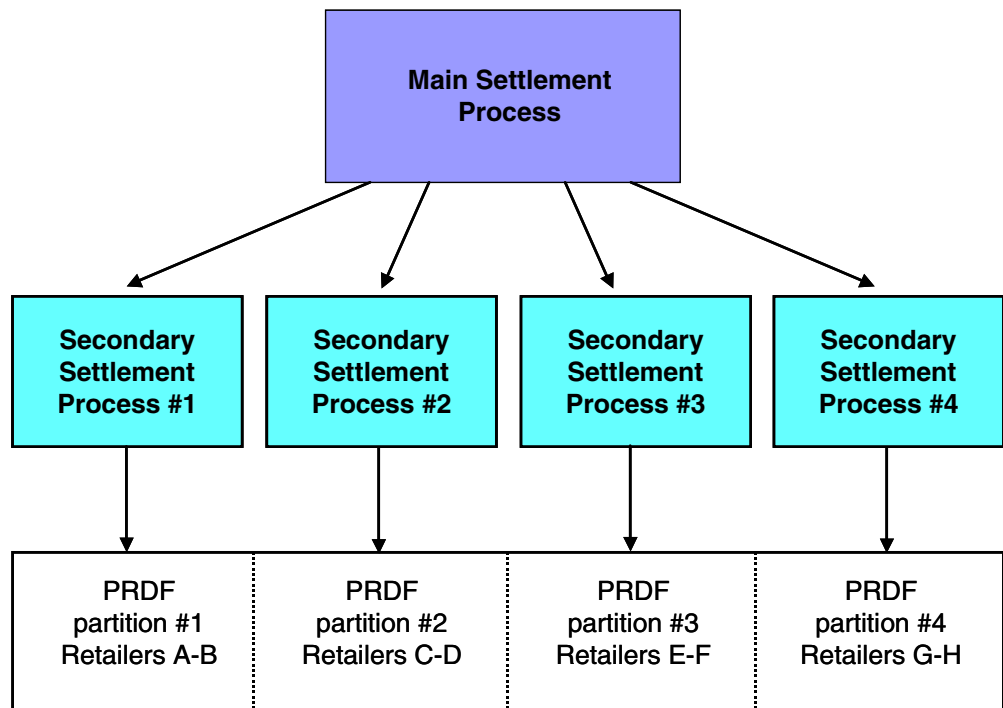
To facilitate cutting over retailers quickly and efficiently, the POS Retailer Definition File (PRDF) can be partitioned, ranges of retailers assigned to each partition, and one or more Secondary Settlement processes assigned to each partition. The configuration of the Main Settlement process and the Secondary Settlement processes, along with the number of PRDF partitions, determines which Logical Network Configuration File (LCONF) assigns and params are used during cutover and reporting.

Possible configuration options include the following:

- One Main Settlement process, no Secondary Settlement processes, and the PRDF is not partitioned. This is an issuer-only configuration in which no retailers are defined to the BASE24-pos system. A PRDF must exist for the Main Settlement process to initialize successfully; however, the PRDF is an empty file.
- One Main Settlement process, one Secondary Settlement process, and the PRDF is not partitioned. The Secondary Settlement process is responsible for all the retailers defined in the logical network.
- One Main Settlement process, two or more Secondary Settlement processes, and the PRDF is partitioned. Each Secondary Settlement process is responsible for a specific range of retailers in the logical network and more than one Secondary Settlement process can access the same partition. For example, Secondary Settlement processes 1 and 2 access the same partition. The partition holds records 1 through 1000. Secondary Settlement process 1 is responsible for records 1 through 500, while Secondary Settlement process 2 is responsible for records 501 through 1000.
- One Main Settlement process, two or more Secondary Settlement processes, and the PRDF contains one partition for each Secondary Settlement process. Each Secondary Settlement process is responsible for a specific range of retailers and accesses its own partition for the records in that range.

Note: When the PRDF is partitioned, at least one Secondary Settlement process must be assigned to access each partition.

The diagram below is an example of the last option in the list above. This is only an example and does not indicate a recommendation.



LCONF Assigns and Params

The assigns and params listed below are used by the Main and Secondary Settlement processes and the report programs. The Main and Secondary Settlement processes read these assigns and params into memory at initialization. (Refer to appendix A for initialization procedures.) If any of the required assigns and params are not configured correctly in the LCONF, the settlement processes abend during initialization.

The roles of each of these assigns and params have in cutover and reporting are explained further in this section. Refer to the ***BASE24 Logical Network Configuration File Manual*** for more information on LCONF assigns and params.

Note: LCONF assigns and params involved in interchange cutover and reporting are read by the Interchange Interface processes during initialization. These assigns and params are discussed in this section but are not listed below.

Assigns	
CRT-AUTH-PRF	POS-RPT-22
IDF (required)	POS-RPT-2R
LMCF (required)	POS-RPT-35
POS-ALTKEY-SCRATCH-FILE (required)	POS-RPT-ERR-FILE (required)
POS-CARD-OBEYFILE	POS-SETL-PTD-EXTMEM-SWAPVOL
POS-PIRF	POS-SETL-RPT-1
POS-PRDF (required)	POS-SETL-RPT-1R
POS-PRRF	POS-SETL-RPT-PROG-1-NAME
POS-PTD-DYN-GNRL *	POS-SETL-RPT-PROG-2-NAME
POS-PTDF*	POS-SETL-RPT-PROG-30-NAME
POS-PTLF (required)	POS-SETL-RPT-PROG-35-NAME
POS-RPT-2	POS-SETL-RPT-PROG-4-NAME
POS-RPT-21	POS-SETL-RPT-PROG-5-NAME

* Either the POS-PTDF or POS-PTD-DYN-GNRL assign is required.

Params	
LOGICAL-NET (required)	POS-SETL-PTLF-EXTENTS
POS-LN-CUTOVER (required)	POS-SETL-PTLF-FILE-RETENTION
POS-NOTIFY-INTERVAL	POS-SETL-REPORTS
POS-NOTIFY-PERCENTAGE	POS-SETL-RPT-CPU
POS-PRF-RECSIZE	POS-SETL-RPT-FILE-RETENTION
POS-PTLF-RECSIZE	POS-SETL-RPT-PRI
POS-SETL-ALIVE-TIMEOUT	POS-SETL-SENIOR-SLAVE
POS-SETL-FI-CUTOVER-LEAD-TIME	POS-SETL-TERM-CUTOVER-DELAY-TIME
POS-SETL-LN-CUTOVER-LEAD-TIME	POS-SETL-THRESHOLD
POS-SETL-MASTER (required)	POS-SETL-TRACE
POS-SETL-MHI-NOTIFY	POS-SETL-WAIT-INTERVAL
POS-SETL-MIDNIGHT-NOTIFY	POS-SLAVE-KEY
POS-SETL-NOTIFY	

Configuring the Settlement Processes

The cutover and reporting procedures of the BASE24-pos system are controlled and executed by two types of settlement processes. These processes are the Main Settlement process and up to 16 Secondary Settlement processes. When multiple Secondary Settlement processes exist, the institution can designate one of the Secondary Settlement processes as a Senior Secondary Settlement process. Issuer-only networks do not require Secondary Settlement processes because these networks are not responsible for retailer cutover or retailer settlement. The functions that the Main Settlement process, Secondary Settlement processes, and Senior Secondary Settlement processes perform are described in section 1.

Main Settlement Process

Each logical network must have one Main Settlement process. The Main Settlement process is configured by setting several params in the Logical Network Configuration File (LCONF). These params are listed and briefly described below.

LOGICAL-NET — The name of the logical network for the Logical Network Configuration File (LCONF) that the Main Settlement process is accessing.

POS-PRF-RECSIZE — The average size of a record, in bytes, in the POS Referral File (PRF). The BASE24-pos Main Settlement process uses the value of this param to calculate the amount of disk space required to build the alternate keys for the PRF.

POS-PTLF-RECSIZE — The average size of a record, in bytes, in the POS Transaction Log File (PTLF). The BASE24-pos Main Settlement process uses the value of this param to calculate the amount of disk space required to build the alternate keys for the PTLF.

POS-SETL-MASTER — The symbolic name of the Main Settlement process. Each settlement process in the BASE24-pos network uses this param to determine if it is a Main Settlement process or a Secondary Settlement process. This param is required and must contain a valid process name. If the param value is the symbolic name of the process reading the param, the process initializes as the Main Settlement process. Otherwise, the process initializes as a Secondary Settlement process.

POS-SETL-PTLF-EXTENTS — The number of extents the Main Settlement process must reserve, or allocate, when creating new POS Transaction Log Files (PTLFs).

POS-SETL-PTLF-FILE-RETENTION — The number of days worth of inactive PTLF and PRF files the BASE24-pos product maintains on the system.

POS-SETL-THRESHOLD — The number of times the Main Settlement process can attempt to send a startup message to a Secondary Settlement process before entering a wait state. Refer to the initialization procedures in appendix A for information about the processing involved in sending startup messages.

POS-SETL-TRACE — A code indicating whether the Main and Secondary Settlement processes will log trace messages indicating processing activity. Trace messages should be logged only in a test environment. Logging trace messages in a production environment could negatively impact performance.

POS-SETL-WAIT-INTERVAL — The number of minutes the Main Settlement process remains in a wait state after the threshold has been reached before attempting to start a Secondary Settlement process again. Refer to the initialization procedures in appendix A for information about the processing involved in sending startup messages.

Secondary Settlement Processes

If the logical network is an acquirer or issuer/acquirer network, it must have at least one Secondary Settlement process. The maximum number of Secondary Settlement processes that can be configured in one logical network is 16. Issuer-only networks do not require Secondary Settlement processes as these networks are not responsible for retailer cutover or retailer settlement.

Each Secondary Settlement process is configured by setting several params in the Logical Network Configuration File (LCONF). These params are listed and briefly described below.

POS-NOTIFY-INTERVAL — The number of times the BASE24-pos system can write to the PTLF between calculating the percentage of file full.

POS-NOTIFY-PERCENTAGE — The percentage of file full at which the Secondary Settlement process logs a message indicating the PTLF is becoming full.

POS-SLAVE-KEY — The range of retailer IDs assigned to the indicated Secondary Settlement process.

Note: The number of retailers that should be assigned to one Secondary Settlement process depends on the configuration and performance issues of the specific customer site. When changing the range of retailer IDs assigned to a Secondary Settlement process, it may be necessary to stop and restart the Main and Secondary Settlement processes in order to repartition the POS Retailer Definition File (PRDF).

Senior Secondary Settlement Process

If the network has Secondary Settlement processes, one of those processes may be configured as the Senior Secondary Settlement process. The Senior Secondary Settlement process is configured just as a Secondary Settlement process, with the addition of the following LCONF params.

POS-SETL-SENIOR-SLAVE — The symbolic name of the Secondary Settlement process that will also perform the responsibilities of a Senior Secondary Settlement process. If a Secondary Settlement process is not identified in this param, no Secondary Settlement process will verify that the Main Settlement process is alive during processing.

POS-SETL-ALIVE-TIMEOUT — How often, in minutes, the Senior Secondary Settlement process must query the Main Settlement process to verify that the Main Settlement process is alive.

Configuring Cutover

The BASE24-pos product supports terminal, retailer, institution, and logical network cutover. These cutovers are configured using assigns and params in the Logical Network Configuration File (LCONF) and fields in the Institution Definition File (IDF), POS Retailer Definition File (PRDF), and the BASE24-pos Terminal Data files.

Interchange cutovers are controlled by the cutover procedures dictated by each individual interchange. The Main Settlement process and Secondary Settlement processes do not affect or control interchange cutover. However, the cutover information provided by the interchange must be configured within the BASE24 system. Interchange cutover information is configured within the BASE24 system in the Interchange Configuration File (ICF) or Enhanced Interchange Configuration File (ICFE).

Terminal Cutover

The BASE24-pos product supports several methods for cutting a terminal over. An operator can cut a terminal over by initiating a balancing and cutover transaction, the Device Handler module can force a terminal to cut over, or the Secondary Settlement process can force a terminal to cut over. No matter how a terminal is cut over, each terminal must be cut over during a time frame, the retailer window, defined by the retailer.

Terminal cutover is configured using assigns in the Logical Network Configuration File (LCONF) and fields in the POS Retailer Definition File (PRDF) and BASE24-pos Terminal Data files.

Logical Network Configuration File (LCONF)

The following assigns control terminal cutover and must be configured in the LCONF:

POS-PRDF — The fully qualified file name of the POS Retailer Definition File (PRDF). The PRDF contains fields that control retailer cutover processing.

POS-PTD-DYN-GNRL — The fully qualified file name of the BASE24-pos Terminal Data Dynamic File—general data (PTDD1). The PTDD1 contains fields that control terminal cutover processing for standard POS and Hypercom devices.

POS-PTDF — The fully qualified file name of the POS Terminal Data File (PTDF). The PTDF contains fields that control terminal cutover processing for all terminal types except for standard POS and Hypercom terminals, which use the POS-PTD-DYN-GNRL assign instead.

POS-SETL-PTD-EXTMEM-SWAPVOL — The fully qualified file name of the BASE24-pos Terminal Data files (PTD) extended memory swap volume. The Secondary Settlement processes use this assign to temporarily store data when building extended memory.

POS Retailer Definition File (PRDF)

The retailer controls the period of time during which all of its terminals are cut over. This period of time is set in the RETAILER CUTOVER/BALANCE START and RETAILER CUTOVER/BALANCE END fields on PRDF screen 1 and is called the *retailer cutover window*. If a terminal is not cut over by the end of this window, the Secondary Settlement process automatically forces the terminal to cut over. At the end of the balancing window, the Secondary Settlement process increases the retailer's PRDF posting dates so the retailer is ready for the next day.

BASE24-pos Terminal Data Files (PTD)

The BASE24-pos Terminal Data files (PTD) contain screens with information impacting how terminals are cut over and the processing steps that the Secondary Settlement process takes during terminal cutover.

On PTD screen 3, the TERMINAL CUTOVER field indicates the method by which a terminal is cut over. Cutover methods include end-of-day, end-of-shift, and end-of-batch cutover transactions entered by the operator; force cutover by the Device Handler module; and force cutover by the Secondary Settlement process.

Also on PTD screen 3, the BALANCE SUPPORT field indicates the type of balancing transactions this terminal supports. The LOG TOTALS fields (BATCH, SHIFT, DAY, NETWORK, and SERVICE) indicate which settlement totals the Secondary Settlement process logs to the PTLF during a terminal forced cutover.

The CLERK TOTALS FLAG field on PTD screen 5 indicates whether the Device Handler module accumulates operator totals for this terminal.

Retailer Cutover

The Secondary Settlement process automatically cuts each retailer over to a new business day. Retailer cutover is controlled using assigns and params in the Logical Network Configuration File (LCONF) and fields in the POS Retailer Definition File (PRDF).

Logical Network Configuration File (LCONF)

The following assigns and params must be configured for retailer cutover.

POS-PRDF assign — The fully qualified file name of the POS Retailer Definition File (PRDF) from which the Secondary Settlement process obtains cutover and settlement processing information. The PRDF can be partitioned and multiple Secondary Settlement processes can access the same PRDF partition. Each partition contains a range of retailer records assigned to a specific Secondary Settlement process in the POS-SLAVE-KEY param. If the network has only one Secondary Settlement process, the PRDF is not partitioned.

POS-SETL-FI-CUTOVER-LEAD-TIME param — The minimum number of minutes the retailer can cut over before the institution cuts over.

POS-SETL-MHI-NOTIFY param — The name of the Merchant Host Interface (MHI) process that the Secondary Settlement process must notify at retailer cutover. If this param is not present, the Secondary Settlement process assumes that no MHI processes must be notified.

POS-SETL-TERM-CUTOVER-DELAY-TIME param — The minimum number of minutes after logical network cutover that a retailer can cut over.

POS Retailer Definition File (PRDF)

Screen 1 of the POS Retailer Definition File (PRDF) contains several fields that control retailer cutover and settlement. The RETAILER CUTOVER/BALANCE START and RETAILER CUTOVER/BALANCE END fields define the retailer cutover window. During this window, all retailer terminals must be cut over. The Secondary Settlement process cuts the retailer over at the time specified in the RETAILER CUTOVER/BALANCE END field.

The POSTING DATES PREVIOUS, POSTING DATES CURRENT, and POSTING DATES NEXT fields are also on PRDF screen 1. The POSTING DATES PREVIOUS field contains the posting date, in YYMMDD format, that the retailer used before the last cutover. The POSTING DATES CURRENT field contains the posting date, in YYMMDD format, the retailer is currently using. The POSTING DATES NEXT field contains the posting date, in YYMMDD format, the retailer will use following the next cutover. The Secondary Settlement process increases the dates in these fields automatically at retailer cutover.

Institution Cutover

The Main Settlement process automatically cuts over each institution at the end of the institution's business day. All institutions must have cut over before logical network cutover. The Main Settlement process cuts each institution over according to a param in the Logical Network Configuration File (LCONF) and fields in the Institution Definition File (IDF).

Logical Network Configuration File (LCONF)

The Main Settlement process uses the value in the POS-SETL-LN-CUTOVER-LEAD-TIME param in the LCONF to determine the minimum number of minutes before logical network cutover that all institutions must be cut over. If an institution is not configured to cut over this many minutes before the logical network cuts over, the Main Settlement process automatically cuts the institution over this number of minutes before logical network cutover regardless of the institution's configured cutover time.

Institution Definition File (IDF)

Institutions control their cutover times individually using the BALANCE AND CUTOVER TIME WINDOW on Institution Definition File (IDF) screen 17. At institution cutover, if any of the institution's retailers have not been cut over, the Secondary Settlement process performs force cutover actions which automatically cut the retailers over. In this way, institution cutover ensures that all retailers have been cut over to the new business day.

If the institution's cutover time is not set properly in the BALANCE AND CUTOVER TIME WINDOW field, the Main Settlement process uses the POS-SETL-LN-CUTOVER-LEAD-TIME param in the LCONF and automatically cuts the institution over this number of minutes prior to logical network cutover.

At the end time specified in the BALANCE AND CUTOVER TIME WINDOW, the Main Settlement process increases the dates in the following fields that also appear on IDF screen 17: CURRENT BUSINESS DATE, CUSTOMER PROCESSING DATE, NEXT BUSINESS DATE, and REPORT DATE. The Main Settlement process automatically increases the dates in these fields as necessary.

Institutions must also control the time of day at which the system purges the Usage Accumulation File (UAF) and the data in the usage accumulation fields of the Cardholder Authorization File (CAF). The BASE24 system performs these functions as indicated in the FIELD CUTOVER field on IDF screen 3.

On IDF screen 4, institutions establish withdrawal period parameters. These parameters define the usage accumulation periods for the institution and the holidays on which the institution does not process. The BEGINNING DATE field indicates the starting date, in YYMMDD format, of the current usage accumulation period for all of the institution's cardholders. The NEXT BEGINNING DATE field indicates the starting date of institution's next usage accumulation period. Once these fields are set at initialization or startup, the BASE24 system increases the dates automatically.

Also on IDF screen 4, the PERIOD LENGTH field and the WORK DAY CODE field indicate the length of the institution's usage accumulation period. The BASE24 system uses only one of these fields. The HOLIDAYS fields contain the dates, in YYMMDD format, on which the institution does not process. The BASE24 system uses the HOLIDAYS fields based on the settings in the PERIOD LENGTH and WORK DAY CODE fields. Refer to the ***BASE24 Base Files Maintenance Manual*** for information on how these fields interact.

Logical Network Cutover

The Main Settlement process automatically cuts the logical network over according to several assigns and params in the Logical Network Configuration File (LCONF). These assigns and params are listed and briefly described below.

CRT-AUTH-PRF assign — The fully qualified name of the POS Referral File (PRF). The Main Settlement process uses this to create the PRF at initialization if CRT Authorization is used.

POS-LN-CUTOVER param — The time of day that the Main Settlement process is to cut over the logical network. All institutions in the logical network must be cut over by the time in this param.

POS-PTLF assign — The naming convention of the POS Transaction Log File (PTLF) and the location of the PTLF template file. The Main Settlement process uses the PTLF template file to create the PTLF at initialization and logical network cutover.

POS-SETL-MIDNIGHT-NOTIFY param — The names of the Device Handler/Router/Authorization processes the Main Settlement process must notify at midnight cutover. When setting this param, all applicable Device Handler/Router/Authorization processes must be in this list. The BASE24 system does not verify that all applicable processes are included in this param.

Note: For BASE24 networks using both the BASE24-pos and BASE24-atm products and the Usage Accumulation File (UAF), this param should include all BASE24-pos Device Handler/Router/Authorization processes and all BASE24-atm Authorization processes. This ensures correct purge processing of the UAF during midnight cutover. The UAF is used by the BASE24-atm and BASE24-pos products with the Negative Authorization with Usage Accumulation method. The first settlement process to cut over after midnight performs the UAF purge processing. This process then needs to notify all processes that use the UAF.

POS-SETL-NOTIFY param — The names of processes the Main Settlement process must notify at logical network cutover. When setting this param, all applicable Device Handler/Router/Authorization processes must be in this list. If the ACI Commerce Gateway product is used with the BASE24-pos product, the symbolic name of an ISO Host Interface process used to communicate with the ACI Commerce Gateway product must also be in this list. The BASE24 system does not verify that all applicable processes are included in these params.

Interchange Cutover

The BASE24-pos product does not control interchange cutover. Interchange cutovers are dictated by each individual interchange. The BASE24-pos product does, however, enable institutions and retailers to reconcile their business days in relation to the interchange's business day.

Each interchange associated with the BASE24-pos network provides cutover information that the BASE24-pos product uses to aid its retailers and institutions in reconciling its interchange transaction activity. This cutover information must be configured in the BASE24-pos system on screen 2 of the Interchange Configuration File (ICF) or Enhanced Interchange Configuration File (ICFE). Because not all the fields on ICF and ICFE screen 2 function for all interchanges,

refer to the ***BASE24 Base Files Maintenance Manual*** and the interchange documentation specific to the particular interchange when configuring this information.

The SETTLEMENT HOUR and SETTLEMENT MINUTE fields contain the hour and minute at which the interchange performs cutover and settlement. The SETTLEMENT DAYS field indicates whether the interchange processes on a 5- or 7-day schedule. This field may not function for all interchanges.

The SWITCH POSTING DATE field contains the posting date, in YYMMDD format, of the interchange. Once this field is set during initial configuration procedures, the Interchange Interface process automatically increases the date in this field at interchange cutover.

The HOLIDAY DATES field contains the holiday dates, in YYMMDD format, on which the interchange does not process. This field may not function for all interchanges.

Configuring Reports

The BASE24-pos daily and periodic reporting functions are configured using information in the Logical Network Configuration File (LCONF), the POS Retailer Definition File (PRDF), and the Institution Definition File (IDF). The BASE24-pos interchange reporting functions are configured using information in the Interchange Configuration File (ICF) or Enhanced Interchange Configuration File (ICFE).

PTLF Alternate Keys

The POS Transaction Log File (PTLF) can have up to five alternate keys: the card number key, the retailer ID key, the terminal ID/retailer ID key, the retailer ID/clerk ID key, and the terminal ID key. The card number key, terminal ID/retailer ID key, retailer ID/clerk ID key, and the terminal ID key are used for both perusing the PTLF and reporting transactions for settlement. The retailer key is used only for reporting.

The PTLF alternate keys are configured at installation. To produce reports B24POS01, B24POS02, B24POS04, B24POS05, and B24POS35, certain alternate keys must be configured. These keys and the reports that require them are shown in the table below. The other reports do not require PTLF alternate keys.

PTLF Alternate Key	Report
Card Key (CRD)	B24POS01
Retailer Key (RETL)	B24POS02
Terminal/Retailer Key (RKEY)	B24POS04
Retailer Key (RETL)	B24POS05
Retailer Key (RETL)	B24POS35

Logical Network Configuration File

The BASE24-pos product uses a number of LCONF assigns and params when running reports. These assigns and params must be configured before the BASE24-pos system can produce reports. Each assign and param the BASE24-pos product uses for reporting is identified and briefly defined below. Refer to the ***BASE24 Logical Network Configuration File Manual*** for more information on each assign and param.

The Main Settlement process produces the BASE24-pos daily reports automatically when configured to do so. The Main Settlement process also produces the BASE24-pos periodic reports at operator request when configured to do so. The interchange reports, however, are produced by the respective Interchange Interface process for each interchange. Refer to the appropriate interchange documentation for more information about Interchange Interface processes.

CRT-AUTH-PRF assign — The fully qualified name of the POS Referral File (PRF). The Main Settlement process uses the PRF when creating the Network Referral Statistical Report (B24POS30).

ILF-LOGGING-ENABLED param — Indicates whether an Interchange Interface process logs transactions to the ILF. If the Interchange Interface process does not log transactions to the ILF, transactions cannot be reported on interchange reports.

Note: Not all Interchange Interface processes use this param. Refer to the interchange-specific documentation for more information.

POS-ALTKEY-SCRATCH-FILE assign — The fully qualified name of the alternate key scratch file. The Main Settlement process uses the alternate key scratch file when creating the obey file that starts the BASE24-pos daily report programs.

POS-CARD-OBEYFILE assign — The naming convention for the obey file that the Main Settlement process creates at logical network cutover. This obey file starts the BASE24-pos daily report programs.

POS-EXPT-RPT-FLG param — A code indicating whether the obey file generates the Daily Cardholder Exception Report (B24POS01-EXPT) and the Daily Retailer Exception Detail Report (B24POS12-EXPT).

POS-PIRF assign — The fully qualified file name of the POS Institution Reporting File (PIRF). The report obey file uses this file to produce the Periodic Institution Summary Report (B24POS21).

POS-PRRF assign — The fully qualified name of the POS Retailer Reporting File (PRRF). The PIRF contains information used to generate the periodic institution summary report. The BASE24-pos report programs use this assign. This assign is required only if periodic reports are to be produced.

POS-RPT-2 assign — The spooler location to which the report program sends the Daily Retailer Summary Report (B24POS11) and Daily Retailer Activity Report (B24POS12). The default spooler location for this assign is specified in the PRINT LOCATION field on PRDF screen 4. If this assign is not configured or is invalid, the report is not generated.

POS-RPT-21 assign — The spooler location to which the report program sends the Periodic Institution Summary Report (B24POS21). This assign has a default spooler location of \$\$.#RPT21. If this assign is not configured or is invalid, the report is not generated.

POS-RPT-22 assign — The spooler location to which the report program sends the Periodic Retailer Summary Report (B24POS22). This assign has a default spooler location of \$\$.#RPT22. If this assign is not configured or is invalid, the report is not generated.

POS-RPT-2R assign — The spooler location to which the report program sends the Daily Retailer Exception Detail Report (B24POS12-EXPT). This assign has a default spooler location of \$\$.#EXPT12. If this assign is not configured or is invalid, the report is not generated.

POS-RPT-30 assign — The spooler location to which the report program sends the Network Referral Statistical Report (B24POS30). This assign has a default spooler location of \$\$.#RPT30. If this assign is not configured or is invalid, the report is not generated.

POS-RPT-35 assign — The spooler location to which the report program sends the Daily Retailer Reconciliation Report (B24POS35). This assign has a default spooler location of \$\$.#RPT35. If this assign is not configured or is invalid, the report is not generated.

POS-RPT-ERR-FILE assign — The network logging location to which all report programs must send their event messages.

POS-SETL-REPORTS param — The time the Main Settlement process creates the report obey file and starts the BASE24-pos daily reports. If this param is not configured or is invalid, the Main Settlement process does not produce BASE24-pos daily reports.

POS-SETL-RPT-1 assign — The spooler location to which the report program sends the Daily Network Settlement Report (B24POS01). This assign has a default spooler location of \$S.#RPT01. If this assign is not configured or is invalid, the report is not generated.

POS-SETL-RPT-1R assign — The spooler location to which the report program sends the Daily Cardholder Exception Report (B24POS02-EXPT). This assign has a default spooler location of \$S.#EXPT2. If this assign is not configured or is invalid, the report is not generated.

POS-SETL-RPT-CPU param — The central processing unit (CPU) in which the Main Settlement process executes the BASE24-pos daily report programs. If this param is not configured or is invalid, the Main Settlement process executes the BASE24-pos daily report programs in the CPU in which the Main Settlement process is running.

POS-SETL-RPT-FILE-RETENTION param — The number of days worth of inactive daily report files the Main Settlement process maintains on the system.

POS-SETL-RPT-PRI param — The priority at which the Main Settlement process must execute the BASE24-pos daily report programs.

POS-SETL-RPT-PROG-1-NAME assign — The fully qualified file name of the BASE24-pos daily report program that generates the Daily Network Settlement Report (B24POS01), the Daily Cardholder Activity Report (B24POS02), and the Daily Cardholder Exception Report (B24POS02-EXPT).

POS-SETL-RPT-PROG-2-NAME assign — The fully qualified file name of the BASE24-pos daily report program that generates the Daily Retailer Summary Report (B24POS11), the Daily Retailer Activity Report (B24POS12), and the Daily Retailer Exception Detail Report (B24POS12-EXPT).

POS-SETL-RPT-PROG-30-NAME assign — The fully qualified file name of the BASE24-pos daily report program that generates the Network Referral Statistical Report (B24POS30).

POS-SETL-RPT-PROG-35-NAME assign — The fully qualified file name of the BASE24-pos daily report program that generates the Daily Retailer Reconciliation Report (B24POS35).

POS-SETL-RPT-PROG-4-NAME assign — The fully qualified file name of the BASE24-pos daily report program that generates the Daily Clerk Totals Report (B24POS04).

POS-SETL-RPT-PROG-5-NAME assign — The fully qualified file name of the BASE24-pos daily report program that generates the Daily Disputed Transactions Report (B24POS05).

POS-SETL-TRACE assign — A code indicating whether the Main Settlement and Secondary Settlement processes log terminal, retailer, and network cutover messages.

SW-RPT01-PROG-NAME assign — The fully qualified file name of the BASE24 interchange report program that generates the Clearings Statement Report (SETL01).

SW-RPT09-PROG-NAME assign — The fully qualified file name of the BASE24 interchange report program that generates the Transaction Detail Report (STAT09).

SW-RPT-OBEY-FILE assign — The fully qualified file name of the Interchange Interface process' report obey file. This report obey file generates the BASE24 interchange reports.

SW-RPT-STARTUP param — The time the Interchange Interface process starts the BASE24 interchange reports. If this param is not configured or is invalid, the Interchange Interface process does not produce BASE24 interchange reports.

POS Retailer Definition File

The POS Retailer Definition File (PRDF) contains one record for each retailer in the logical network. The settings in these records determine how transactions initiated by the retailer are processed and reported.

The fields on PRDF screen 4 impact transaction reporting. The PRINT LOCATION field indicates where the report programs print the daily retailer reports. The REPORT SET field indicates how the report programs are to print the retailer reports. For example, the report programs could print the reports in full, print only the totals pages, or not print the reports at all.

Finally, the RETAIN REPORTS field indicates the number of days periodic data the report programs retain in the Periodic Retailer Report File (PRRF). This field must be set appropriately for the Periodic Retailer Summary Report (B24POS22) to contain the correct information. For example, if the RETAIN REPORTS field is set to 7 days and the institution runs monthly periodic reports, the Periodic Retailer Summary Report (B24POS22) will contain insufficient data for the month.

Institution Definition File

The Institution Definition File (IDF) contains one record for each institution in the logical network. The settings in these records determine how transactions are processed and reported for each institution.

The fields on IDF screen 21 impact transaction reporting. The CARDHOLDER ACTIVITY REPORT PRINT LOCATION field indicates where the report programs print the Cardholder Activity Report (B24POS02). The REPORT SET field indicates whether the report programs produce both parts 1 and 2 of the Cardholder Activity Report (B24POS02) as one report, parts 1 and 2 as separate reports, only part 2, or does not produce the report at all.

The REPORT RETENTION PERIOD field, also on IDF screen 21, indicates the number of days periodic data the report programs retain in the Periodic Institution Report File (PIRF). This field must be set appropriately for the Periodic Institution Summary Report (B24POS21) to contain the correct information. For example, if the REPORT RETENTION PERIOD field is set to 7 days and the institution runs monthly periodic reports, the Periodic Institution Summary Report (B24POS21) will contain insufficient data for the month.

The MERCHANT SETTLEMENT REPORT PRINT LOCATION field on IDF screen 21 indicates where the report programs print the Daily Disputed Transactions Report (B24POS05).

Interchange Configuration Files

The Interchange Configuration File (ICF) or Enhanced Interchange Configuration File (ICFE) contains one record for each interchange communicating with the BASE24 logical network. The settings in the ICF and ICFE records indicate how BASE24 processes and reports transactions from each respective interchange.

The fields on ICF and ICFE screen 2 impact transaction reporting. The REPORT PRIORITY field indicates the priority at which the Interchange Interface process produces the BASE24 interchange reports. The REPORT CPU field indicates the central processing unit (CPU) in which the Interchange Interface process produces the interchange reports.

COBNAMES File

Some information that appears on the BASE24-pos reports can be customized for a particular logical network. For example, ACI assigns descriptions to response codes that appear on BASE24-pos reports where applicable. These descriptions can be changed to better fit the needs of the logical network. Such changes are made in the COBNAMES file.

The COBNAMES file contains tables of information and is used by all BASE24 products during processing. Each product accesses particular tables in the COBNAMES file to complete certain processing. The information tables that the BASE24-pos product uses in settlement and reporting contain device types and response code descriptions.

After making changes to the COBNAMES file, the requesters that access the COBNAMES file must be recompiled and the object code must be placed in the appropriate volume and subvolume.

6: Cutover and Reporting Operations

This section describes the tasks that an operator must perform to manage cutover and reporting from day to day. These tasks include generating daily, periodic, and interchange reports.

Producing BASE24-pos Daily Reports

The Main Settlement process automatically produces the BASE24-pos daily reports after logical network cutover. Once the system is correctly configured for reporting, an operator does not need to perform any tasks to produce the BASE24-pos daily reports.

At logical network cutover, the Main Settlement process creates an obey file that contains commands necessary to run report programs. The report obey files have the naming convention *RCyymmdd*, where *yymmdd* is the report date. The Main Settlement process creates the obey files in the volume and subvolume where the institution's data files are located. The POS-CARD-OBEYFILE assign in the Logical Network Configuration File (LCONF) specifies the fully qualified name of the obey file. (Obey files are explained in detail later in this section.)

The Main Settlement process obtains the names of the report programs from the LCONF. The report programs start at the time specified in the POS-SETL-REPORTS param. Each day at that time, the Main Settlement process runs the obey file to start the BASE24-pos daily report programs. If the POS-SETL-REPORTS param is not configured or contains an invalid value, the Main Settlement process runs the obey file one hour after logical network cutover.

Creating the BASE24-pos Daily Report Obey File

The report obey file is an edit file, or text file, that operators can access and modify when necessary. The report obey file has the naming convention *RCyymmdd*, where *yymmdd* is the report date. The Main Settlement process creates the obey file in the volume and subvolume where the institution's data files are located. The POS-CARD-OBEYFILE assign in the LCONF specifies the fully qualified file name of the obey file. The BASE24-pos report obey file is configured in the BASE24-pos system when the system is installed. An example report obey file is shown on the following page.

Because the report obey file is an edit file, operators can modify the contents of the file as necessary. Changes that an operator may be required to make regard the Retailer Summary Report (B24POS11) and the Retailer Activity Report (B24POS12) and are made in the highlighted fields shown in the example obey file. The operator can modify the obey file to create these reports for the entire logical network or for a single institution by entering either the logical network identifier or the appropriate FIID. An operator can also reprint these reports for a particular retailer by adding the retailer's group, region, and retailer ID and rerunning the report obey file.

The Main Settlement process creates and runs the report obey file according to the value in the POS-SETL-REPORTS param in the LCONF. Refer to section 5 or the *BASE24 Logical Network Configuration File Manual* for more information about this param.

Example Report Obey File

An example of a report obey file used to generate BASE24-pos reports is shown below. Shaded text indicates information that an operator may be required to change when running the Retailer Summary Report (B24POS11) and the Retailer Activity Report (B24POS12).

```
FUP ALTER \B24.$DATA.PRO1PTLF.POYMMDD,ALTFILE (1 , \B24.$DATA.
PRO1PTLF.POYMMDD,ALTKEY ("CR",FILE 1,KEYOFF 016, KEYLEN 27)
FUP ALTER\B24.$DATA.PRO1PTLF.POYMMDD, ALTKEY ("RT",FILE 1,KEYOFF
016, KEYLEN 27)
FUP LOADALTFILE 1,\B24.$DATA.PRO1PTLF.POYMMDD,MAX 0000000000,
SCRATCH \B24.$DATA.PRO1DATA.FILE
FUP ALTER \B24.$DATA.PRO1DATA.FFYMMDD,ALTFILE (0, \B24.$DATA.
PRO1DATA.FILE1>), ALTKEY ("FI",FILE 1,KEYOFF 447, KEYLEN 04)
FUP LOADALTFILE 0,\B24.$DATA.PRO1DATA.FFYMMDD,MAX
0000000000,SCRATCH \B24.$DATA.PRO1DATA.FILE2
ASSIGN LCONF-REC, \B24.$DATA.PRO1DATA.LCONF
RUN \B24.$DATA.PS50OBJ.B24PS1 /IN \B24.$DATA.PRO1PTLF.POYMMDD,
CPU 3 , PRI 125, NOWAIT/<LNET or FIID>
CLEAR ALL
ASSIGN LCONF-REC, \B24.$DATA.PRO1DATA.LCONF
RUN \B24.$DATA.PS50OBJ.B24PS2 /IN \B24.$DATA.PRO1PTLF.POYMMDD,
CPU 3 , PRI 125, NOWAIT/<LNET OR FIID><group region retailerID>
CLEAR ALL
ASSIGN LCONF-REC, \B24.$DATA.PRO1DATA.LCONF
RUN \B24.$DATA.PS50OBJ.B24PS4, CPU 3 , PRI 125, NOWAIT/<LNET or
FIID>/<yymmdd>
CLEAR ALL
ASSIGN LCONF-REC, \B24.$DATA.PRO1DATA.LCONF
RUN \B24.$DATA.PS50OBJ.B24PS5/IN \B24.$DATA.PRO1PTLF.POYMMDD, CPU
3 , PRI 125, NOWAIT/<LNET or FIID>
CLEAR ALL
ASSIGN LCONF-REC, \B24.$DATA.PRO1DATA.LCONF
RUN \B24.$DATA.BP50OBJ.B24PS30 /IN \B24.$DATA.PRO1DATA.FILE, CPU 3
, PRI 125, NOWAIT/<LNET or FIID>
CLEAR ALL
ASSIGN LCONF-REC, \B24.$DATA.PRO1DATA.LCONF
RUN \B24.$DATA.PS50OBJ.B24PS35 /IN \B24.$DATA.PRO1PTLF.POYMMDD,
CPU 3 , PRI 125, NOWAIT/<LNET or FIID>
CLEAR ALL
```

Manually Starting BASE24-pos Daily Reports

An operator must manually run reports when the Main Settlement process encounters a problem while producing the reports, the reports must be rerun, or the value in the POS-SETL-REPORTS param is a -1 (create the report obey file but do not run the reports). An operator can start the report programs by running the obey file for the day required from a TACL prompt. An operator can also produce additional sets of reports by manually rerunning the obey file.

Suppressing BASE24-pos Daily Reports

To prevent the Main Settlement process from producing BASE24-pos daily reports, the operator must enter the value -2 (do not create the report obey file) in the POS-SETL-REPORTS param in the LCONF. To prevent the Main Settlement process from producing a specific report, the operator must replace the report program name in the LCONF with a *dummy* name. Refer to the topic “Suppressing BASE24 Interchange Reports” provided later in this section for more information on this procedure.

Producing BASE24-pos Periodic Reports

BASE24-pos periodic reports must be requested by an operator. The Main Settlement process does not produce BASE24-pos periodic reports automatically. An operator requests these periodic report using the POS Report Generator (PRG) screen. The PRG screen is described later in this section.

Information that appears on the Periodic Institution Summary Report (B24POS21) and the Periodic Retailer Summary Report (B24POS22) is from the POS Institution Reporting File (PIRF) and the POS Retailer Reporting File (PRRF), respectively. Each time the Main Settlement process generates the Cardholder Activity Report (B24POS02), the report program that produces the B24POS02 report creates a record of the report totals and writes it to the PIRF. Likewise, each time the Main Settlement process generates the Retailer Activity Report (B24POS12), the report program that produces the B24POS12 report creates a record of the report totals and writes it to the PRRF.

Deleting PIRF and PRRF Records

The report programs automatically delete records in the PIRF and PRRF after a specified period of time. Operators do not perform any manual tasks to delete these records.

Each time the B24POS02 and B24POS12 reports are generated, the report programs delete the PIRF and PRRF records that have been on the system for a specified number of days. The report programs delete the PIRF records that have been in the system for more than the number of days configured in the REPORT RETENTION PERIOD field on Institution Definition File (IDF) screen 21. The report programs delete the PRRF records that have been in the system for more than the number of days configured in the RETAIN REPORTS field on POS Retailer Definition File (PRDF) screen 5.

POS Report Generator (PRG) Screen

Operators generate BASE24-pos periodic reports using the POS Report Generator (PRG) screen. An operator must specify the report, the time frame desired, and, for retailer reports, the level at which totals must be presented. The PRG screen is shown on the next page followed by descriptions of its fields. The spooler location where the periodic reports are printed must be configured in the LCONF.

Operators can specify the time frame for the report by entering the date range on the PRG screen or by pressing a function key. If the operator specifies the time frame on the PRG screen, the operator must enter the appropriate dates in the REPORT STARTING DATE and REPORT ENDING DATE fields. If the operator specifies the time frame by pressing a function key, reports can be produced weekly, biweekly, monthly, or quarterly.

Note: The REPORT RETENTION PERIOD field on IDF screen 21 and the RETAIN REPORTS field on PRDF screen 5 must be set to retain the appropriate number of days worth of data requested for a periodic report. Refer to section 5 for information on configuring periodic reports.

```
BASE24-POS      POS RPT GENERATOR      LLLL      YY/MM/DD  HH:MM  01 OF 01
```

```
TO GENERATE PERIODIC REPORTS, PLEASE ENTER THE FOLLOWING
DESIRED PARAMETERS.
```

```
REPORT NUMBER: 21  ( 21 = INSTITUTION REPORT  OR  22 = RETAILER REPORT )
```

```
FIID:           GROUP:           REGION:           RETAILER:
```

```
TERMINAL ID:
```

```
REPORT STARTING DATE:           (YYMMDD)  REPORT ENDING DATE:           (YYMMDD)
```

```
***** BASE24 *****
FILE DESTINATION:           NEW LOGICAL NETWORK ID:
F12 - HELP
```

REPORT NUMBER — The number identifying the report to be generated.

Valid values are:

21 = Periodic Institution Summary Report (B24POS21)

22 = Periodic Retailer Summary Report (B24POS22)

Field Length: 2 numeric characters

Required Field: Yes

Default Value: 21

Data Name: Not applicable

FIID — The FIID of the financial institution whose activity is to be included on the report.

Field Length: 1–4 alphanumeric characters
Required Field: Yes
Default Value: ALL
Data Name: Not applicable

GROUP — An identifier indicating the retailer group whose activity is to be included on the report. The group must match a group established in the POS Retailer Definition File (PRDF).

Field Length: 1–4 alphanumeric characters
Required Field: Yes, if the REPORT NUMBER field contains the value 22.
Default Value: All
Data Name: Not applicable

REGION — An identifier indicating the region whose activity is to be included on the report. The region has to match a region established in the POS Retailer Definition File (PRDF).

Field Length: 1–4 alphanumeric characters
Required Field: Yes, if the REPORT NUMBER field contains the value 22.
Default Value: All
Data Name: Not applicable

RETAILER — An identifier indicating the retailer whose activity is to be included on the report. The retailer IDs are established in the POS Retailer Definition File (PRDF).

Field Length: 1–19 alphanumeric characters
Required Field: Yes, if the REPORT NUMBER field contains the value 22.
Default Value: All
Data Name: Not applicable

TERMINAL ID — The identification number of the terminal whose activity is to be included on the report. The terminal IDs are established in the BASE24-pos Terminal Data files.

Field Length: 1–16 alphanumeric characters
Required Field: Yes, if the REPORT NUMBER field contains the value 22.
Default Value: All
Data Name: Not applicable

REPORT STARTING DATE — The date, in YYMMDD format, of the first activity to be included on this report. This field is used in conjunction with the REPORT ENDING DATE field. Specifying dates in these two fields enables an operator to produce periodic reports for any time period desired. When dates are specified, the report program uses only those records on file with the specified starting date up through and including the specified ending date to produce the periodic report.

Field Length: 6 numeric characters
Required Field: No
Default Value: No default value
Data Name: Not applicable

REPORT ENDING DATE — The date, in YYMMDD format, of the last activity to be included on this report. This field is used in conjunction with the REPORT STARTING DATE field. Specifying dates in these two fields enables an operator to produce periodic reports for any time period desired. When dates are specified, the report program uses only those records on file with the specified starting date up through and including the specified ending date to produce the periodic report.

Field Length: 6 numeric characters
Required Field: No
Default Value: No default value
Data Name: Not applicable

Requesting a Periodic Report by Date

To produce a periodic report by date:

1. Enter information in the REPORT NUMBER, FIID, GROUP, REGION, RETAILER, and TERMINAL ID fields if needed.
2. Enter the date range the report should include in the REPORT STARTING DATE and REPORT ENDING DATE fields.
3. Press the **F2** key to start the report.

Requesting a Weekly Periodic Report

Weekly periodic reports include information for the last full week. One week is considered to be Monday through Sunday. Records for the current week are not included on weekly periodic reports. To produce a weekly periodic report:

1. Enter information in the REPORT NUMBER, FIID, GROUP, REGION, RETAILER, and TERMINAL ID fields if needed.
2. Press the **F3** key to start the report.

Requesting a Biweekly Periodic Report

Biweekly periodic reports include information for the last full two-week period. One week is considered to be Monday through Sunday. Records for the current week are not included on biweekly periodic reports. To produce a biweekly periodic report:

1. Enter information in the REPORT NUMBER, FIID, GROUP, REGION, RETAILER, and TERMINAL ID fields if needed.
2. Press the **F4** key to start the report.

Requesting a Monthly Periodic Report

Monthly periodic reports include information for the last full month. For example, if the current month is August, the monthly periodic report included records for July 1 through July 31. Records for the current month are not included on monthly periodic reports. To produce a monthly periodic report:

1. Enter information in the REPORT NUMBER, FIID, GROUP, REGION, RETAILER, and TERMINAL ID fields if needed.
2. Press the **F5** key to start the report.

Requesting a Quarterly Periodic Report

Quarterly periodic reports include information for the last full three-month period. For example, if the current month is August, the quarterly periodic report includes records for May 1 through July 31. Records for the current month are not included on quarterly periodic reports. To produce a quarterly periodic report:

1. Enter information in the REPORT NUMBER, FIID, GROUP, REGION, RETAILER, and TERMINAL ID fields if needed.
2. Press the **F6** key to start the report.

Producing BASE24 Interchange Reports

The Interchange Interface process automatically produces BASE24 interchange reports for its respective interchange each day. If interchange reporting is configured correctly in the BASE24 system, the operator does not need to perform any tasks to generate the BASE24 interchange reports. The Main Settlement process, which is responsible for producing BASE24-pos daily and periodic reports, is not involved in producing BASE24 interchange reports.

Each BASE24 interchange report is produced by a report program. These report programs are started automatically by the Interchange Interface process for each interchange. The RP01-xxx Interchange Clearings Statement report program produces the Clearings Statement Report (SETL01). The RP09-xxx Interchange Transaction Detail report program produces the Interchange Transaction Detail Report (STAT09). The xxx suffix in the report program names differentiates the report programs for the different interchanges. For example, the report programs for the CIRRUS interchange are RP01-CIR and RP09-CIR.

The report programs are configured in the BASE24 system using assigns in the LCONF. The LCONF assigns for the interchange report programs are SW-RPT01-PROG-NAME and SW-RPT09-PROG-NAME for the SETL01 and STAT09 reports, respectively.

The Interchange Interface process starts the report programs for an interchange at the time configured in the SW-RPT-STARTUP param in the LCONF. One SW-RPT-STARTUP param exists in the LCONF for each interchange. At the specified time, the Interchange Interface process creates and runs an obey file to start the report programs for the specific interchange.

Manually Starting BASE24 Interchange Reports

Under normal circumstances, the Interchange Interface process starts and stops report programs automatically. However, if the Interchange Interface process encounters a problem while producing the reports or if the reports must be rerun, an operator can start the report programs by running the obey file for the day required from a TACL prompt. The report programs then overwrite anything previously written to the spooler location. An operator can also produce additional sets of reports by manually rerunning the obey file.

Suppressing BASE24 Interchange Reports

To prevent the Interchange Interface process from producing BASE24 interchange reports, the operator must replace the report program name in the LCONF with a *dummy* name. When the Interchange Interface process attempts to start the report program, the Interchange Interface is unable to find the corresponding report program and, therefore, is unable to produce the report. Replacing the report program name with a *dummy* name causes the BASE24 system to generate event messages; these event messages are informational only and processing is not impacted.

A: Processing Information

This appendix includes the following processing information pertaining to the BASE24-pos Main and Secondary Settlement processes:

- Supplementary information for process configuration
- A description of initialization processing
- A description of warmboot processing
- Descriptions of the processing that occurs when the timers used in cutover and settlement expire
- A list and brief descriptions of text commands that the Main or Secondary Settlement processes can receive

Process Configuration

The following configuration issues relate to the BASE24-pos Main and Secondary Settlement processes.

Multiple Copies

Each logical network requires at least one BASE24-pos settlement process. This process is called the Main Settlement process. If configured to do so, additional settlement processes can exist in the system. These other settlement processes are called Secondary Settlement processes. Up to 16 Secondary Settlement processes can be configured in one logical network.

The POS-SETL-MASTER param in the LCONF specifies the Main Settlement process. The symbolic name of the Main Settlement process must be specified in this param. During initialization, each settlement process checks the POS-SETL-MASTER param in the LCONF. If its name is not in the param, that settlement process is a Secondary Settlement process.

Startup Logic

The Main Settlement process is designed to run continuously and should be assigned a startup logic of AUTOMATIC in the XPNET system. A startup logic of AUTOMATIC means that the process starts automatically when the XPNET node is initiated. The Main Settlement process restarts automatically should it ever fail.

The Secondary Settlement processes must have a startup logic of DEMAND in the XPNET system. A startup logic of DEMAND means that the Secondary Settlement process starts when it receives a startup message from the Main Settlement process.

Event Messages and Product Module ID

The BASE24-pos Main and Secondary Settlement processes generate event messages using a standard routing code of 2. Refer to the ***BASE24 Event Message Reference Guide*** for more information about event messages.

The settlement processes have a product module ID of PSSETL within the BASE24 system.

Initialization

The Main Settlement process initializes when the XPNET node is started. During its initialization, the Main Settlement process sends startup messages to the Secondary Settlement processes in the logical network. These startup messages cause the Secondary Settlement processes to begin initializing.

When an operator enters a SET PROCESS or RESET PROCESS command from a network control facility with the STARTOPTIONS attribute indicating the name of the Logical Network Configuration File (LCONF) used by the Main Settlement process at system start up, the XPNET process sends a startup message containing the name of the LCONF to the Main Settlement process. The Main Settlement process reads the name of the LCONF, reads the POS-SETL-MASTER param from the LCONF to verify that it is the Main Settlement process, and initializes as follows:

1. Reads the params and assigns it uses in processing from the LCONF and then closes the LCONF.

The required assigns and params are listed in section 5. Refer to the ***BASE24 Logical Network Configuration File Manual*** for a complete list of assigns and params each settlement process uses, descriptions of each assign and param, and possible valid values for each assign and param.

2. Builds the slave table using the POS-SLAVE-KEY param from the LCONF and verifies that each Secondary Settlement process has its own unique range of retailers.

If no Secondary Settlement processes are defined in the system, as in the case of issuer-only institutions, the Main Settlement process does not build a slave table. If Secondary Settlement processes are defined in the system and the Main Settlement process is unable to build the slave table, the Main Settlement process abends.

3. Initializes the network logging process. If the Main Settlement process is unable to initialize the network logging process, the Main Settlement process abends.
4. Determines the current and next business dates.

To do this, the Main Settlement process uses the current Compaq NonStop system date for the current business processing date and adds one to this date to determine the next business processing date.

5. Opens the Institution Definition File (IDF). If the Main Settlement process is unable to open the IDF, the process abends.

6. Forces yesterday's cutover to occur if the Main Settlement process went down before yesterday's cutover took place. If the next day's POS Transaction Log File (PTLF) does not exist, the Main Settlement process went down before yesterday's cutover.
7. Creates the POS Transaction Log File (PTLF) and PTLF alternate key file for the current and next business day.

The Main Settlement process names the files POYYMMDD and P0YYMMDD, P1YYMMDD, P2YYMMDD, P3YYMMDD, and P4YYMMDD as specified in the template of the PTLF assign. POYYMMDD is the PTLF; P0YYMMDD, P1YYMMDD, P2YYMMDD, P3YYMMDD, and P4YYMMDD are the alternate key files for the PTLF. The Main Settlement process creates one of each of these files for the current and next business dates.

8. Creates the POS Referral File (PRF) and the PRF alternate key file for the next business date and, if necessary, the current business date.

The Main Settlement process names the files FFYYMMDD and F0YYMMDD, as specified in the template of the PRF assign. FFYYMMDD is the name of the PRF and F0YYMMDD is the name of the alternate key file for the PRF. The Main Settlement process creates one of each of these files for both the current date and the next business date.

9. Initializes its timer tables. The Main Settlement process does not set any timers at this point.
10. Calculates the extended memory needed for the IDF and initializes its IDF extended memory table.
11. Sends startup messages, using the XPNET process, to each Secondary Settlement process, if they exist. The Main Settlement process also sets a 60 second startup timer during which all Secondary Settlement processes must perform their initialization steps and send startup complete messages back to the Main Settlement process.

When a Secondary Settlement process receives a startup message from the Main Settlement process it performs as follows:

- a. Reads the params and assigns it uses in processing from the LCONF and then closes the LCONF.
- b. Determines the current and next business dates.

To do this, the Secondary Settlement process uses the current date known to the system for the current date and adds one to this date to determine the next business date.

- c. Opens the IDF.

The Secondary Settlement process can read a locked IDF, allowing multiple Secondary Settlement processes to access the IDF simultaneously. If the Secondary Settlement process cannot access the IDF, the process abends.

- d. Opens the POS Retailer Definition File (PRDF). If the Secondary Settlement process is unable to open the PRDF, the process abends.
- e. Opens the BASE24-pos Terminal Data files. If the POS-PTDF assign is present in the LCONF, the process opens the POS Terminal Data File (PTDF). If the POS-PTD-DYN-GNRL assign is present in the LCONF, the process opens the BASE24-pos Terminal Data Dynamic File—general data (PTDD1). If the Secondary Settlement process is unable to open either the PTDF or PTDD1, the process abends.
- f. Opens the PTLF. If the Secondary Settlement process is unable to open the PTLF, the process abends.
- g. Initializes its timer tables. The Secondary Settlement process does not set any timers at this point.
- h. Reads the primary PRDF and retrieves the retailer range that identifies its partition. The Secondary Settlement process then closes the primary PRDF and opens its individual partition. If the Secondary Settlement process cannot open its partition, the process abends.
- i. Calculates the extended memory needed for its individual partition of the PRDF and for the PTDF if any Device Handler processes exist for an MHI device in host control mode.
- j. Closes its individual partition of the PRDF and opens the primary PRDF. If the Secondary Settlement process is unable to open the primary PRDF, the process abends.
- k. Forces retailers and terminals to cut over if they have not been cut over already.
- l. Allocates the extended memory needed for the IDF and PRDF and initializes its extended memory tables.
- m. Sets its timers.

The Secondary Settlement process sets the retailer cutover timers. If this is the Senior Secondary Settlement process, as determined by the POS-SETL-SENIOR-SLAVE param, it sets the retailer cutover timers and the alive timer.

- n. Sends a startup complete message containing its PPD name to the Main Settlement process using the XPNET process.

12. When the Main Settlement process receives a startup complete message, it opens each Secondary Settlement process with the PPD name it received in the startup complete message. Once the Secondary Settlement process is open, the Main Settlement process and the Secondary Settlement process can communicate directly with one another.

If the Main Settlement process has not received startup complete messages from all Secondary Settlement processes by the time the 60 second startup timer has expired, the Main Settlement process compares the number of times the 60 second startup timer has expired (the threshold counter) to the value in the POS-SETL-THRESHOLD param in the LCONF and performs according to the following:

- If the threshold counter is less than the value in the POS-SETL-THRESHOLD param, the Main Settlement process increases the threshold counter by 1 and sends another startup message to each of the Secondary Settlement processes that did not send startup complete messages. The Main Settlement process then sets another 60 second startup timer and performs step 12 again.
 - If the threshold counter is equal to the value in the POS-SETL-THRESHOLD param, the Main Settlement process sets its threshold counter to zero, logs a message indicating that the Main Settlement process is unable to start the Secondary Settlement processes, and waits for the amount of time indicated in the POS-SETL-WAIT-INTERVAL param in the LCONF. After waiting for this amount of time, the Main Settlement process performs step 11 again.
13. Once the Main Settlement process has received startup complete messages from all the Secondary Settlement processes, the Main Settlement process determines whether logical network cutover has passed for the current day. To do this, the Main Settlement process checks the current time against the POS-LN-CUTOVER param in the LCONF and performs according to the following situations:
 - If the logical network cutover time has not passed, the Main Settlement process checks the FIELD CUTOVER field on IDF screen 3 and the CUSTOMER PROCESSING DATE field on IDF screen 17. If the value in the FIELD CUTOVER field is 2 or 5 (clear the CAF usage or purge the UAF data at midnight) and the date in the CUSTOMER PROCESSING DATE field is less than or equal to the date in the CURRENT BUSINESS DATE field on IDF screen 17, the Main Settlement process takes the same action for the IDF record as it would upon expiration of the MIDNIGHT-CUTOVER-TIMER.

- If the logical network cutover time has passed, the Main Settlement process verifies that all IDF records have been cut over. If cutover has passed and an IDF record has not been cut over, the Main Settlement process performs institution cutover for these IDF records. Once all IDF records have been cut over, the Main Settlement process goes through its logical network cutover procedure and begins processing on the next date.
14. Sets its timers.

The Main Settlement process sets the financial institution cutover timer, the logical network cutover timer, the midnight timer, and the start reports timer.

Warmboot Processing

The BASE24-pos settlement processes can be warmbooted by issuing a warmboot command from a network control facility. Several warmboot commands are available to warmboot the settlement processes.

An operator must initiate a warmboot command when configuration changes have been made to the BASE24-pos system. The command that an operator uses to perform a warmboot depends on the specific configuration changes. An operator can warmboot:

- Only the Main Settlement process
- A specific Secondary Settlement process
- All Secondary Settlement processes
- Both the Main and Secondary Settlement processes

For more information about the warmboot commands, including detailed descriptions and the command syntax, refer to the ***BASE24 Text Command Reference Manual***.

Warmbooting the Main Settlement Process

The Main Settlement process must be warmbooted when an LCONF assign or param that only the Main Settlement process uses has been modified. For example, when the POS-SETL-PTLF-FILE-RETENTION or POS-SETL-LN-CUTOVER-LEAD-TIME param is changed, the Main Settlement process must be warmbooted.

When only the Main Settlement process is warmbooted, the Main Settlement process performs as it would during initialization except it:

- Does not send warmboot messages to the Secondary Settlement processes and, therefore, the Secondary Settlement processes do not go through a warmboot procedure.
- Does not rebuild the slave table.
- Does not retrieve a new LCONF name.
- Closes and reopens any files it opened during initialization.

Warmbooting One Secondary Settlement Process

A specific Secondary Settlement process must be warmbooted when a single PRDF or PTDD1 (if the terminal is an MHI device in host control mode) record has been modified from a file maintenance screen. Only the Secondary Settlement process responsible for the partition that contains the modified record must be warmbooted.

When a specific Secondary Settlement process is warmbooted, the Main Settlement process sends a warmboot command to the specified Secondary Settlement process. The Secondary Settlement process then performs as it would during initialization, except that it does not retrieve a new LCONF name and it closes and reopens any files it opened during initialization. The Main Settlement process does not perform any of its initialization or warmboot processing besides sending the warmboot message to the Secondary Settlement process.

Warmbooting All Secondary Settlement Processes

All Secondary Settlement processes must be warmbooted when an LCONF assign or param that all Secondary Settlement processes read has been modified. For example, when the POS-SETL-FI-CUTOVER-LEAD-TIME param is changed, all Secondary Settlement processes must be warmbooted.

When all Secondary Settlement processes are warmbooted, the Main Settlement process sends warmboot commands to all the Secondary Settlement processes that are listed in its slave table. Each Secondary Settlement process then performs as it would during initialization, except it does not retrieve a new LCONF name and it closes and reopens any files it opened during initialization. The Main Settlement process does not perform any of its initialization or warmboot processing besides sending warmboot messages to the Secondary Settlement processes.

Warmbooting the Main and Secondary Settlement Processes

Both the Main and Secondary Settlement processes must be warmbooted when the IDF has been modified, an LCONF assign or param used by both the Main and Secondary Settlement processes has been modified, or the slave table must be rebuilt due to a change in the high or low retailer key values for a Secondary Settlement process partition.

When both the Main and Secondary Settlement process are warmbooted, the processes perform as they would during initialization, except they do not retrieve a new LCONF name and they close and open any files that were opened during initialization.

Timer Processing

The BASE24-pos cutover and reporting process is controlled by timers. These timers are listed below, and the actions they trigger are described on the following pages.

- Retailer cutover timer
- Financial institution cutover timer
- Logical network cutover timer
- Midnight timer
- Start reports timer
- Alive timer

Retailer Cutover Timer

When the retailer cutover timer expires, the Secondary Settlement process begins retailer cutover. The Secondary Settlement process maintains a separate timer for each unique retailer cutover time in its partition. For example, if four retailers have the same cutover time, the Secondary Settlement process maintains only one retailer cutover timer for all four retailers.

The retailer cutover timer is set from the RETAILER CUTOVER/BALANCE END field on PRDF screen 1. The retailer must be cut over by the time indicated in this field.

All retailers that process on the same day as the logical network must have cutover times that precede financial institution cutover by at least the number of minutes specified in the POS-SETL-FI-CUTOVER-LEAD-TIME param in the LCONF. If the value in the PRDF does not allow for the time indicated in this param, the Secondary Settlement process sets the retailer cutover timer so that it meets the requirements.

All retailers that process one day ahead of the network must wait for a specified interval after logical network cutover before they can begin cutting over. This time interval is specified in the POS-SETL-TERM-CUTOVER-DELAY-TIME param. If the value in the PRDF does not allow for the time interval indicated in this param, the Secondary Settlement process sets the retailer cutover timer so that it meets the requirements.

When the retailer cutover timer expires, the Secondary Settlement process performs as follows:

1. Checks the TERM STATUS field on BASE24-pos Terminal Data files (PTD) screen 1 for each terminal associated with the retailer being cut over.
 - a. If the value in the TERM STATUS field is D (deactivated), the terminal has been deactivated and will not be cut over. The Secondary Settlement process continues with step 2.
 - b. If the value in the TERM STATUS field is A (activated) or P (pending deactivate), and the terminal has not been cut over to the new day, the Secondary Settlement process cuts the terminal over by performing the following steps:
 - 1) Creates terminal balancing records as indicated in the LOG TOTALS fields on PTD screen 3 and logs them to the PTLF. These balancing records can include the terminal totals for batch, shift, day, service, and network, or no totals need be logged at all.

Each record created and logged is given a RECORD-TYPE of 04 (administrative transaction) and contains the current transaction totals from the BASE24-pos Terminal Data files.
 - 2) Checks the TERMINAL CUTOVER field on PTD screen 3 to determine if the terminal is to be forced cut over by the Secondary Settlement process, forced cut over by the Device Handler module, or cut over by an operator-initiated cutover transaction (end-of-batch, end-of-shift, or end-of-day).

If the Secondary Settlement process or Device Handler module must force the terminal to cut over, the batch, shift, day, service, or network totals are cleared and the BALANCE FLAG field on PTD screen 4 is set to 9 (force balanced).

If the terminal must cut over when a balancing and cutover transaction is received from an operator but that transaction has not yet been received, the Secondary Settlement process forces the terminal to cut over. The Secondary Settlement process clears only the network totals and sets the BALANCE FLAG field on PTD screen 4 to 8 (force balance—incomplete batch).
 - 3) Increases the date in the POST DATE field on PTD screen 4 by one calendar day.
 - 4) If the TERM STATUS field is set to P and the batch, shift, daily, and network totals are all zeros, the Secondary Settlement process sets the TERM STATUS field to a value of D (deactivated).

2. Reads the next BASE24-pos Terminal Data files record, repeating step 1 until all terminals belonging to the retailer have been cutover.
3. Adds one calendar day to the dates in the POSTING DATES PREVIOUS, POSTING DATES CURRENT, and POSTING DATES NEXT fields on PRDF screen 1.

Financial Institution Cutover Timer

When the financial institution cutover timer expires, the Main Settlement process begins institution cutover. The Main Settlement process maintains a separate timer for each institution in the logical network.

An institution's cutover time is based on the institution's balancing window, specified in the BALANCE AND CUTOVER TIME WINDOW fields on IDF screen 17. The financial institution cutover timer expires at the end of the cutover window.

All institution cutovers must precede logical network cutover by at least the number of minutes specified in the POS-SETL-LN-CUTOVER-LEAD-TIME param in the LCONF. If the cutover time in the IDF does not allow for the time indicated in the POS-SETL-LN-CUTOVER-LEAD-TIME param, the Main Settlement process sets the financial institution cutover timer so that it meets the requirements.

When the financial institution cutover timer expires, the Main Settlement process:

1. Sends a message to all the Secondary Settlement processes to verify that all the retailers and their terminals have cut over.

When the Secondary Settlement process receives the message, it performs the following steps:

- a. Uses the FIID of the retailer to read the PRDF. If the retailer has been cut over, processing continues with step 2. If the retailer has not cut over, processing continues with step 1b.
- b. Uses the PTDF or PTDD1 alternate key of TERM-OWNER to locate the retailer's terminal records. The Secondary Settlement process then checks the POST DATE field on BASE24-pos Terminal Data files (PTD) screen 4 for each terminal belonging to the retailer.
 - If the terminal has been cut over to a new day or the TERM STATUS field is set to D (deactivated), the Secondary Settlement process moves on to the next BASE24-pos Terminal Data files record or, if there are no more BASE24-pos Terminal Data files records, the Secondary Settlement process continues with step 1c.
 - If the date in the POST DATE field indicates the terminal has not been cut over to a new day and the TERM STATUS field is set to either A (activated) or P (pending), the Secondary Settlement process cuts the terminal over as it does when the retailer cutover timer expires. The Secondary Settlement process then checks the

POST DATE field for the next terminal and repeats this step. If there are no more BASE24-pos Terminal Data files records, the Secondary Settlement process continues with step 1c.

- c. When all the terminals have been cutover for all of the Secondary Settlement process' retailers, the Secondary Settlement process sends an FIID reply message to the Main Settlement process.
2. When the Main Settlement process receives FIID reply messages from all the Secondary Settlement processes indicating that all the retailers that belong to the institution have been cut over, the Main Settlement process performs institution cutover.
3. Determines if a new customer business date should be computed.

If the value in the FIELD CUTOVER field on IDF screen 3 specifies to clear the UAF at logical network cutover, the Main Settlement process computes a new customer business date and places it in memory. In doing so, the Main Settlement process checks the WORK DAY CODE field on IDF screen 4 and performs according to the following situations:

- If the value in the WORK DAY CODE field is 0, the Main Settlement process increases the date in the CUSTOMER PROCESSING DATE field on IDF screen 17 by one day. This date should equal the date in the CURRENT BUSINESS DATE field on IDF screen 17 plus one calendar day.
 - If the value in the WORK DAY CODE field is not 0, the Main Settlement process calculates the date for the CUSTOMER PROCESSING DATE field on IDF screen 17 based on the value in the WORK DAY CODE field and the holidays listed in the HOLIDAYS fields on IDF screen 4. For example, if the institution is set up not to process on weekends or holidays, the customer business date is set to the next non-weekend, non-holiday date.
4. Purges data in the UAF, if required according to the value in the FIELD CUTOVER field on IDF screen 3.

To do this, the Main Settlement process verifies that a UAF is being used by checking the POS ROUTING TABLE fields on IDF screen 16. If the institution is using the Negative Authorization with Usage Accumulation method, a UAF exists, and the FIELD CUTOVER field is set to 1 or 2, then the Main Settlement process purges the data in the UAF. If the institution is not using the Negative Authorization with Usage Accumulation method, a UAF does not exist, and the FIELD CUTOVER field is set to 0, 4, or 5, then the Main Settlement process continues with step 5.

5. Advances the date in the REPORT DATE field on IDF screen 17 to the date in CURRENT BUSINESS DATE field on IDF screen 17.

6. Advances the date in the CURRENT BUSINESS DATE field to the date in NEXT BUSINESS DATE field.
7. Advances the date in the NEXT BUSINESS DATE field by one calendar day.
8. Establishes a new withdrawal period by resetting the BEGINNING DATE and NEXT BEGINNING DATE fields on IDF screen 4, if the value in the FIELD CUTOVER field on IDF screen 3 specifies anything other than to clear the UAF at midnight.

If the CUSTOMER PROCESSING DATE field on IDF screen 17 is equal to or greater than the date in the NEXT BEGINNING DATE field on IDF screen 4, the Main Settlement process must generate new withdrawal period dates. To do this, the Main Settlement process checks the WORK DAY CODE field on IDF screen 4 and performs according to the following situations:

- If the value in the WORK DAY CODE field is 0, the Main Settlement process computes a new next begin date based on the length of the institution's withdrawal period set in the PERIOD LENGTH field on IDF screen 4. It also checks the new next begin date against the date currently in the NEXT BEGINNING DATE field on IDF screen 4. If these dates are the same, the Main Settlement process continues with step 9.

If these dates are not the same, the Main Settlement process moves the date in the NEXT BEGINNING DATE field to the BEGINNING DATE field and sets the date in the NEXT BEGINNING DATE field to its new next begin date.

- If the value in the WORK DAY CODE field is not 0, the Main Settlement process moves the date in the NEXT BEGINNING DATE field to the BEGINNING DATE field. It then calculates the new next beginning date based on the value in the WORK DAY CODE field and the holidays listed in the HOLIDAYS field on IDF screen 4. For example, if the institution is set up not to process on weekends or holidays, the date in the NEXT BEGINNING DATE field is set to the next non-weekend or the next non-holiday date.
9. Writes the IDF record back to the file.

While the date fields have been changed, BASE24-pos processes do not reread them until logical network cutover. Consequently, the new dates take effect at logical network cutover, not at institution cutover.

Logical Network Cutover Timer

When the logical network cutover timer expires, the Main Settlement process begins logical network cutover.

The Main Settlement process sets its logical network cutover timer so that it expires at the time specified in the POS-LN-CUTOVER param in the LCONF. The Main Settlement process sets this timer at initialization and at logical network cutover.

When the logical network cutover timer expires, the Main Settlement process:

1. Purges old PTLFs in the system according to the POS-SETL-PTLF-FILE-RETENTION param in the LCONF.
2. Purges old PRFs in the system, if they exist, according to the POS-SETL-PTLF-FILE-RETENTION param in the LCONF.
3. Creates the day-after-next's PTLFs.

The Main Settlement process creates both a primary key file and the alternate key files for the day after next. The naming convention for the primary key PTLF is POYYMMDD, taken from the PTLF assign in the LCONF. The Main Settlement process gets the file's extent size from the PTLF template file. The naming convention for the alternate key PTLFs is PxYYMMDD, where *x* equals 0, 1, 2, 3, or 4. This naming convention is taken from the PTLF template file in the LCONF. The Main Settlement process also gets the file's extent size from this template file.

Note: Institutions must determine which alternate key files they will be using and configure only those alternate keys.)

The date (YYMMDD) that the Main Settlement process uses for creating these files is one day greater than the date in the NEXT BUSINESS DATE field on IDF screen 17 for the institution.

When these files are created, the Main Settlement process specifies in the file creation that ENSCRIBE is to create the PTLF alternate keys automatically. The POS-SETL-PTLF-EXTENTS param in the LCONF specifies the number of extents to use in creating the PTLF. The Main Settlement process retrieves this param at initialization.

4. Creates the day-after-next's PRFs, if needed.

The Main Settlement process creates both a primary key file and an alternate key file for the day after next. The naming convention for the primary key PRF is FFYYMMDD, taken from the PRF assign in the LCONF. The Main Settlement process gets the file's extent size from the PRF template file. The

naming convention for the alternate key PRF is F0YYMMDD, taken from the PRF template file in the LCONF. The Main Settlement process also gets the file's extent size from this template file.

The date (YYMMDD) that the Main Settlement process uses for creating these files is one day greater than the date in the NEXT BUSINESS DATE field on IDF screen 17 for the institution.

5. Broadcasts a cutover (9502) message to all Secondary Settlement processes and to those processes specified in the POS-SETL-NOTIFY params in the LCONF. The Main Settlement process reads the LCONF at initialization and maintains the broadcast list and slave table in memory.

The cutover message contains the current date, which is the day just ending. The Main Settlement process uses this message to notify each process to go through its cutover procedures (e.g., close files, open files, reinitialize internal memory for the new day).

When each Secondary Settlement process receives a cutover message from the Main Settlement process it:

- a. Closes today's PTLF and opens the next day's PTLF.
 - b. Increases the current and next business dates by one day.
 - c. Sets the retailer cutover timer.
 - d. Returns a message to the Main Settlement process indicating that the Secondary Settlement process has completed its logical network cutover procedures.
6. Purges the old obey files used for running the BASE24-pos report programs. The Main Settlement process purges these files according to the POS-SETL-RPT-FILE-RETENTION param in the LCONF.
 7. Creates the TACL obey file that begins POS reports, if reports are to be run.

The Main Settlement process also names the obey file according to the value in the POS-CARD-OBEYFILE assign. This assign contains the naming convention for the obey file created by the Main Settlement process. The obey file runs the report programs named in the following assigns:

POS-SETL-RPT-PROG-1-NAME
POS-SETL-RPT-PROG-2-NAME
POS-SETL-RPT-PROG-4-NAME
POS-SETL-RPT-PROG-5-NAME
POS-SETL-RPT-PROG-30-NAME
POS-SETL-RPT-PROG-35-NAME

8. Deallocates the unused file extents for the PTLF just closed.

9. Generates the event message “Cutover completed for YYMMDD.”
10. Resets the following timers:
 - Financial institution cutover timer
 - Logical network cutover timer
 - Midnight timer
 - Start reports timer

Midnight Timer

When the midnight timer expires, the Main Settlement process purges the data in the Usage Accumulation File (UAF) if required and clears the usage accumulation fields in the Cardholder Account File (CAF). Only one midnight timer exists for the entire logical network. If one or more institutions are set up to clear their usage accumulation at midnight instead of at logical network cutover, the Main Settlement process sets the midnight timer.

The Main Settlement process sets the midnight timer at initialization or logical network cutover if both of the following conditions exist:

- The value in the FIELD CUTOVER field on IDF screen 3 equals 2 (purge UAF at midnight; begin clearing CAF usage accumulation fields at midnight) or 5 (do not purge the UAF; begin clearing the CAF usage accumulation fields at midnight).
- The date in the CUSTOMER PROCESSING DATE field on IDF screen 17 is less than or equal to the current calendar date.

When the midnight timer expires, the Main Settlement process reads the first record in the IDF and performs the following steps:

1. Determines whether midnight processing is necessary. To do this, the Main Settlement process performs the following steps:
 - a. Verifies that the UAF name is present in the IDF. If so, the Main Settlement process assumes the institution uses the Negative Authorization with Usage Accumulation method.
 - b. Checks the FIELD CUTOVER field on IDF screen 3. If that field contains a value of 2 or 5 and the date in the CUSTOMER PROCESSING DATE field on IDF screen 17 is earlier than or equal to the date in the CURRENT BUSINESS DATE field on IDF screen 17, the Main Settlement process continues with step 2. Otherwise, the Main Settlement process continues with step 4.
2. Updates the CUSTOMER PROCESSING DATE field on IDF screen 17. To do this, the Main Settlement process performs the following steps:
 - a. Computes a new customer business date by checking the WORK DAY CODE field on IDF screen 4 and performing according to the following situations:
 - If the value in the WORK DAY CODE field is 0, the Main Settlement process increases the date in the CUSTOMER PROCESSING DATE field in the IDF by one.

- If the value in the WORK DAY CODE field is not 0, the Main Settlement process calculates the customer processing date based on the value in the WORK DAY CODE field and the holidays listed in the HOLIDAYS fields on IDF screen 4. For example, if the institution is set up not to process on weekends or holidays, the customer processing date is set to the next non-weekend, non-holiday date.
 - b. Moves the newly computed customer processing date to the CUSTOMER PROCESSING DATE field on IDF screen 17.
- 3. Determines whether the data in the UAF must be purged for the institution. To do this, the Main Settlement process performs the following steps:
 - a. Verifies that the newly computed customer processing date is equal to or greater than the date in the NEXT BEGINNING DATE field on IDF screen 4. If so, the Main Settlement process continues with step 3b. If not, the Main Settlement process continues with step 4.
 - b. Opens the UAF using the filename retrieved in step 1 and purges the data in the file.
 - c. Generates a new withdrawal period according to the following situations:
 - If the value in the WORK DAY CODE field is 0, the Main Settlement process computes the new beginning date based on the length of the institution's withdrawal period set in the PERIOD LENGTH field on IDF screen 4. It also checks the new next beginning date against the date currently in the NEXT BEGINNING DATE field on IDF screen 4. If these dates are the same, the Main Settlement process proceeds with step 4.

If these dates are not the same, the Main Settlement process moves the date in the NEXT BEGINNING DATE field to the BEGINNING DATE field. It also places the new next beginning date in the NEXT BEGINNING DATE field.
 - If the value in the WORK DAY CODE field is not 0, the Main Settlement process moves the date in the NEXT BEGINNING DATE field to the BEGINNING DATE field. The new date for the NEXT BEGINNING DATE field is calculated based on the value in the WORK DAY CODE field and the holidays listed in the HOLIDAY fields in the IDF. For example, if the institution is set up not to process on weekends or holidays, the date in the NEXT BEGINNING DATE field would be set to the next non-weekend or next non-holiday date.
- 4. Reads the next IDF and repeats steps 1 through 3.

Start Reports Timer

When the start reports timer expires, the Main Settlement process begins producing reports. The Main Settlement process sets the start reports timer at logical network cutover. The timer is set to expire at the time specified in the POS-SETL-REPORTS param in the LCONF.

The reports produced when this timer expires are:

- Daily Network Settlement Report (B24POS01)
- Daily Cardholder Activity Report (B24POS02)
- Daily Cardholder Exception Report (B24POS02-EXPT)
- Daily Retailer Summary Report (B24POS11)
- Daily Retailer Activity Report (B24POS12)
- Daily Retailer Exception Detail Report (B24POS12-EXPT)
- Daily Clerk Totals Report (B24POS04)
- Daily Disputed Transaction Report (B24POS05)
- Daily Referral Statistical Report (B24POS30)
- Daily Retailer Reconciliation Report (B24POS35)

When the start reports timer expires, the Main Settlement process:

1. Retrieves the name of the obey file that runs the reports from the POS-CARD-OBEYFILE assign in the LCONF.
2. Runs the obey file.

This obey file starts the BASE24-pos report programs that actually create the BASE24-pos reports. The report programs send the formatted reports to a spooler location.

3. Delays 30 seconds and generates an event message indicating that the report programs are running. If the report programs are not running, the Main Settlement process generates an event message with this information.

Alive Timer

When the alive timer expires, the Senior Secondary Settlement process sends an *alive* message to the Main Settlement process. The *alive* message verifies that the Main Settlement process is running. The Senior Secondary Settlement process sets the alive timer when it receives a startup message from the Main Settlement process at initialization. The timer is set to expire at the time interval specified in the POS-SETL-ALIVE-TIMEOUT param in the LCONF.

When the alive timer expires, the Senior Secondary Settlement process:

1. Attempts to open the Main Settlement process if the Main Settlement process is not open already.
2. Sends an *alive* message to the Main Settlement process and sets an acknowledgment timer. The length of this acknowledgment timer is always 60 seconds.
3. Waits for a protocol-level response from the Main Settlement process and processes as follows:
 - If the Senior Secondary Settlement process receives a protocol-level response from the Main Settlement process, the Main Settlement process is alive. Processing continues with the next step.
 - If the Senior Secondary Settlement process does not receive a protocol-level response from the Main Settlement process, the Senior Secondary Settlement process closes the Main Settlement process and logs a message indicating that the Senior Secondary Settlement process was unable to send an “alive” message to the Main Settlement process. Processing then continues with the next step.
4. Sets another alive timer based on the POS-SETL-ALIVE-TIMEOUT param.

Secondary Settlement Process Cutover Timer

When the Secondary Settlement process cutover (9502) timer expires the Main Settlement process verifies that all Secondary Settlement processes have cut over. To verify that the Secondary Settlement processes have cut over, the Main Settlement process performs as follows:

1. Checks the slave table in memory to verify that all Secondary Settlement processes have cut over.
2. Sends a cutover (9502) message to each Secondary Settlement process which, according to the slave table, has not been cut over.
3. Resets the Secondary Settlement process cutover (9502) timer.

Text Commands

The Main and Secondary Settlement processes can receive and process several text commands sent using a network control facility. Because the responsibilities of cutover, settlement, and reporting are divided between the Main Settlement process and one or more Secondary Settlement processes, operators must be aware of which text commands can be sent to the Main Settlement process and which text commands can be sent to the Secondary Settlement processes.

All text commands can be sent to the Main Settlement process. The Main Settlement process can also receive commands intended for the Secondary Settlement processes and pass them on to the Secondary Settlement processes. Only certain text commands can be sent and processed by the Secondary Settlement processes.

The text commands that the Main and Secondary Settlement processes can receive are listed below. Each command has several variations, each of which is intended for either the Main Settlement process or the Secondary Settlement process. The command name variations for each command and the processes that can use each variation are listed in a table following each command description. For more information about each command, including command variations, descriptions, command syntax, and command processing, refer to the ***BASE24 Text Command Reference Manual***. Commands in the ***BASE24 Text Command Reference Manual*** are documented according to the command name variations listed here.

Create Log Files — Causes the Main Settlement process to create either the PTLF or PRF as specified in the command.

Command Variations	Process Use
CREATEPTLF	Main
CREATEPRF	Main

Display IDF — Causes the Main Settlement process to display the FIID, cutover end time, current business date, and customer business date for each record in the IDF. The syntax for this command is DUMPIDF.

Display IDF Extended Memory Table — Causes the Main and Secondary Settlement processes to display the contents of their IDF extended memory tables. The slave table contains the FIID, cutover time, indicators of which totals are sent

to the host and logged to the PTLF, and the symbolic name of the Host Interface process. The Main Settlement process table contains the FIID, cutover time, unique flag, midnight flag, query flag, and cutover flag.

Command Variations	Process Use
DUMPIDFTBL	All Secondary (process-generated by Main)
DUMPIDFTBLALL	Main
DUMIDFTBL SECONDARY	Individual Secondary

Display PRDF — Displays the retailer ID, FIID, end of retailer window, and PRDF posting date for each PRDF record.

Command Variations	Process Use
DUMPPRDF	All Secondary (process-generated by Main)
DUMPPRDFALL	Main
DUMPPRDF SECONDARY	Individual Secondary

Display PRDF Extended Memory Table — Displays the contents of the PRDF extended memory table. This table contains the retailer ID, FIID, cutover time, unique flag, and cutover flag.

Command Variations	Process Use
DUMPPRDFTBL	All Secondary (process-generated by Main)
DUMPPRDFTBLALL	Main
DUMPPRDFTBL SECONDARY	Individual Secondary

Display PTD — Displays the terminal ID, retailer ID, Device Handler process name, deactivate flag value, and posting date for each BASE24-pos Terminal Data files record within a Secondary Settlement process' retailer range.

Command Variations	Process Use
DUMPPTD	All Secondary (process-generated by Main)
DUMPPTDALL	Main
DUMPPTD SECONDARY	Individual Secondary

Display PTD Extended Memory Table — Displays the contents of the BASE24-pos Terminal Data files extended memory table if the records in the BASE24-pos Terminal Data files are in host control mode for the Merchant Host Interface (MHI) Device Handler process. This table contains the retailer ID and the name of the Device Handler process.

Command Variations	Process Use
DUMPPTDTBL	All Secondary (process-generated by Main)
DUMPPTDTBLALL	Main
DUMPPTDTBL SECONDARY	Individual Secondary

Display Slave Table — Causes the Main Settlement process to display the contents of the slave table. The slave table contains the Secondary Settlement process symbolic name, Secondary Settlement process PPD name, process number, low retailer key, high retailer key, and status flag. The syntax for this command is DUMPSLAVETBL.

Display Timers — Causes either the Main or Secondary Settlement processes to display the timer type, expiration date and time, and the user buffer for each entry in the timer table.

Command Variations	Process Use
DUMPTIMERTBL	Main or all Secondary (process-generated by Main)

Command Variations	Process Use
DUMPTIMERTBLALL	Main
DUMPTIMERTBL SECONDARY	Individual Secondary

Display Times — Causes the Secondary Settlement process to display the POS cutover time, institution cutover lead time, terminal cutover lead time, current business date, next business date, and trace flag. Causes the Main Settlement process to display the POS cutover time, logical network cutover lead time, report time, current business date, next business date, and trace flag.

Command Variations	Process Use
DUMPTIMES	Main or all Secondary (process-generated by Main)
DUMPTIMESALL	Main
DUMPTIMES SECONDARY	Individual Secondary

Dump Notify Tables — Causes the Main Settlement process to display the contents of the logical network cutover table, midnight table, and MHI notify table. The syntax for this command is DUMPNOTIFY.

Midnight Processing — Causes the Main Settlement process to perform midnight processing as if the MIDNIGHT-TIMER had expired. The syntax for this command is MIDNIGHT.

Reset Dates — Resets the PRDF, BASE24-pos Terminal Data files, or IDF dates to a specified date.

Command Variations	Process Use
RESETIDFDATES	Main
RESETRDFDATES	All Secondary (process-generated by Main)
RESETPTDDATES	All Secondary (process-generated by Main)

Command Variations	Process Use
RESETALLDATES	Main and all Secondary (process-generated by Main)

Reset PRDF End of Window — Causes the Secondary Settlement process to reset the RETAILER CUTOVER/BALANCE END field on PRDF screen 1.

Command Variations	Process Use
RESETRDFTIMESALL	All Secondary (process-generated by Main)
RESETRDFTIMES	Individual Secondary

Resync — Causes the Main Settlement process to resync the IDF. This command also causes the Secondary Settlement processes to resync their retailers and terminals. The syntax for this command is RESYNC.

Trace — Causes the BASE24-pos system to generate event messages for the following actions: terminal cutover, retailer cutover, institution cutover, purging the UAF, resyncing the institutions and retailers, sending a logical network cutover message, sending a MHI notify message, sending totals to the Host Interface process, sending PTLF information to the Authorization process, and purging files.

Command Variations	Process Use
TRACEON	Main or Secondary
TRACEOFF	Main or Secondary

Warmboot — Causes the Main or Secondary Settlement processes to perform various reinitialization tasks.

Command Variations	Process Use
WARMBOOT ALL	Main and all Secondary (process-generated by Main)
WARMBOOT MAIN	Main

Command Variations	Process Use
WARMBOOT NOTIFY	Main
WARMBOOTRDFKEY	All Secondary (process-generated by Main)
WARMBOOT SECONDARY	Individual Secondary

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B: Recovery Procedures

This appendix includes recovery procedures that must be followed if the Main Settlement process or Secondary Settlement processes go down during the business day. The procedure that must be performed depends on the time at which the Main Settlement process or Secondary Settlement process went down.

Downed Secondary Settlement Process

The scenarios described below involve a downed Secondary Settlement process. Recovery procedures follow each scenario.

Scenario 1

The Secondary Settlement process has a startup logic of DEMAND in the XPNET system; however, the Secondary Settlement process does not start when it receives a startup message from the Main Settlement process.

Recovery Steps

The Main Settlement process attempts to send a startup message to the Secondary Settlement process three times. If this limit is exceeded, the Secondary Settlement process is marked down and can no longer be started automatically by the Main Settlement process.

To recover, start the downed Secondary Settlement process from a network control facility. Warmbooting is not necessary.

Scenario 2

The Secondary Settlement process goes down and is restarted before logical network cutover.

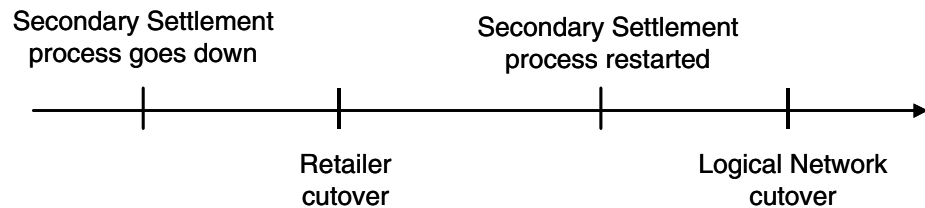


Recovery Steps

To recover, start the downed Secondary Settlement process from a network control facility. Warmbooting is not necessary.

Scenario 3

The Secondary Settlement process goes down before retailer cutover and logical network cutover, and is restarted after the retailer cutover window but before logical network cutover.

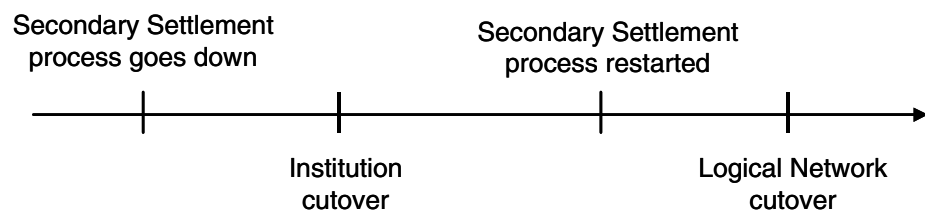


Recovery Steps

To recover, start the downed Secondary Settlement process from a network control facility. Warmbooting is not necessary.

Scenario 4

The Secondary Settlement process goes down before institution cutover and logical network cutover. It is restarted after institution cutover should have occurred but before logical network cutover.

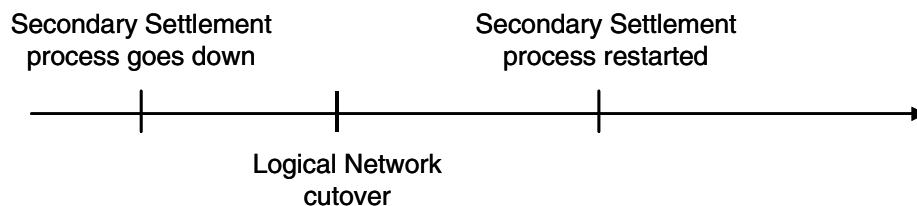


Recovery Steps

To recover, start the downed Secondary Settlement process from a network control facility. Then, send a WARMBOOT ALL (9501****) command to the Main Settlement process.

Scenario 5

The Secondary Settlement process goes down before logical network cutover and is restarted after logical network cutover.

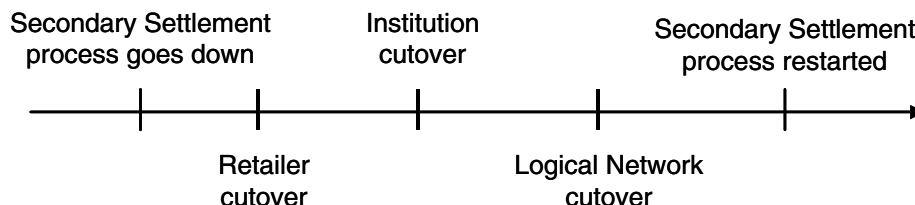


Recovery Steps

To recover, start the downed Secondary Settlement process from a network control facility. Warmbooting is not necessary.

Scenario 6

The Secondary Settlement process goes down before retailer cutover, institution cutover, and logical network cutover. It is restarted after logical network cutover.



Recover Steps

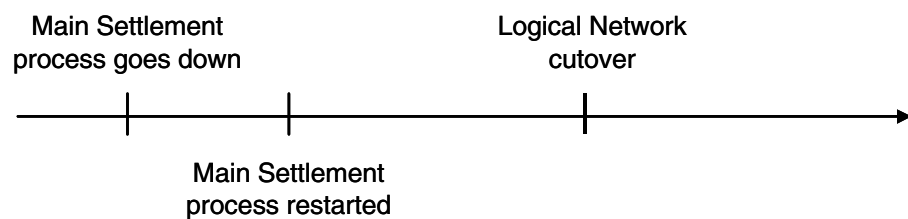
To recover, start the downed Secondary Settlement process from a network control facility. Then, send a WARMBOOT ALL (9501****) command to the Main Settlement process.

Downed Main Settlement Process

The scenarios described below involve a downed Main Settlement process. Recovery procedures follow each scenario.

Scenario 1

The Main Settlement process goes down and is restarted before logical network cutover.

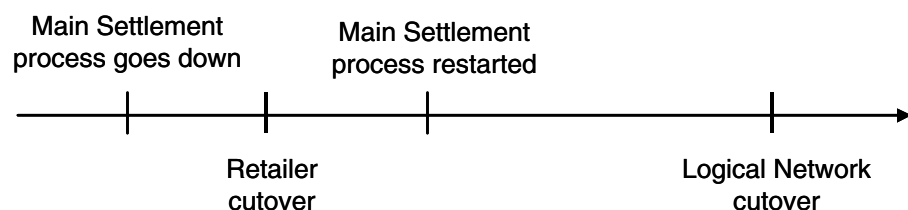


Recovery Steps

Because the Main Settlement process has a startup logic of AUTOMATIC, the XPNET process should automatically restart the process. If the Main Settlement process does not restart automatically, start the Main Settlement process from a network control facility.

Scenario 2

The Main Settlement process goes down before retailer cutover and logical network cutover, and is restarted after the end of the retailer cutover window but before logical network cutover.

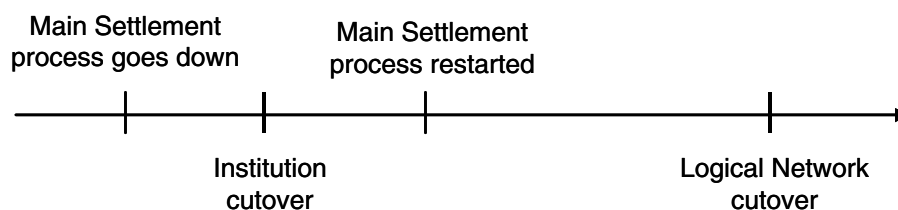


Recovery Steps

To recover, start the Main Settlement process from a network control facility. Warmbooting is not necessary.

Scenario 3

The Main Settlement process goes down before institution cutover and logical network cutover. It is restarted after institution cutover should have occurred but before logical network cutover.

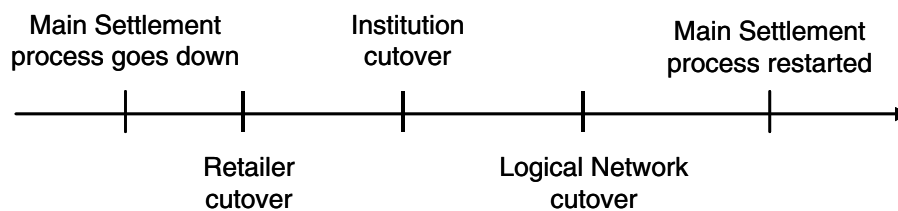


Recovery Steps

To recover, start the Main Settlement process from a network control facility. Warmbooting is not necessary.

Scenario 4

The Main Settlement process goes down before retailer cutover, institution cutover, and logical network cutover. It is restarted after logical network cutover.



Recovery Steps

To recover, start the Main Settlement process from a network control facility. Warmbooting is not necessary.

C: BASE24-pos Terminal Data Files Totals

This appendix provides examples of how BASE24-pos Terminal Data files totals are increased, written to the PTLF, and cleared in various terminal balancing and cutover situations.

One factor that guides how the BASE24-pos product handles BASE24-pos Terminal Data files totals is the terminal's cutover option established in the TERMINAL CUTOVER field on BASE24-pos Terminal Data files (PTD) screen 3. The value in this field indicates how and by what transaction, if any, the terminal must cut over each day. The value in the TERMINAL CUTOVER field also indicates whether the terminal is balanced at cutover.

Another factor that guides how the BASE24-pos product handles BASE24-pos Terminal Data files totals is the balance support option established in the BALANCE SUPPORT field on PTD screen 3. The value in this field identifies the types of balancing transactions the terminal supports. The BALANCE SUPPORT field indicates which BASE24-pos Terminal Data files totals are cleared at terminal cutover.

A third factor that guides how the BASE24-pos product handles BASE24-pos Terminal Data files totals is the multiple currency totaling option established in the MC TOTALLING TYPE field on PTD screen 2. The value in this field indicates whether transaction amounts are converted from the transaction currency to the primary terminal currency before they are stored and accumulated in the BASE24-pos Terminal Data files. This option is available only when the BASE24-pos Multiple Currency add-on product is installed. The examples in this appendix do not depict currency conversion. For detailed information on currency conversion processing, refer to the *BASE24-pos Multiple Currency Support Manual*.

Note: When a terminal is cut over by a balancing transaction initiated at the terminal and BASE24-pos Terminal Data files totals are to be cleared, only the BASE24-pos Terminal Data files totals on the screen are cleared. BASE24-pos stores the BASE24-pos Terminal Data files totals on disk until it receives another transaction from the terminal. This is a precautionary measure to prevent out-of-

balance situations. For example, if totals are cleared from disk and sent to the terminal for balancing, the totals could be lost if communications go down between the BASE24-pos system and the terminal. When the terminal does not receive the totals, the operator sends a duplicate balancing transaction. The BASE24-pos system receives this duplicate balancing transaction but cannot provide totals because they were cleared. Holding the totals on disk enables the BASE24-pos system to resend the totals in these situations.

Terminal Cutover Options

The following pages contain examples of how the BASE24-pos product handles BASE24-pos Terminal Data files totals based on the terminal cutover option and the types of balancing transactions the terminal supports. The cutover options are:

- Network force cutover

The Secondary Settlement process always forces the terminal to cut over at the end of the retailer's cutover window. Close batch, close shift, and close day transactions do not affect terminal cutover.

- Device Handler force cutover

The Device Handler module forces the terminal to cut over when it receives the first financial transaction initiated within the Device Handler cutover window. If the Device Handler module does not receive a financial transaction within the Device Handler cutover window, the Secondary Settlement process forces the terminal to cut over at the end of the retailer's cutover window.

- Close batch transaction

The Device Handler module cuts the terminal over when it receives the first close batch transaction initiated within the retailer's cutover window. If the Device Handler module does not receive a close batch transaction within the retailer's cutover window, the Secondary Settlement process forces the terminal to cut over at the end of the retailer's cutover window.

- Close shift transaction

The Device Handler module cuts the terminal over when it receives the first close shift transaction initiated within the retailer's cutover window. If the Device Handler module does not receive a close shift transaction within the retailer's cutover window, the Secondary Settlement process forces the terminal to cut over at the end of the retailer's cutover window. Close shift transactions, when entered, must be immediately preceded by a close batch transaction. If not, the Device Handler module denies the close shift transaction.

- Close day transaction

The Device Handler module cuts the terminal over when it receives the first close day transaction initiated within the retailer's cutover window. If the Device Handler module does not receive a close day transaction within the retailer's cutover window, the Secondary Settlement process forces the terminal to cut over at the end of the retailer's cutover window. Close day transactions, when entered, must be immediately preceded by a close shift

transaction. If not, the Device Handler module denies the close day transaction. Close day transactions should be entered only once during a reporting day to maintain reporting integrity.

Note: Throughout this appendix, service totals are described as being logged to the PTLF in a single record. This is true when 17 services or less are defined in the BASE24-pos Terminal Data files. However, if more than 17 services are defined in the BASE24-pos Terminal Data files, an additional service total record is logged in the PTLF. Refer to section 2 for more information about BASE24-pos Terminal Data files service totals.

Example 1 - Network Force Cutover

The following example describes how BASE24-pos Terminal Data files totals are handled when the Secondary Settlement process is to force a terminal to cut over each day (TERMINAL CUTOVER field on PTD screen 3 is set to a value of 3). In this case, close batch, close shift, and close day transactions do not affect terminal cutover. The terminal in this example does not support any balancing transactions and all totals in the BASE24-pos Terminal Data files are cleared when the Secondary Settlement process forces the terminal to cut over (BALANCE SUPPORT field on PTD screen 3 is set to a value of 9).

The example takes a three-day period, with transactions being posted each day, and describes how the BASE24-pos Terminal Data files totals are accumulated, written to the PTLF, and cleared.

Note	Actions
1	Terminal cutover is forced. a) Service totals, batch totals, shift totals, daily totals, and current network totals records are logged to the PTLF according to the LOG TOTALS fields on PTD screen 3. b) Current network totals are moved to previous totals. c) Batch, shift, daily, service, and current network totals are cleared.
2	Terminal cutover is forced. a) Service totals, batch totals, shift totals, daily totals, and current network totals records are logged to the PTLF according to the LOG TOTALS fields on PTD screen 3. b) Current network totals are moved to previous totals. c) Batch, shift, daily, service, and current network totals are cleared.
3	Terminal cutover is forced. a) Service totals, batch totals, shift totals, daily totals, and current network totals records are logged to the PTLF according to the LOG TOTALS fields on PTD screen 3. b) Current network totals are moved to previous totals. c) Batch, shift, daily, service, and current network totals are cleared.

The Actions column in this table describes the Events column in the following table.

Note	Events	Batch	BASE24-pos Terminal Data Files Terminal Totals				
			Batch	Shift	Daily	Net	Prev
1	Day One	1					
	\$10 tran		\$10	\$10	\$10	\$10	\$00
	\$25 tran		\$35	\$35	\$35	\$35	\$00
	\$15 tran		\$50	\$50	\$50	\$50	\$00
	\$42 tran		\$92	\$92	\$92	\$92	\$00
	\$20 tran		\$112	\$112	\$112	\$112	\$00
	Force cutover		\$00	\$00	\$00	\$00	\$112
2	Day Two	1					
	\$6 tran		\$6	\$6	\$6	\$6	\$112
	\$30 tran		\$36	\$36	\$36	\$36	\$112
	\$15 tran		\$51	\$51	\$51	\$51	\$112
	\$17 tran		\$68	\$68	\$68	\$68	\$112
	\$10 tran		\$78	\$78	\$78	\$78	\$112
	Force cutover		\$00	\$00	\$00	\$00	\$78
3	Day Three	1					
	\$10 tran		\$10	\$10	\$10	\$10	\$78
	\$15 tran		\$25	\$25	\$25	\$25	\$78
	\$30 tran		\$55	\$55	\$55	\$55	\$78
	\$25 tran		\$80	\$80	\$80	\$80	\$78
	\$20 tran		\$100	\$100	\$100	\$100	\$78
	\$10 tran		\$110	\$110	\$110	\$110	\$78
	Force cutover		\$00	\$00	\$00	\$00	\$110

Example 2 - Device Handler Force Cutover

The following example describes how BASE24-pos Terminal Data files totals are handled when the Device Handler module is to force a terminal to cut over each day (TERMINAL CUTOVER field on PTD screen 3 is set to a value of 4). The terminal in this example does not support any balancing transactions and all totals in the BASE24-pos Terminal Data files are cleared when the Device Handler module forces the terminal to cut over (BALANCE SUPPORT field on PTD screen 3 is set to a value of 9).

The example takes a three-day period, with transactions being posted each day, and describes how the BASE24-pos Terminal Data files totals are accumulated, written to the PTLF, and cleared.

Note	Actions
1	Terminal cutover is forced. a) Service totals, batch totals, shift totals, daily totals, and current network totals records are logged to the PTLF if configured to be done in the LOG TOTALS fields on PTD screen 3. b) Current network totals are moved to previous totals. c) Batch, shift, daily, service, and current network totals are cleared. d) The \$10 financial transaction that caused the terminal to cut over is posted to the new day.
2	Terminal cutover is forced. a) Service totals, batch totals, shift totals, daily totals, and current network totals records are logged to the PTLF is configured to be done in the LOG TOTALS fields on PTD screen 3. b) Current network totals are moved to previous totals. c) Batch, shift, daily, service, and current network totals are cleared. d) The \$10 financial transaction that caused the terminal to cut over is posted to the new day.

The Actions column in this table describes the Events column in the following table.

Note	Events	Batch	BASE24-pos Terminal Data Files Terminal Totals				
			Batch	Shift	Daily	Net	Prev
1	Day One	1					
	\$10 tran		\$10	\$10	\$10	\$10	\$00
	\$25 tran		\$35	\$35	\$35	\$35	\$00
	\$15 tran		\$50	\$50	\$50	\$50	\$00
	\$42 tran		\$92	\$92	\$92	\$92	\$00
	\$20 tran		\$112	\$112	\$112	\$112	\$00
	Force cutover		\$10	\$10	\$10	\$10	\$112
2	Day Two	1					
	\$6 tran		\$6	\$6	\$6	\$6	\$112
	\$30 tran		\$36	\$36	\$36	\$36	\$112
	\$15 tran		\$51	\$51	\$51	\$51	\$112
	\$17 tran		\$68	\$68	\$68	\$68	\$112
	\$10 tran		\$78	\$78	\$78	\$78	\$112
	Force cutover		\$10	\$10	\$10	\$10	\$78

Note	Actions
3	Terminal cutover is forced. a) Service totals, batch totals, shift totals, daily totals, and current network totals records are logged to the PTLF if configured to be done in the LOG TOTALS fields on PTD screen 3. b) Current network totals are moved to previous totals. c) Batch, shift, daily, service, and current network totals are cleared. d) The \$10 financial transaction that caused the terminal to cut over is posted to the new day.

The Actions column in this table describes the Events column in the following table.

Note	Events	Batch	BASE24-pos Terminal Data Files Terminal Totals				
			Batch	Shift	Daily	Net	Prev
3	Day Three	1					
	\$10 tran		\$10	\$10	\$10	\$10	\$78
	\$15 tran		\$25	\$25	\$25	\$25	\$78
	\$30 tran		\$55	\$55	\$55	\$55	\$78
	\$25 tran		\$80	\$80	\$80	\$80	\$78
	\$20 tran		\$100	\$100	\$100	\$100	\$78
	\$10 tran		\$110	\$110	\$110	\$110	\$78
	Force cutover		\$10	\$10	\$10	\$10	\$110

Example 3 - Close-Batch Cutover with Close-Batch Balancing

The following example describes how BASE24-pos Terminal Data files totals are handled when a terminal is set up to be cut over using close batch transactions (i.e., the TERMINAL CUTOVER field on PTD screen 3 is set to a value of 2).

The terminal in this example supports close-batch balancing transactions (the BALANCE SUPPORT field on PTD screen 3 is set to a value of 0).

The example takes a three-day period, with transactions posting each day, and describes how the BASE24-pos Terminal Data files totals are accumulated, written to the PTLF, and cleared. In day two of the example, the Device Handler module does not receive a close batch transaction and, therefore, forces the terminal to cut over.

Note	Actions
1	A close batch transaction is received outside the retailer window, closing batch 1. a) Service and batch totals records are logged to the PTLF if configured to be done in the LOG TOTALS fields on PTD screen 3. b) Batch and service totals are cleared.
2	A close batch transaction is received within the retailer cutover window, closing batch 2 and cutting over the terminal to a new day. a) Service, batch, and current network totals records are logged to the PTLF if configured to be done in the LOG TOTALS fields on PTD screen 3. b) Current network totals are moved to previous totals. c) Batch, shift, daily, service and current network totals are cleared.
3	A close batch transaction is received outside the retailer window, closing batch 3. a) Service and batch totals records are logged to the PTLF if configured to be done in the LOG TOTALS fields on PTD screen 3. b) Batch and service totals are cleared.

The Actions column in this table describes the Events column in the following table.

Note	Events	Batch	BASE24-pos Terminal Data Files Terminal Totals				
			Batch	Shift	Daily	Net	Prev
1	Day One	1					
	\$10 tran		\$10	\$10	\$10	\$10	\$00
	\$25 tran		\$35	\$35	\$35	\$35	\$00
	\$15 tran		\$50	\$50	\$50	\$50	\$00
	Close batch tran		\$00	\$50	\$50	\$50	\$00
	\$42 tran		\$42	\$92	\$92	\$92	\$00
	\$20 tran		\$62	\$112	\$112	\$112	\$00
2	Close batch tran	2	\$00	\$00	\$00	\$00	\$112
3	Day Two	3					
	\$6 tran		\$6	\$6	\$6	\$6	\$112
	\$30 tran		\$36	\$36	\$36	\$36	\$112
	Close batch tran		\$00	\$36	\$36	\$36	\$112

Note	Actions
4	No close batch transaction is received within the retailer cutover window, so terminal cutover is forced. a) Service, batch (incomplete), shift, daily, and current network totals records are logged to the PTLF if configured to be done in the LOG TOTALS fields on PTD screen 3. b) Current network totals are moved to previous totals. c) Current network totals are cleared (because the BASE24-pos reporting day is ending).
5	A close batch transaction is received outside the retailer window, closing batch 4. a) Service and batch totals records are logged to the PTLF if configured to be done in the LOG TOTALS fields on PTD screen 3. b) Batch and service totals are cleared.
6	A close batch transaction is received outside the retailer window, closing batch 5. a) Service and batch totals records are logged to the PTLF if configured to be done in the LOG TOTALS fields on PTD screen 3. b) Batch and service totals are cleared.
7	A close batch transaction is received within the retailer cutover window, closing batch 6 and cutting over the terminal to a new day. a) Service, batch, and current network totals records are logged to the PTLF if configured to be done in the LOG TOTALS fields on PTD screen 3. b) Current network totals are moved to previous totals. c) Batch, shift, daily, service and current network totals are cleared.

The Actions column in this table describes the Events column in the following table.

Note	Events	Batch	BASE24-pos Terminal Data Files Terminal Totals				
			Batch	Shift	Daily	Net	Prev
4	Day Two <i>continued</i>	4					
	\$15 tran		\$15	\$51	\$51	\$51	\$112
	\$17 tran		\$32	\$68	\$68	\$68	\$112
	\$10 tran		\$42	\$78	\$78	\$78	\$112
	Force cutover		\$42	\$78	\$78	\$00	\$78
5	Day Three	4					
	\$10 tran		\$52	\$88	\$88	\$10	\$78
	Close batch tran		\$00	\$88	\$88	\$10	\$78
	\$15 tran		\$15	\$103	\$103	\$25	\$78
	\$30 tran		\$45	\$133	\$133	\$55	\$78
6	\$25 tran	5	\$70	\$158	\$158	\$80	\$78
	Close batch tran		\$00	\$158	\$158	\$80	\$78
	\$20 tran		\$20	\$178	\$178	\$100	\$78
	\$10 tran		\$30	\$188	\$188	\$110	\$78
	Close batch tran		\$00	\$00	\$00	\$00	\$110
7		6					

Example 4 - Close-Batch Cutover with Batch, Shift, and Day Balancing

The following example describes how BASE24-pos Terminal Data files totals are handled when a terminal is set up to be cut over using close batch transactions (i.e., the **TERMINAL CUTOVER** field on PTD screen 3 is set to a value of 2). The terminal in this example supports close batch, close shift, and close day balancing transactions (**BALANCE SUPPORT** field on PTD screen 3 is set to a value of 3).

The example takes a three-day period, with transactions being posted each day, and describes how the BASE24-pos Terminal Data files totals are accumulated, written to the PTLF, and cleared. Note that in day two of the example, a close batch transaction is not entered and the system forces the terminal to cut over.

Note	Actions
1	A close batch transaction is received, closing batch 1. a) Service and batch totals records are logged to the PTLF if configured to be done in the LOG TOTALS fields on PTD screen 3. b) Batch and service totals are cleared.
2	A close shift transaction is received, closing the shift. a) Shift totals record is logged to the PTLF if configured to be done in the LOG TOTALS fields on PTD screen 3. b) Shift totals are cleared.
3	A close batch transaction is received within the retailer cutover window, closing batch 2 and cutting the terminal over to a new day. a) Service, batch, and current network totals records are logged to the PTLF if configured to be done in the LOG TOTALS fields on PTD screen 3. b) Current network totals are moved to previous totals. c) Batch, service, and current network totals are cleared.
4	A close shift transaction is received, closing the shift. a) Shift totals record is logged to the PTLF if configured to be done in the LOG TOTALS fields on PTD screen 3. b) Shift totals are cleared.

The Events column in the following table is described by the Actions column in the table on the preceding page.

Note	Events	Batch	BASE24-pos Terminal Data Files Terminal Totals				
			Batch	Shift	Daily	Net	Prev
1	Day One	1					
	\$15 tran		\$10	\$10	\$10	\$10	\$00
	\$25 tran		\$35	\$35	\$35	\$35	\$00
	\$10 tran		\$50	\$50	\$50	\$50	\$00
	Close batch tran		\$00	\$50	\$50	\$50	\$00
	Close shift tran		\$00	\$00	\$50	\$50	\$00
	\$20 tran		\$20	\$20	\$70	\$70	\$00
	\$22 tran		\$42	\$42	\$92	\$92	\$00
	\$11 tran		\$53	\$53	\$103	\$103	\$00
	Close batch tran		\$00	\$53	\$103	\$00	\$103
2	Close shift tran	2	\$00	\$00	\$103	\$00	\$103
3	Close batch tran		\$00	\$00	\$103	\$00	\$103
4	Close shift tran						

Note	Actions
5	A close day transaction is received, closing the day. a) Daily totals record is logged to the PTLF if configured to be done in the LOG TOTALS field on PTD screen 3. b) Daily totals are cleared.
6	A close batch transaction is received, closing batch 3. a) Service and batch totals records are logged to the PTLF if configured to be done in the LOG TOTALS fields on PTD screen 3. b) Batch and service totals are cleared.

Note	Actions
7	A close shift transaction is received, closing the shift. a) Shift totals record is logged to the PTLF if configured to be done in the LOG TOTALS field on PTD screen 3. b) Shift totals are cleared.
8	No close batch transaction is received within the retailer cutover window, so terminal cutover is forced. a) Service, batch, shift, daily, and current network totals records are logged to the PTLF if configured to be done in the LOG TOTALS fields on PTD screen 3. b) Current network totals are moved to previous totals. c) Current network totals are cleared because the BASE24-pos business day is ending.

The Actions column in this table describes the Events column in the following table.

Note	Events	Batch	BASE24-pos Terminal Data Files Terminal Totals				
			Batch	Shift	Daily	Net	Prev
5	Day One <i>continued</i>						
	Close day tran		\$00	\$00	\$00	\$00	\$103
6	Day Two						
	\$14 tran		\$14	\$14	\$14	\$14	\$103
	\$32 tran		\$46	\$46	\$46	\$46	\$103
	\$8 tran		\$54	\$54	\$54	\$54	\$103
	Close batch tran	3	\$00	\$54	\$54	\$54	\$103
	Close shift tran		\$00	\$00	\$54	\$54	\$103
	\$15 tran		\$15	\$15	\$69	\$69	\$103
	\$17 tran		\$32	\$32	\$86	\$86	\$103
7	\$10 tran		\$42	\$42	\$96	\$96	\$103

Note	Events	Batch	BASE24-pos Terminal Data Files Terminal Totals				
			Batch	Shift	Daily	Net	Prev
8	Force cutover		\$42	\$42	\$96	\$00	\$96

Note	Actions
9	A close batch transaction is received, closing batch 4. a) Service and batch totals records are logged to the PTLF if configured to be done in the LOG TOTALS fields on PTD screen 3. b) Batch and service totals are cleared.
10	A close batch transaction is received within the retailer cutover window, closing batch 5 and cutting the terminal over to a new day. a) Service, batch, and current network totals records are logged to the PTLF if configured to be done in the LOG TOTALS fields on PTD screen 3. b) Current network totals are moved to previous totals. c) Batch, service, and current network totals are cleared.
11	A close shift transaction is received, closing the shift. a) Shift totals record is logged to the PTLF if configured to be done in the LOG TOTALS fields on PTD screen 3. b) Shift totals are cleared.
12	A close day transaction is received, closing the day. a) Daily totals record is logged to the PTLF if configured to be done in the LOG TOTALS fields on PTD screen 3. b) Daily totals are cleared.

The Actions column in this table describes the Events column in the following table.

Note	Events	Batch	BASE24-pos Terminal Data Files Terminal Totals				
			Batch	Shift	Daily	Net	Prev
9	Day Three	4					
	\$17 tran		\$59	\$59	\$113	\$17	\$96
	\$20 tran		\$79	\$79	\$133	\$37	\$96
	\$15 tran		\$94	\$94	\$148	\$52	\$96
	\$5 tran		\$99	\$99	\$153	\$57	\$96
	Close batch tran		\$00	\$99	\$153	\$57	\$96
	\$20 tran		\$20	\$119	\$173	\$77	\$96
10	\$10 tran	5	\$30	\$129	\$183	\$87	\$96
	Close batch tran		\$00	\$129	\$183	\$00	\$87
	Close shift tran		\$00	\$00	\$183	\$00	\$87
12	Close day tran		\$00	\$00	\$00	\$00	\$87

Example 5 - Close-Day Cutover with Batch, Shift, and Day Balancing

The following example describes how BASE24-pos Terminal Data files totals are handled when a terminal is set up to be cut over using close day transactions (i.e., the TERMINAL CUTOVER field on PTD screen 3 is set to a value of 0).

The terminal supports close batch, close shift, and close day balancing transactions in this example (the BALANCE SUPPORT field on PTD screen 3 is set to a value of 3).

The example takes a three-day period, with transactions being posted each day, and describes how the BASE24-pos Terminal Data files totals are accumulated, written to the PTLF, and cleared. Note that in day two of the example, a close day transaction is not received and the system forces the terminal to cut over.

Note	Actions
1	A close batch transaction is received, closing batch 1. a) Service and batch totals records are logged to the PTLF if configured to be done in the LOG TOTALS fields on PTD screen 3. b) Batch and service totals are cleared.
2	A close batch transaction is received, closing batch 2. a) Service and batch totals records are logged to the PTLF if configured to be done in the LOG TOTALS fields on PTD screen 3. b) Batch and service totals are cleared.
3	A close shift transaction is received, closing the shift. a) Shift totals record is logged to the PTLF if configured to be done in the LOG TOTALS fields on PTD screen 3. b) Shift totals are cleared.
4	A close batch transaction is received, closing batch 3. a) Service and batch totals records are logged to the PTLF if configured to be done in the LOG TOTALS fields on PTD screen 3. b) Batch and service totals are cleared.

The Actions column in this table describes the Events column in the following table.

Note	Events	Batch	BASE24-pos Terminal Data Files Terminal Totals				
			Batch	Shift	Daily	Net	Prev
1	Day One	1					
	\$10 tran		\$10	\$10	\$10	\$10	\$00
	\$30 tran		\$40	\$40	\$40	\$40	\$00
	\$20 tran		\$60	\$60	\$60	\$60	\$00
	Close batch tran		\$00	\$60	\$60	\$60	\$00
2	\$42 tran	2	\$42	\$102	\$102	\$102	\$00
	\$20 tran		\$62	\$122	\$122	\$122	\$00
	Close batch tran		\$00	\$122	\$122	\$122	\$00
3	Close shift tran	2	\$00	\$00	\$122	\$122	\$00
	\$15 tran		\$15	\$15	\$137	\$137	\$00
	\$12 tran		\$27	\$27	\$149	\$149	\$00
	\$11 tran		\$38	\$38	\$160	\$160	\$00
4	\$10 tran	3	\$48	\$48	\$170	\$170	\$00
	Close batch tran		\$00	\$48	\$170	\$170	\$00

Note	Actions
5	A close shift transaction is received, closing the shift. a) Shift totals record is logged to the PTLF if configured to be done in the LOG TOTALS fields on PTD screen 3. b) Shift totals are cleared.
6	A close day transaction is received within the retailer cutover window, closing the daily totals and cutting over the terminal to a new day. a) Daily and current network totals records are logged to the PTLF if configured to be done in the LOG TOTALS fields on PTD screen 3. b) Current network totals are moved to previous totals. c) Daily and current network totals are cleared.
7	A close batch transaction is received, closing batch 4. a) Service and batch totals records are logged to the PTLF if configured to be done in the LOG TOTALS fields on PTD screen 3. b) Batch and service totals are cleared.
8	A close shift transaction is received, closing the shift. a) Shift totals record is logged to the PTLF if configured to be done in the LOG TOTALS fields on PTD screen 3. b) Shift totals are cleared.
9	A close batch transaction is received, closing batch 5. a) Service and batch totals records are logged to the PTLF if configured to be done in the LOG TOTALS fields on PTD screen 3. b) Batch and service totals are cleared.
10	A close shift transaction is received, closing the shift. a) Shift totals record is logged to the PTLF if configured to be done in the LOG TOTALS fields on PTD screen 3. b) Shift totals are cleared.

The Actions column in this table describes the Events column in the following table.

Note	Events	Batch	BASE24-pos Terminal Data Files Terminal Totals				
			Batch	Shift	Daily	Net	Prev
	Day One <i>continued</i>						
5	Close shift tran		\$00	\$00	\$170	\$170	\$00
6	Close day tran		\$00	\$00	\$00	\$00	\$170
	Day Two						
	\$8 tran		\$8	\$8	\$8	\$8	\$170
	\$12 tran		\$20	\$20	\$20	\$20	\$170
	\$18 tran		\$38	\$38	\$38	\$38	\$170
7	Close batch tran	4	\$00	\$38	\$38	\$38	\$170
8	Close shift tran		\$00	\$00	\$38	\$38	\$170
	\$12 tran		\$12	\$12	\$50	\$50	\$170
	\$5 tran		\$17	\$17	\$55	\$55	\$170
9	Close batch tran	5	\$00	\$17	\$55	\$55	\$170
10	Close shift tran		\$00	\$00	\$55	\$55	\$170

Note	Actions
11	No close day transaction is received within the retailer cutover window, so the terminal cutover is forced. a) Service, batch (incomplete), shift, daily, and current network totals records are logged to the PTLF if configured to be done in the LOG TOTALS fields on PTD screen 3. b) Current network totals are moved to previous totals. c) Current network totals are cleared because the BASE24-pos reporting day is ending.
12	A close batch transaction is received, closing batch 6. a) Service and batch totals records are logged to the PTLF if configured to be done in the LOG TOTALS fields on PTD screen 3. b) Batch and service totals are cleared.
13	A close shift transaction is received, closing the shift. a) Shift totals record is logged to the PTLF if configured to be done in the LOG TOTALS fields on PTD screen 3. b) Shift totals are cleared.
14	A close batch transaction is received, closing batch 7. a) Service and batch totals records are logged to the PTLF if configured to be done in the LOG TOTALS fields on PTD screen 3. b) Batch and service totals are cleared.
15	A close shift transaction is received, closing the shift. a) Shift totals record is logged to the PTLF if configured to be done in the LOG TOTALS fields on PTD screen 3. b) Shift totals are cleared.
16	A close day transaction is received within the retailer cutover window, closing the daily totals and cutting over the terminal to a new day. a) Daily and current network totals records are logged to the PTLF if configured to do so in the LOG TOTALS fields on PTD screen 3. b) Current network totals are moved to previous totals. c) Daily and current network totals are cleared.

The Actions column in this table describes the Events column in the following table.

Note	Events	Batch	BASE24-pos Terminal Data Files Terminal Totals				
			Batch	Shift	Daily	Net	Prev
11	Day Two <i>continued</i>						
	\$17 tran		\$17	\$17	\$72	\$72	\$170
	\$10 tran		\$27	\$27	\$82	\$82	\$170
	Force cutover		\$27	\$27	\$82	\$00	\$82
12	Day Three						
	\$10 tran		\$37	\$37	\$92	\$10	\$82
	\$15 tran		\$52	\$52	\$107	\$25	\$82
	\$18 tran		\$70	\$70	\$125	\$43	\$82
12	Close batch tran	6	\$00	\$70	\$125	\$43	\$82
13	Close shift tran		\$00	\$00	\$125	\$43	\$82
	\$20 tran		\$20	\$20	\$145	\$63	\$82
	\$10 tran		\$30	\$30	\$155	\$73	\$82
14	Close batch tran	7	\$00	\$30	\$155	\$73	\$82
15	Close shift tran		\$00	\$00	\$155	\$73	\$82
16	Close day tran		\$00	\$00	\$00	\$00	\$73

Index

A

ACI Commerce Gateway product support, [1-27](#)
Alive timer, [1-23](#)

B

Balancing transactions
 close batch transaction, [2-8](#)
 close day transaction, [2-9](#)
 close shift transaction, [2-8](#)
 subtotal transaction, [2-8](#)
BASE24-pos report generation
 by Main Settlement process, [6-2](#)
 manually by operator, [6-4](#)
 suppressing, [6-4](#)
BASE24-pos Terminal Data Dynamic File—general data (PTDDI), [1-19](#), [2-2](#)
BASE24-pos Terminal Data files (PTD)
 clerk totals, [4-20](#)
 description, [1-18](#)
 terminal cutover, [1-10](#)
BASE24-pos Terminal Data files totals
 close-batch cutover with batch, shift, and day balancing example, [C-14](#)
 close-batch cutover with close-batch balancing example, [C-10](#)
 close-day cutover with batch, shift, and day balancing example, [C-19](#)
 Device Handler force cutover example, [C-7](#)
 multiple currency totaling options, [C-1](#)
 network force cutover example, [C-5](#)
 terminal cutover options, [C-3](#)
Batch
 definition, [2-7](#)
 incomplete batches, [2-24](#)
 out-of-balance batches, [2-26](#)
Broadcast notification, [1-28](#)
Business day
 institution cutover, [1-9](#), [1-13](#)
 interchange cutover, [1-14](#)
 logical network cutover, [1-9](#), [1-14](#)
 retailer cutover, [1-9](#), [1-13](#)
 terminal balancing, [1-10](#), [2-2](#)
 terminal cutover, [1-9](#), [1-10](#)

C

Card issuers
 definition, [1-7](#)
 posting cardholder accounts, [3-6](#)
 settlement, [3-6](#)
 settling with acquirers, [3-6](#)
 settling with retailer sponsors, [3-8](#)
Cardholders, definition, [1-4](#)
Chargeback
 definition, [1-4](#), [4-24](#)
 report for disputed transactions, [4-24](#)
Clearing BASE24-pos Terminal Data files totals
 APACS 30/40 devices, [2-18](#)
 BALANCE SUPPORT field, [2-16](#)
 Hypercom devices, [2-18](#)
 MHI devices, [2-19](#)
 NCR/NDP devices, [2-18](#)
 SPDH devices, [2-19](#)
 TERMINAL CUTOVER field, [2-16](#)
 Visa I devices, [2-17](#)
 Visa II devices, [2-18](#)
Clearings Statement Report (SETL01)
 definition, [1-26](#)
 description of, [4-71](#)
 field descriptions, [4-74](#)
 sample of, [4-71](#)
Clerk totals, [2-4](#)
CLERK TOTALS FLAG field
 clerk totals reporting, [4-20](#)
 terminal totals, [2-4](#)
COBNAMES table, [5-22](#)
Configuration
 assigns and params, [5-4](#)
 institution cutover, [5-12](#)
 interchange cutover, [5-14](#)
 logical network cutover, [5-13](#)
 Main Settlement process, [5-6](#), [A-2](#)
 reports, [5-16](#)
 retailer cutover, [5-11](#)
 Secondary Settlement process, [5-7](#), [A-2](#)
 Senior Secondary Settlement process, [5-8](#)
 terminal cutover, [5-9](#)
CRT Administration, [4-24](#)
CRT Authorization, [4-47](#)

Cutover

- assigns and params, [5-4](#)
- configuration, [5-9](#)
- Device Handler cutover window, [2-12](#)
- financial institution cutover timer, [1-22](#)
- hierarchy diagram, [1-10](#)
- institution cutover, [1-9](#), [1-13](#), [5-12](#), [A-15](#)
- interchange cutover, [1-9](#), [1-14](#), [5-14](#)
- logical network cutover, [1-9](#), [1-14](#), [5-13](#), [A-18](#)
- logical network cutover timer, [1-22](#)
- midnight timer, [1-22](#)
- of intelligent terminals, [2-12](#)
- of nonintelligent terminals, [2-12](#)
- process configuration, [5-2](#)
- retailer cutover, [1-9](#), [1-13](#), [1-22](#), [5-11](#), [A-12](#)
- retailer cutover window, [2-12](#)
- Secondary Settlement process cutover, [A-25](#)
- Secondary Settlement process cutover timer, [1-23](#)
- start reports timer, [1-23](#)
- terminal cutover, [1-9](#), [1-10](#), [2-11](#), [5-9](#)
- timer processing, [A-12](#)

Cutover timers

- financial institution cutover timer, [A-15](#)
- logical network cutover timer, [A-18](#)
- retailer cutover timer, [A-12](#)
- Secondary Settlement process cutover timer, [A-25](#)

Cutover transactions

- close batch transaction, [2-14](#)
- close day transaction, [2-13](#)
- close shift transaction, [2-14](#)

D**Daily Cardholder Activity Report (B24POS02)**

- definition, [1-25](#)
- description of, [4-7](#)
- field descriptions, [4-10](#), [4-15](#)
- print location, [4-7](#)
- sample of, [4-7](#), [4-14](#)

Daily Cardholder Exception Report (B24POS02-EXPT)

- definition, [1-25](#)
- description of, [4-18](#)
- print location, [4-19](#)
- sample of, [4-19](#)

Daily Clerk Totals Report (B24POS04)

- definition, [1-25](#)
- description of, [4-20](#)
- field descriptions, [4-22](#)
- print location, [4-20](#)
- sample of, [4-20](#)

Daily Disputed Transactions Report (B24POS05)

- definition, [1-25](#)
- description of, [4-24](#)
- field descriptions, [4-26](#)
- print location, [4-24](#)
- sample of, [4-24](#)
- use in crediting retailers, [3-2](#)

Daily Network Settlement Report (B24POS01)

- definition, [1-25](#)
- description of, [4-3](#)
- field descriptions, [4-5](#)
- print location, [4-3](#)
- sample of, [4-3](#)
- use in settlement, [3-7](#), [3-8](#)

Daily reports

- Daily Cardholder Activity Report (B24POS02), [1-25](#), [4-7](#)
- Daily Cardholder Exception Report (B24POS02-EXPT), [1-25](#), [4-18](#)
- Daily Clerk Totals Report (B24POS04), [1-25](#), [4-20](#)
- Daily Disputed Transactions Report (B24POS05), [1-25](#), [3-2](#), [4-24](#)
- Daily Network Settlement Report (B24POS01), [1-25](#), [3-4](#), [3-7](#), [3-8](#), [4-3](#)
- Daily Retailer Activity Report (B24POS12), [1-25](#), [3-2](#), [4-34](#)
- Daily Retailer Exception Detail Report (B24POS12-EXPT), [1-25](#), [4-45](#)
- Daily Retailer Summary Report (B24POS11), [1-25](#), [3-2](#), [4-29](#)
- Network Referral Statistical Report (B24POS30), [1-26](#), [4-47](#)
- Retailer Reconciliation Report (B24POS35), [1-26](#), [2-22](#), [3-2](#), [4-54](#)

Daily Retailer Activity Report (B24POS12)

- definition, [1-25](#)
- description of, [4-34](#)
- field descriptions, [4-38](#), [4-43](#)
- print location, [4-34](#)
- sample of, [4-35](#), [4-42](#)
- use in crediting retailers, [3-2](#)

Daily Retailer Exception Detail Report (B24POS12-EXPT)

- definition, [1-25](#)
- description of, [4-45](#)
- print location, [4-46](#)
- sample of, [4-46](#)

Daily Retailer Summary Report (B24POS11)

- definition, [1-25](#)
- description of, [4-29](#)
- field descriptions, [4-32](#)
- print location, [4-29](#)
- sample of, [4-29](#)
- use in crediting retailers, [3-2](#)

Database

- BASE24-pos Terminal Data Dynamic File—general data (PTDD1), [1-19](#), [2-2](#)
- BASE24-pos Terminal Data files, [1-10](#)
- Institution Definition File (IDF), [1-13](#), [1-19](#)
- Interchange Log File (ILF), [1-8](#), [1-18](#), [4-71](#), [4-76](#)
- Log Message Configuration File (LMCF), [1-19](#)
- Logical Network Configuration File (LCONF), [1-19](#), [5-2](#)
- POS Institution Reporting File (PIRF), [1-17](#), [4-63](#)
- POS Referral File (PRF), [1-19](#), [4-47](#)
- POS Retailer Definition File (PRDF), [1-11](#), [1-13](#), [1-18](#), [5-2](#)

Database *continued*

- POS Retailer Reporting File (PRRF), [1-18](#), [4-67](#)
- POS Terminal Data File (PTDF), [1-19](#), [2-2](#)
- POS Transaction Log File (PTLF), [1-2](#), [1-4](#), [1-6](#), [1-7](#), [1-8](#), [1-9](#), [1-14](#), [1-16](#), [4-3](#), [4-7](#), [4-18](#), [4-29](#), [4-34](#), [4-54](#), [5-16](#)
- Usage Accumulation File (UAF), [1-19](#), [1-22](#)

Day, definition, [2-7](#)

Device Handler cutover window

- defining, [1-11](#)
- illustration, [1-11](#)
- use in retailer cutover, [2-12](#)

Device Handler module force cutover, [2-15](#)

Draft capture options

- authorization and draft capture, [2-29](#)
- authorization only, [2-28](#)
- authorization with electronic follow-up, [2-29](#)
- terminal-defined draft capture, [2-30](#)

Draft capture transaction

- crediting retailers, [3-2](#)
- description, [1-5](#)
- in settlement, [2-28](#)

E

Exception processing

- Daily Cardholder Exception Report (B24POS02-EXPT), [1-7](#), [4-18](#)
- Daily Retailer Exception Detail Report (B24POS12-EXPT), [1-6](#), [4-45](#)
- transactions resulting in, [4-18](#), [4-45](#)

F

Financial institution cutover timer, [1-22](#)

I

Institution cutover

- configuration, [5-12](#)
- description, [1-13](#)
- financial institution cutover timer, [1-22](#), [A-15](#)

Institution Definition File (IDF)

- description, [1-19](#)
- institution cutover, [1-13](#)

Interchange cutover

- configuration, [5-14](#)
- description, [1-14](#)
- Enhanced Interchange Configuration File (ICFE), [5-14](#)
- Interchange Configuration File (ICF), [5-14](#)

Interchange Log File (ILF)

- Clearings Statement Report, [4-71](#)
- description, [1-18](#)
- interchange reports, [1-8](#)
- Transaction Detail Report, [4-76](#)

Interchange report generation

- by Interchange Interface process, [6-11](#)
- manually by operator, [6-11](#)
- suppressing, [6-12](#)

Interchange reports

- Clearings Statement Report (SETL01), [1-26](#), [4-71](#)
- generation of, [6-11](#)
- Transaction Detail Report (STAT09), [1-26](#), [4-76](#)

Interchanges

- definition, [1-7](#)
- functional types in settlement, [3-4](#)
- settlement, [3-10](#)

L

Log Message Configuration File (LMCF), [1-19](#)

Log messages, routing code, [A-3](#)

Logical Network Configuration File (LCONF)

- description, [1-19](#)
- for ACI Commerce Gateway product support, [1-27](#)
- process configuration, [A-2](#)
- process configuration options, [5-2](#)

Logical network cutover

- configuration, [5-13](#)
- description, [1-14](#)
- Logical Network Configuration File (LCONF), [5-13](#)
- logical network cutover timer, [1-22](#), [A-18](#)
- start reports timer, [1-23](#)

Logical network cutover timer, [1-22](#)

Logical network settlement, [3-8](#)

M

Main Settlement process

- alive timer, [1-23](#)
- communication with Secondary, [1-20](#)
- initialization, [A-4](#)
- log messages, [A-3](#)
- number defined, [1-20](#)
- params, [5-6](#)
- product module ID, [A-3](#)
- recovery procedures, [B-5](#)
- responsibilities, [1-20](#)
- warmboot processing, [A-9](#)

Midnight timer

- introduction to, [1-22](#)
- processing for, [A-21](#)

Movement of funds, [1-3](#)

N

Network participants

- card issuers, [1-7](#), [3-6](#), [3-8](#)
- cardholders, [1-4](#)
- defining, [1-4](#)
- interchanges, [1-7](#), [3-4](#), [3-10](#)
- retailer sponsors, [1-6](#), [3-4](#), [3-8](#)
- retailers, [1-5](#)
- terminals, [1-5](#)

Network Referral Statistical Report (B24POS30)
definition, [1-26](#)
description of, [4-47](#)
field descriptions, [4-49](#), [4-53](#)
print location, [4-47](#)
sample of, [4-47](#), [4-52](#)

Non-draft capture transaction
crediting retailers, [3-2](#)
description, [1-5](#)
in settlement, [2-28](#)

O

Obey file
creation of, [6-2](#)
example of, [6-3](#)
naming convention, [6-2](#)

Operations
automatic BASE24-pos report generation, [6-2](#)
automatic interchange report generation, [6-11](#)
manual BASE24-pos report generation, [6-4](#)
manual interchange report generation, [6-11](#)
suppressing BASE24-pos reports, [6-4](#)
suppressing interchange reports, [6-12](#)

Operator-initiated terminal cutover, [1-12](#)

P

Periodic Institution Summary Report (B24POS21)
definition, [1-26](#)
description of, [4-63](#)
field descriptions, [4-65](#)
print location, [4-63](#)
sample of, [4-63](#)

Periodic reports
Periodic Institution Summary Report
(B24POS21), [1-26](#), [4-63](#)
Periodic Retailer Summary Report
(B24POS22), [1-26](#), [4-67](#)
POS Report Generator (PRG), [4-63](#), [4-67](#)

Periodic Retailer Summary Report (B24POS22)
definition, [1-26](#)
description of, [4-67](#)
field descriptions, [4-69](#)
print location, [4-67](#)
sample of, [4-67](#)

Point-of-sale devices, [1-5](#)

POS Institution Reporting File (PIRF)
description, [1-17](#)
Periodic Institution Summary Report, [4-63](#)

POS Referral File (PRF)
description, [1-19](#)
Network Referral Statistical Report, [4-47](#)

POS Report Generator (PRG) screen
Periodic Institution Summary Report, [4-63](#)
Periodic Retailer Summary Report, [4-67](#)

POS Retailer Definition File (PRDF)
description, [1-18](#)
partitioning, [5-2](#)
retailer cutover, [1-13](#)
retailer cutover window, [1-11](#)

POS Retailer Reporting File (PRRF)
description, [1-18](#)
Periodic Retailer Summary Report, [4-67](#)

POS Terminal Data File (PTDF)
description, [1-19](#)
terminal totals, [2-2](#)

POS Transaction Log File (PTLF)
alternate keys, [5-16](#)
contents, [1-16](#)
creation, [1-16](#)
deletion, [1-16](#)
extraction, [1-16](#)
multiple currency impacts, [1-17](#)
perusal, [1-16](#)
use in creating reports, [4-3](#), [4-7](#), [4-18](#), [4-20](#), [4-29](#),
[4-34](#), [4-45](#), [4-54](#)

PTD
see BASE24-pos Terminal Data files (PTD)

PTDD1
see BASE24-pos Terminal Data Dynamic File—
general data (PTDD1)

PTLF keys for reports
card number key, [5-16](#)
retailer ID key, [5-16](#)
retailer ID/clerk ID key, [5-16](#)
terminal ID key, [5-16](#)
terminal ID/retailer ID key, [5-16](#)

R

Report configuration
COBNAMEs table, [5-22](#)
Enhanced Interchange Configuration File
(ICFE), [5-22](#)
Institution Definition File (IDF), [5-21](#)
Interchange Configuration File (ICF), [5-22](#)
Logical Network Configuration File (LCONF), [5-17](#)
POS Retailer Definition File (PRDF), [5-21](#)

Report obey files
creation of, [6-2](#)
example of, [6-3](#)
naming convention, [6-2](#)

Report programs, [6-11](#)

Report types
daily reports, [1-24](#)
interchange reports, [1-24](#)
periodic reports, [1-24](#)
settlement reports, [1-24](#)
statistical reports, [1-24](#)

Reporting operations
automatic BASE24-pos report generation, [6-2](#)
automatic interchange report generation, [6-11](#)
manual BASE24-pos report generation, [6-4](#)
manual interchange report generation, [6-11](#)

- Reporting operations *continued*
 suppressing BASE24-pos reports, 6-4
 suppressing interchange reports, 6-12
- Reports
 Clearings Statement Report (SETL01), 1-26, 4-71
 configuration of, 5-16
 Daily Cardholder Activity Report (B24POS02), 1-25, 4-7
 Daily Cardholder Exception Report (B24POS02-EXPT), 1-25, 4-18
 Daily Clerk Totals Report (B24POS04), 1-25, 4-20
 Daily Disputed Transactions Report (B24POS05), 1-25, 3-2, 4-24
 Daily Network Settlement Report (B24POS01), 1-25, 3-4, 3-7, 3-8, 4-3
 Daily Retailer Activity Report (B24POS12), 1-25, 3-2, 4-34
 Daily Retailer Exception Detail Report (B24POS12-EXPT), 1-25, 4-45
 Daily Retailer Summary Report (B24POS11), 1-25, 3-2, 4-29
 interchange reports, 6-11
 Network Referral Statistical Report (B24POS30), 1-26, 4-47
 Periodic Institution Summary Report (B24POS21), 1-26, 4-63
 periodic reports, 6-5
 Periodic Retailer Summary Report (B24POS22), 1-26, 4-67
 producing BASE24-pos daily reports, 6-2
 report obey files, 6-2, 6-3
 report programs, 6-11
 Retailer Reconciliation Report (B24POS35), 1-26, 2-22, 3-2, 4-54
 timer processing, A-12
 Transaction Detail Report (STAT09), 1-26, 4-76
- Representation
 definition, 1-4, 4-24
 report for disputed transactions, 4-24
- Retailer cutover
 configuration, 5-11
 PRDF partitioning, 5-2
 retailer cutover timer, 1-22, A-12
 retailers processing day ahead of LN, 1-13
 retailers processing on same day as LN, 1-13
- Retailer cutover timer, 1-22
- Retailer cutover window
 defining, 1-11
 illustration, 1-11
- Retailer Reconciliation Report (B24POS35)
 definition, 1-26
 description of, 4-54
 field descriptions, 4-59
 print location, 4-54
 sample of, 2-22, 4-54
 use in crediting retailers, 3-2
- Retailer settlement
 clearing retailer items, 3-4
 crediting retailers, 3-2
 definition, 3-2
- Retailer sponsor
 as a network participant, 1-6
 clearing retailer items, 3-4
 logical network settlement, 3-8
- Retailers, 1-5
- ## S
- Secondary Settlement process
 params, 5-7
 recovery procedures, B-2
 responsibilities, 1-21
- Secondary Settlement process cutover timer, 1-23
- Secondary Settlement process force cutover, 2-14
- Secondary Settlement processes
 alive timer, 1-23
 communication with Main, 1-20
 initialization, A-5
 log messages, A-3
 number defined, 1-21
 product module ID, A-3
 Senior Secondary Settlement process, 1-21
 startup logic, A-2
 warmboot processing, A-9
 XPNET configuration, A-2
- Senior Secondary Settlement process
 params, 5-8
 responsibilities, 1-21
- Service totals, 2-2
- Settlement
 BASE24-pos system role in, 1-2
 definition of, 1-3
 institution's role, 1-2
 movement of funds, 1-3
- Settlement database, 1-16
- Settlement operations, 6-1
- Settlement processes
 Logical Network Configuration File (LCONF), A-2
 Main Settlement process, 1-20
 multiple copies, A-2
 NCS commands, A-26
 Secondary Settlement processes, 1-20
- Settlement reports
 Clearings Statement Report (SETL01), 1-8, 1-26, 4-71
 Daily Network Settlement Report (B24POS01), 1-6, 1-7, 1-25, 3-4, 3-7, 3-8, 4-3
 Retailer Reconciliation Report (B24POS35), 1-6, 1-26, 2-22, 3-2, 4-54
- Shift, definition, 2-7
- Start reports timer, 1-23
- Statistical reports
 Daily Cardholder Activity Report (B24POS02), 1-7, 1-25, 4-7
 Daily Cardholder Exception Report (B24POS02-EXPT), 1-7, 1-25, 4-18
 Daily Clerk Totals Report (B24POS04), 1-6, 1-25, 4-20

Statistical reports *continued*

- Daily Disputed Transactions Report (B24POS05), [1-6](#), [1-25](#), [3-2](#), [4-24](#)
- Daily Retailer Activity Report (B24POS12), [1-6](#), [1-25](#), [3-2](#), [4-34](#)
- Daily Retailer Exception Detail Report (B24POS12-EXPT), [1-6](#), [1-25](#), [4-45](#)
- Daily Retailer Summary Report (B24POS11), [1-6](#), [1-25](#), [3-2](#), [4-29](#)
- Network Referral Statistical Report (B24POS30), [1-7](#), [1-26](#), [4-47](#)
- Periodic Institution Summary Report (B24POS21), [1-7](#), [1-26](#), [4-63](#)
- Periodic Retailer Summary Report (B24POS22), [1-6](#), [1-26](#), [4-67](#)
- Transaction Detail Report (STAT09), [1-8](#), [1-26](#), [4-76](#)

T

Terminal balancing

- balancing scenarios, [2-9](#)
- balancing transactions, [2-7](#)
- definition, [1-10](#), [2-7](#)
- intelligent terminals, [2-10](#)
- nonintelligent terminals, [2-10](#)

Terminal cutover

- clearing BASE24-pos Terminal Data files totals, [2-16](#)
- close batch transaction, [2-14](#)
- close day transaction, [2-13](#)
- close shift transaction, [2-14](#)
- configuration, [5-9](#)
- cutover methods, [1-11](#), [1-12](#), [2-12](#)
- Device Handler cutover window, [1-11](#)
- Device Handler force cutover, [1-12](#), [2-15](#)
- posting date, [2-11](#)
- retailer cutover window, [1-11](#), [5-9](#)
- Secondary Settlement process force cutover, [1-12](#), [2-14](#)
- terminal cutover options, [2-13](#)

Terminal reconciliation, [2-21](#)

Terminal totals

- batch totals, [2-3](#)
- clearing PTD totals, [2-5](#), [2-16](#)
- clerk totals, [2-4](#)
- current network totals, [2-3](#)
- daily totals, [2-3](#)
- previous network totals, [2-3](#)
- service totals, [2-2](#)
- shift totals, [2-3](#)
- terminal totals, [2-3](#)
- updating service/terminal/clerk totals, [2-5](#)

Terminal types

- intelligent terminals, [2-7](#), [2-10](#)
- nonintelligent terminals, [2-7](#), [2-10](#)

Terminals, [1-5](#)Text commands, [A-26](#)

Timers

- alive timer, [1-23](#), [A-24](#)
- financial institution cutover timer, [1-22](#), [A-15](#)
- logical network cutover timer, [1-22](#), [A-18](#)

Timers *continued*

- midnight timer, [1-22](#), [A-21](#)
- retailer cutover timer, [1-22](#), [A-12](#)
- Secondary Settlement process cutover timer, [1-23](#), [A-25](#)
- start reports timer, [1-23](#), [A-23](#)

Transaction Detail Report (STAT09)

- definition, [1-26](#)
- description of, [4-76](#)
- field descriptions, [4-79](#)
- sample of, [4-76](#)

Transaction profile, [2-30](#)

Transactions

- chargeback, [1-4](#), [4-24](#)
- draft capture, [1-5](#), [2-28](#), [3-2](#)
- non-draft capture, [1-5](#), [2-28](#), [3-2](#)
- representation, [1-4](#), [4-24](#)

U

Usage Accumulation File (UAF)

- description, [1-19](#)
- midnight timer processing, [1-22](#)